

# JavaScript — Learn Everything

The complete course, from `var/let/const` to algorithms, i18n & production.

52 lessons · 5 projects · generated 2026-06-08

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# Lessons

## 01 · VARIABLES — var, let, const

---

```
/* =====
 * 01 · VARIABLES – var, let, const
 * =====
 * Run:  node lessons/01-variables.js
 *
 * WHAT YOU'LL LEARN
 *   • How to declare variables with var, let, and const
 *   • The difference between declaration, initialization, and reassignment
 *   • Scope: function-scope (var) vs block-scope (let/const)
 *   • Hoisting and the "temporal dead zone"
 *
 * RULE OF THUMB
 *   Use `const` by default. Use `let` only when you must reassign.
 *   Avoid `var` – it has confusing scoping and is considered legacy.
 *
 * NOTE: reserved keywords (var, let, const, if, else, for, while, function,
 *       return, ...) cannot be used as variable names.
 * ===== */

// — 1. Declaration & initialization —————
// "Declaration" = create the variable. "Initialization" = give it a value.

let count;           // declaration only – value is `undefined` for now
count = 12;          // initialization
console.log(count);  // => 12

const TAX_RATE = 0.2; // const MUST be declared AND initialized together
console.log(TAX_RATE); // => 0.2

// const RATE;       // ❌ SyntaxError: Missing initializer in const declaration

// — 2. Reassignment —————
let score = 10;
score = 25;          // ✅ let can be reassigned
console.log(score);  // => 25

// TAX_RATE = 0.3;    // ❌ TypeError: Assignment to constant variable
console.log(TAX_RATE); // => 0.2 (const can never be reassigned)

// IMPORTANT: `const` means the *binding* can't change – but if it holds an
// object or array, the contents can still be mutated:
```

```

const user = { name: 'Sam' };
user.name = 'Alex';    // ✅ allowed – we mutate the object, not the binding
console.log(user);     // => { name: 'Alex' }

// — 3. Redeclaration —————
// "Redeclaration" = declaring a variable with the SAME name twice in the SAME
// scope. This is different from reassignment (which only changes the value).

// `var` ALLOWS redeclaration – the second `var` is silently merged into the
// first. This is a common source of bugs, since you can clobber a variable
// without realising it:
var total = 1;
var total = 2;          // ✅ no error – `var` lets you redeclare
console.log(total);     // => 2

// `let` and `const` FORBID redeclaration in the same scope:
// let count = 1;
// let count = 2;       // ❌ SyntaxError: Identifier 'count' has already been declared
//
// const PI = 3.14;
// const PI = 3.15;     // ❌ SyntaxError: Identifier 'PI' has already been declared

// NOTE: redeclaring in a *different* (nested) block is fine – that's a brand new
// variable that "shadows" the outer one, not a redeclaration:
let level = 'outer';
{
  let level = 'inner'; // ✅ separate variable in its own block
  console.log(level);  // => inner
}
console.log(level);    // => outer

// — 4. Scope —————
// `var` is FUNCTION-scoped. `let`/`const` are BLOCK-scoped ({ ... }).

function demoScope() {
  var functionScoped = 'I live in the whole function';
  let blockScoped = 'I live only in my block';
  console.log(functionScoped, '|', blockScoped);
}
demoScope(); // => I live in the whole function | I live only in my block
// console.log(functionScoped); // ❌ ReferenceError – not visible outside

// var ignores block boundaries:
{
  var leaks = 'I escape the block';
}
console.log(leaks); // => I escape the block (this is why var is confusing)

```

```
// let/const respect block boundaries:
{
  let trapped = 'I stay inside';
}
// console.log(trapped); // ❌ ReferenceError: trapped is not defined

// — 5. Hoisting & the Temporal Dead Zone (TDZ) —————
// `var` declarations are "hoisted" to the top and start as `undefined`.
console.log(hoistedVar); // => undefined (declared below, but already exists)
var hoistedVar = 'now I have a value';

// `let`/`const` are also hoisted, but you CANNOT use them before their line.
// That gap is the "Temporal Dead Zone".
try {
  console.log(hoistedLet); // ❌ throws – we're in the TDZ
} catch (err) {
  console.log(err.message); // => Cannot access 'hoistedLet' before initialization
}
let hoistedLet = 'safe to use now';

/* PRACTICE -----
* 1. Declare a const for your name and log it.
* 2. Declare a let counter, set it to 0, then reassign it to 5.
* 3. Try reassigning a const and read the exact error message.
* ----- */
```

## 02 · DATA TYPES

---

```
/* =====
* 02 · DATA TYPES
* =====
* Run: node lessons/02-data-types.js
*
* WHAT YOU'LL LEARN
* • The 7 primitive types and the object type
* • How `typeof` reports each type (and its one famous bug)
* • The difference between value types and reference types
* ===== */

// — 1. The primitive types —————
// Primitives are immutable, single values. There are 7:

const aString = 'hello'; // string
const aNumber = 42; // number (integers AND decimals)
```

```

const aBoolean = true;           // boolean – true / false
const nothing  = null;           // null – an intentional "empty" value
let notSet;                       // undefined – declared but no value yet
const big      = 9007199254740993n; // bigint – integers beyond Number's safe range
const unique   = Symbol('id');    // symbol – a guaranteed-unique value

console.log(typeof aString); // => string
console.log(typeof aNumber); // => number
console.log(typeof aBoolean); // => boolean
console.log(typeof notSet);   // => undefined
console.log(typeof big);      // => bigint
console.log(typeof unique);   // => symbol

// ⚠ FAMOUS BUG: typeof null is "object". It's a 30-year-old mistake, kept for
// backwards compatibility. Just memorize it.
console.log(typeof nothing); // => object (NOT "null!")

// — 2. Objects (everything that isn't a primitive) —————
const obj  = { a: 1 };
const arr  = [1, 2, 3];
const fn   = function () {};

console.log(typeof obj); // => object
console.log(typeof arr); // => object (arrays are objects! use Array.isArray)
console.log(typeof fn);  // => function (functions are callable objects)

console.log(Array.isArray(arr)); // => true ← the correct way to detect arrays
console.log(Array.isArray(obj)); // => false

// — 3. Value vs reference —————
// Primitives are copied BY VALUE – each variable holds its own copy.
let x = 10;
let y = x; // y gets a COPY of 10
y = 99;
console.log(x, y); // => 10 99 (changing y did not affect x)

// Objects/arrays are copied BY REFERENCE – both names point to the SAME data.
const original = { score: 1 };
const alias = original; // alias points to the same object
alias.score = 100;
console.log(original.score); // => 100 (they share one object!)

// To truly copy an object, spread it (shallow copy):
const copy = { ...original };
copy.score = 5;
console.log(original.score, copy.score); // => 100 5

```

```
// — 4. null vs undefined —————
// undefined → "this has not been given a value yet" (usually by JS)
// null      → "this is intentionally empty" (usually set by you)
let unassigned;
console.log(unassigned);    // => undefined
const cleared = null;
console.log(cleared);      // => null

/* PRACTICE -----
* 1. Log the typeof for: "5", 5, true, null, undefined, [], {}.
* 2. Copy an object two ways: by reference and with spread. Mutate each copy
*    and observe which one affects the original.
* ----- */
```

## 03 • OPERATORS

```
/* =====
* 03 • OPERATORS
* =====
* Run:  node lessons/03-operators.js
*
* WHAT YOU'LL LEARN
* • Arithmetic, assignment, comparison, and logical operators
* • Strict (===) vs loose (==) equality – and why you should use ===
* • Modern operators: nullish (??) and optional chaining (?.)
* ===== */

// — 1. Arithmetic —————
console.log(7 + 3); // => 10
console.log(7 - 3); // => 4
console.log(7 * 3); // => 21
console.log(7 / 3); // => 2.333...
console.log(7 % 3); // => 1    (modulo / remainder – great for "is it even?")
console.log(2 ** 3); // => 8    (exponentiation: 2 to the power of 3)

// — 2. Assignment shorthands —————
let n = 10;
n += 5; console.log(n); // => 15    (same as n = n + 5)
n -= 3; console.log(n); // => 12
n *= 2; console.log(n); // => 24
n /= 4; console.log(n); // => 6

// Increment / decrement
let i = 0;
console.log(i++); // => 0    (post: returns THEN increments → i is now 1)
```



```
console.log(++i); // => 2 (pre: increments THEN returns)
```

```
// — 3. Comparison —————
```

```
console.log(5 > 3); // => true
console.log(5 <= 5); // => true
```

```
// === is STRICT equality: compares value AND type, no conversion. Prefer it.
```

```
console.log(5 === 5); // => true
console.log(5 === '5'); // => false (number vs string)
```

```
// == is LOOSE equality: converts types first → surprising results. Avoid it.
```

```
console.log(5 == '5'); // => true (string '5' gets converted to number 5)
console.log(0 == false); // => true (🤔 this is why we avoid ==)
console.log(0 === false); // => false
```

```
// — 4. Logical operators —————
```

```
console.log(true && false); // => false (AND – both must be true)
console.log(true || false); // => true (OR – at least one true)
console.log(!true); // => false (NOT – flips the boolean)
```

```
// && and || actually return one of the OPERANDS, not just true/false:
```

```
console.log('' || 'fallback'); // => fallback (first truthy value)
console.log('hi' && 'world'); // => world (last value if all truthy)
```

```
// — 5. Nullish coalescing (??) —————
```

```
// ?? returns the right side ONLY when the left is null or undefined.
```

```
// Unlike ||, it treats 0 and '' as valid values.
```

```
const volume = 0;
console.log(volume || 5); // => 5 (|| sees 0 as falsy – wrong here!)
console.log(volume ?? 5); // => 0 (?? keeps 0 – correct)
```

```
// — 6. Optional chaining (?.) —————
```

```
// Safely read deep properties without crashing on null/undefined.
```

```
const profile = { name: 'Sam', address: { city: 'Pune' } };
console.log(profile.address?.city); // => Pune
console.log(profile.contact?.email); // => undefined (no crash!)
// console.log(profile.contact.email); // ❌ TypeError without the ?.
```

```
/* PRACTICE -----
```

- \* 1. Use % to check whether 17 is even or odd.
- \* 2. Predict then verify: 0 == '', 0 === '', null == undefined.
- \* 3. Give a setting a default with ?? where 0 should be a valid value.
- \* ----- \*/

## 04 · TYPE CONVERSION & COERCION

---

```
/* =====
 * 04 · TYPE CONVERSION & COERCION
 * =====
 * Run: node lessons/04-type-conversion.js
 *
 * WHAT YOU'LL LEARN
 *   • Explicit conversion (you do it) vs implicit coercion (JS does it)
 *   • Truthy and falsy values
 *   • The classic gotchas that confuse every beginner
 * ===== */

// — 1. Explicit conversion (recommended) —————
// You clearly state the target type. Predictable and readable.

console.log(Number('42'));    // => 42
console.log(Number('42px'));  // => NaN   ("Not a Number" – conversion failed)
console.log(parseInt('42px')); // => 42   (parseInt grabs the leading number)
console.log(parseFloat('3.14rad')); // => 3.14

console.log(String(42));       // => "42"
console.log(String(true));     // => "true"
console.log((42).toString()); // => "42"

console.log(Boolean(1));       // => true
console.log(Boolean(0));       // => false

// — 2. Truthy & Falsy —————
// Every value is either "truthy" or "falsy" in a boolean context (if, &&, etc).
// There are exactly 8 FALSY values – memorize these; everything else is truthy.

const falsyValues = [false, 0, -0, 0n, '', null, undefined, NaN];
falsyValues.forEach(v => console.log(Boolean(v))); // => all false

// Common surprises – these are all TRUTHY:
console.log(Boolean('0'));    // => true   (non-empty string!)
console.log(Boolean('false')); // => true   (non-empty string!)
console.log(Boolean([]));      // => true   (empty array is truthy)
console.log(Boolean({}));      // => true   (empty object is truthy)

// — 3. Implicit coercion (happens automatically) —————
// The + operator: if EITHER side is a string, it concatenates.
console.log(1 + '2');          // => "12"   (number 1 becomes string "1")
console.log('3' + 4);          // => "34"
console.log(1 + 2 + '3');      // => "33"  (left-to-right: 1+2=3, then 3+'3'="33")
```

```
// Other math operators (- * / %) convert strings TO numbers:
console.log('6' - 2);    // => 4
console.log('6' * '2');  // => 12
console.log('a' - 2);    // => NaN

// — 4. The == vs === reminder —————
// == coerces types before comparing → unpredictable. Always prefer ===.
console.log('' == 0);      // => true   🤔
console.log('0' == 0);     // => true   🤔
console.log(null == undefined); // => true   (the one == case that's handy)
console.log(NaN === NaN);  // => false  (NaN is never equal to anything!)
console.log(Number.isNaN(NaN)); // => true  (the correct way to check for NaN)

/* PRACTICE -----
* 1. Convert the string "100" to a number three different ways.
* 2. Predict the output of: [] + [], [] + {}, true + 1. Then run them.
* 3. List all 8 falsy values from memory, then check yourself above.
* ----- */
```

## 05 • STRINGS

```
/* =====
* 05 • STRINGS
* =====
* Run:  node lessons/05-strings.js
*
* WHAT YOU'LL LEARN
* • Creating strings and template literals
* • The most-used string methods
* • Strings are immutable – methods return NEW strings
* ===== */

// — 1. Creating strings —————
const single = 'single quotes';
const double = "double quotes";    // identical to single quotes
const name = 'Sam';
const age = 25;

// Template literals (backticks): allow ${expressions} and multi-line text.
// PREFER these – they're cleaner than + concatenation.
const greeting = `Hi, I'm ${name} and I'm ${age} years old.`;
console.log(greeting); // => Hi, I'm Sam and I'm 25 years old.

const multiline = `Line one
```

```

Line two`;
console.log(multiline);

// — 2. Length & accessing characters —————
const text = 'JavaScript';
console.log(text.length);    // => 10
console.log(text[0]);        // => J      (zero-indexed)
console.log(text.at(-1));    // => t      (.at()) supports negative indexes

// — 3. Searching —————
console.log(text.includes('Script')); // => true
console.log(text.startsWith('Java')); // => true
console.log(text.endsWith('pt'));     // => true
console.log(text.indexOf('S'));       // => 4   (-1 if not found)

// — 4. Transforming (returns a NEW string – originals never change) —————
console.log(text.toUpperCase());       // => JAVASCRIPT
console.log(text.toLowerCase());       // => javascript
console.log(' trim me '.trim());       // => "trim me"
console.log(text.replace('Java', 'Type')); // => TypeScript
console.log('a-b-c'.replaceAll('-', '/')); // => a/b/c
console.log('ab'.repeat(3));           // => ababab

// slice(start, end) – extract a portion (end not included)
console.log(text.slice(0, 4));          // => Java
console.log(text.slice(4));             // => Script
console.log(text.slice(-6));            // => Script (negative = from the end)

// Immutability proof: text itself is unchanged
console.log(text); // => JavaScript

// — 5. Splitting & joining —————
const csv = 'red,green,blue';
const colors = csv.split(',');         // string → array
console.log(colors);                   // => [ 'red', 'green', 'blue' ]
console.log(colors.join(' | '));       // array → string => red | green | blue

// — 6. Padding & numbers-as-strings —————
console.log('5'.padStart(3, '0'));      // => 005   (handy for time/IDs)
console.log('5'.padEnd(3, '.'));        // => 5..

/* PRACTICE -----
 * 1. Build a sentence with a template literal using two variables.

```

```
* 2. Take " Hello World ", trim it, lowercase it, and replace the space.
* 3. Split "2024-06-04" by "-" and log the year, month, day separately.
* ----- */
```

## 06 • NUMBERS & MATH

---

```
/* =====
* 06 • NUMBERS & MATH
* =====
* Run: node lessons/06-numbers-math.js
*
* WHAT YOU'LL LEARN
* • How JS represents numbers (one type for ints and decimals)
* • Rounding and formatting
* • The Math object
* • The floating-point precision gotcha
* ===== */

// — 1. One number type —————
console.log(10);    // integer
console.log(10.5);  // decimal – both are the same `number` type
console.log(1e6);   // => 1000000 (scientific notation)
console.log(0xff);  // => 255      (hexadecimal)

// — 2. Formatting & rounding —————
const price = 19.99876;
console.log(price.toFixed(2)); // => "19.99" (string, 2 decimal places)
console.log(Math.round(2.5));  // => 3      (nearest integer)
console.log(Math.floor(2.9));  // => 2      (round DOWN)
console.log(Math.ceil(2.1));   // => 3      (round UP)
console.log(Math.trunc(-2.7)); // => -2     (drop the decimal, no rounding)

// Format with thousands separators:
console.log((1234567.89).toLocaleString('en-US')); // => 1,234,567.89

// Intl.NumberFormat – the full-control formatter (currency, percent, locales).
// Build it once and reuse it; it handles the currency symbol and rounding.
const usd = new Intl.NumberFormat('en-US', { style: 'currency', currency: 'USD' });
console.log(usd.format(1234.5)); // => $1,234.50

const inr = new Intl.NumberFormat('en-IN', { style: 'currency', currency: 'INR' });
console.log(inr.format(1234567)); // => ₹12,34,567.00 (note Indian digit grouping)

const pct = new Intl.NumberFormat('en-US', { style: 'percent', maximumFractionDigits: 1 });
console.log(pct.format(0.1487)); // => 14.9%
```

```
// — 3. The Math object —————
console.log(Math.max(3, 7, 2)); // => 7
console.log(Math.min(3, 7, 2)); // => 3
console.log(Math.abs(-9));      // => 9
console.log(Math.sqrt(81));     // => 9
console.log(Math.pow(2, 10));    // => 1024 (or use 2 ** 10)
console.log(Math.PI);           // => 3.14159...

// Random number between 0 (inclusive) and 1 (exclusive):
// (commented out so the lesson's output stays stable when you re-run it)
// console.log(Math.random());

// Random integer between min and max (inclusive) – a reusable pattern:
function randomInt(min, max) {
  return Math.floor(Math.random() * (max - min + 1)) + min;
}
// console.log(randomInt(1, 6)); // a dice roll: 1–6

// — 4. Parsing & validating —————
console.log(parseInt('42px', 10)); // => 42 (always pass the radix 10)
console.log(parseFloat('3.14m')); // => 3.14
console.log(Number.isInteger(5));   // => true
console.log(Number.isInteger(5.1)); // => false
console.log(Number.isNaN(0 / 0));   // => true (0/0 is NaN)
console.log(Number.isFinite(1/0));  // => false (1/0 is Infinity)

// — 5. ▲ The floating-point gotcha —————
// Computers store decimals in binary, which can't represent some values exactly.
console.log(0.1 + 0.2);              // => 0.30000000000000004 (not 0.3!)
console.log(0.1 + 0.2 === 0.3);      // => false

// Fixes:
console.log((0.1 + 0.2).toFixed(2)); // => "0.30"
console.log(Math.abs(0.1 + 0.2 - 0.3) < 1e-9); // => true (compare with tolerance)
// For money, work in the smallest unit (cents) as integers to avoid this entirely.

/* PRACTICE -----
* 1. Round 7.456 to 1 decimal place as a string.
* 2. Find the largest of [12, 7, 25, 3] using Math.max with the spread (...).
* 3. Write a function that returns a random integer between 1 and 100.
* ----- */
```

## 07 • CONDITIONALS

---

```
/* =====
 * 07 • CONDITIONALS
 * =====
 * Run:  node lessons/07-conditionals.js
 *
 * WHAT YOU'LL LEARN
 *   • if / else if / else
 *   • The ternary operator (? :)
 *   • switch statements
 *   • Guard clauses for cleaner code
 * ===== */

// — 1. if / else if / else —————
const score = 82;

if (score >= 90) {
  console.log('Grade: A');
} else if (score >= 80) {
  console.log('Grade: B'); // => Grade: B
} else if (score >= 70) {
  console.log('Grade: C');
} else {
  console.log('Grade: F');
}

// — 2. Ternary operator – a one-line if/else that RETURNS a value —————
const age = 20;
const status = age >= 18 ? 'adult' : 'minor';
console.log(status); // => adult

// Great for assigning a value; avoid nesting them deeply (hard to read).
const fee = age < 12 ? 0 : age < 65 ? 100 : 50;
console.log(fee); // => 100

// — 3. Logical operators as conditionals —————
// && runs the right side only if the left is truthy:
const isLoggedIn = true;
isLoggedIn && console.log('Welcome back!'); // => Welcome back!

// || provides a fallback value:
const username = '' || 'Guest';
console.log(username); // => Guest

// — 4. switch – clean alternative for many discrete cases —————
```

```

const day = 'SAT';
let kind;

switch (day) {
  case 'SAT':
  case 'SUN':           // cases can share a body (no break between them)
    kind = 'weekend';
    break;              // ⚠ without break, execution "falls through" to the next case
  case 'MON':
    kind = 'start of week';
    break;
  default:              // runs if nothing matched
    kind = 'weekday';
}
console.log(kind); // => weekend

```

// — 5. Guard clauses – return early to avoid deep nesting —————

// ❌ Nested and hard to read:

```

function checkoutNested(cart) {
  if (cart) {
    if (cart.items.length > 0) {
      return 'Proceeding to payment';
    } else {
      return 'Cart is empty';
    }
  } else {
    return 'No cart';
  }
}

```

// ✅ Flat and readable – handle the exits first:

```

function checkout(cart) {
  if (!cart) return 'No cart';
  if (cart.items.length === 0) return 'Cart is empty';
  return 'Proceeding to payment';
}

console.log(checkout({ items: ['book'] })); // => Proceeding to payment
console.log(checkout({ items: [] }));       // => Cart is empty
console.log(checkout(null));                 // => No cart

```

/\* PRACTICE -----

- \* 1. Write an if/else that logs "even" or "odd" for a number (use % 2).
- \* 2. Rewrite it as a ternary.
- \* 3. Use a switch to map 1–7 to weekday names.
- \* ----- \*/



## 08 · LOOPS

---

```
/* =====
 * 08 · LOOPS
 * =====
 * Run:  node lessons/08-loops.js
 *
 * WHAT YOU'LL LEARN
 *   • for, while, do...while
 *   • for...of (values) vs for...in (keys)
 *   • break and continue
 *   • Which loop to reach for
 * ===== */

// — 1. Classic for loop – when you need an index/counter —————
for (let i = 0; i < 3; i++) {
  console.log('for', i); // => 0, 1, 2
}
// Anatomy:  for (initialize; condition; afterEachLoop) { body }
```

```
// — 2. while – loop until a condition becomes false —————
let count = 3;
while (count > 0) {
  console.log('while', count); // => 3, 2, 1
  count--;
}
```

```
// — 3. do...while – runs the body AT LEAST once, then checks —————
let n = 0;
do {
  console.log('do-while runs once even though condition is false');
} while (n > 0);
```

```
// — 4. for...of – iterate over VALUES (arrays, strings, Sets, Maps) —————
// This is the modern go-to for looping over a collection.
const fruits = ['apple', 'banana', 'cherry'];
for (const fruit of fruits) {
  console.log(fruit); // => apple, banana, cherry
}
```

```
// Need the index too? Use .entries():
for (const [index, fruit] of fruits.entries()) {
  console.log(index, fruit); // => 0 apple, 1 banana, 2 cherry
}
```

```
// Strings are iterable too:
for (const char of 'hi') {
```

```

    console.log(char); // => h, i
  }

// — 5. for...in – iterate over KEYS of an object —————
// Use this for objects. (Avoid it for arrays – for...of is better there.)
const user = { name: 'Sam', age: 25, city: 'Pune' };
for (const key in user) {
  console.log(`${key}: ${user[key]}`); // => name: Sam, age: 25, city: Pune
}

// — 6. break & continue —————
for (let i = 1; i <= 5; i++) {
  if (i === 4) break;      // stop the loop entirely
  if (i === 2) continue;   // skip the rest of THIS iteration only
  console.log('flow', i);  // => 1, 3
}

// — 7. Array helper loops (preview of lesson 13) —————
// forEach runs a function for each element – no manual index needed.
fruits.forEach((fruit, i) => console.log(i, fruit));

/* WHICH LOOP? -----
 *   • Looping over array/string values ..... for...of
 *   • Looping over object keys ..... for...in
 *   • Need a numeric counter / step ..... classic for
 *   • Repeat until a condition ..... while
 *   • Transforming an array into a new one ..... .map() (lesson 13)
 * ----- */

/* PRACTICE -----
 *   1. Print 1–10 with a for loop, but skip multiples of 3 (continue).
 *   2. Sum all numbers in [4, 8, 15, 16, 23, 42] with for...of.
 *   3. Loop over an object's keys and values with for...in.
 * ----- */

```

## 09 • FUNCTIONS

---

```

/* =====
 * 09 • FUNCTIONS
 * =====
 * Run:  node lessons/09-functions.js
 *
 * WHAT YOU'LL LEARN

```

```

*   • Function declarations vs expressions vs arrow functions
*   • Parameters: default values, rest parameters
*   • Returning values
*   • Functions are "first-class" – they're values you can pass around
*   • Recursion – a function that calls itself
*   ===== */

// — 1. Function declaration —————
// Hoisted – you can call it BEFORE it's defined in the file.
console.log(add(2, 3)); // => 5 (works even though add is defined below)

function add(a, b) {
  return a + b; // `return` sends a value back to the caller
}
// A function with no `return` returns undefined.

// — 2. Function expression —————
// Stored in a variable. NOT hoisted – must be defined before use.
const multiply = function (a, b) {
  return a * b;
};
console.log(multiply(4, 5)); // => 20

// — 3. Arrow functions – shorter syntax —————
const square = (x) => x * x;           // single expression → implicit return
console.log(square(6));               // => 36

const subtract = (a, b) => a - b;      // multiple params need parentheses
console.log(subtract(10, 4));         // => 6

const greet = () => 'Hello!';         // no params → empty parentheses
console.log(greet());                 // => Hello!

// Need multiple lines? Use a block { } and an explicit return:
const describe = (name, age) => {
  const label = `${name} is ${age}`;
  return label;
};
console.log(describe('Sam', 25));     // => Sam is 25
// NOTE: arrows handle `this` differently – covered in lesson 11.

// — 4. Default parameters —————
function greetUser(name = 'Guest') {
  return `Welcome, ${name}!`;
}
console.log(greetUser('Alex')); // => Welcome, Alex!

```

```
console.log(greetUser());           // => Welcome, Guest! (default kicks in)
```

```
// — 5. Rest parameters – collect "the rest" into an array —————  
function sum(...numbers) {          // numbers is a real array: [1, 2, 3, 4]  
  return numbers.reduce((total, n) => total + n, 0);  
}  
console.log(sum(1, 2, 3, 4));        // => 10  
console.log(sum(5, 5));              // => 10
```

```
// — 6. Functions are values (first-class) —————  
// You can store them, pass them as arguments, and return them.  
function applyTwice(fn, value) {  
  return fn(fn(value));             // call the passed-in function twice  
}  
console.log(applyTwice(square, 2)); // => 16 (square(square(2)) = square(4))
```

```
// A function returned by another function (a "factory"):  
function makeMultiplier(factor) {  
  return (x) => x * factor;  
}  
const triple = makeMultiplier(3);  
console.log(triple(5)); // => 15
```

```
// — 7. Recursion – a function that calls itself —————  
// Every recursion needs TWO things:  
// 1. a BASE CASE that stops it (no base case → "Maximum call stack" crash)  
// 2. a step that moves CLOSER to the base case on each call  
function factorial(n) {  
  if (n <= 1) return 1;             // base case – stop here  
  return n * factorial(n - 1);      // recursive step – n gets smaller each time  
}  
console.log(factorial(5)); // => 120 (5 * 4 * 3 * 2 * 1)
```

```
// Anything you can do with a loop, you can do with recursion:  
function sumTo(n) {  
  if (n === 0) return 0;            // base case  
  return n + sumTo(n - 1);  
}  
console.log(sumTo(4)); // => 10 (4 + 3 + 2 + 1 + 0)
```

```
// Recursion SHINES on nested/tree-shaped data, where loops get awkward.  
// Here we add up every number in an arbitrarily deep nested array:  
function deepSum(arr) {  
  let total = 0;  
  for (const item of arr) {  
    // if it's an array, recurse into it; otherwise add the number
```

```

    total += Array.isArray(item) ? deepSum(item) : item;
  }
  return total;
}
console.log(deepSum([1, [2, [3, 4], 5], [6]])); // => 21

// Mental model: each call waits ("on the call stack") for the deeper call to
// finish, then combines the result – like opening boxes inside boxes, then
// re-stacking them on the way back out.

/* PRACTICE -----
* 1. Write isEven(n) three ways: declaration, expression, arrow.
* 2. Write a function with a default greeting parameter.
* 3. Write average(...nums) that returns the mean of any count of numbers.
* 4. Write a recursive countdown(n) that logs n, n-1, ... down to 0.
* 5. Write a recursive power(base, exp) – no Math.pow, no ** operator.
* ----- */

```

## 10 · SCOPE & CLOSURES

---

```

/* =====
* 10 · SCOPE & CLOSURES
* =====
* Run:  node lessons/10-scope-closures.js
*
* WHAT YOU'LL LEARN
* • Global, function, and block scope
* • Lexical scope (inner functions see outer variables)
* • Closures – the single most important "aha" concept in JS
* ===== */

// — 1. The three scopes —————
const globalVar = 'I am global'; // visible everywhere

function outer() {
  const functionVar = 'I am in outer()'; // visible inside outer() only
  console.log(globalVar); // ✅ inner code can see outer scopes
  console.log(functionVar); // ✅

  if (true) {
    const blockVar = 'I am in this block'; // visible in this { } only
    console.log(blockVar); // ✅
  }
  // console.log(blockVar); // ❌ ReferenceError – block-scoped
}
outer();

```

```
// console.log(functionVar); // ❌ ReferenceError – not visible outside
```

```
// — 2. Lexical scope —————
```

```
// "Lexical" = scope is determined by WHERE code is written, not where it runs.
```

```
// Inner functions always have access to the variables of their parents.
```

```
function grandparent() {  
  const family = 'Smith';  
  function parent() {  
    function child() {  
      return family; // reaches all the way up to grandparent's variable  
    }  
    return child();  
  }  
  return parent();  
}  
console.log(grandparent()); // => Smith
```

```
// — 3. Closures —————
```

```
// A closure = a function that "remembers" the variables from where it was
```

```
// created, EVEN AFTER that outer function has finished running.
```

```
function makeCounter() {  
  let count = 0; // this variable is "enclosed" by the inner function  
  return function () {  
    count++; // it keeps living between calls  
    return count;  
  };  
}
```

```
const counter = makeCounter();  
console.log(counter()); // => 1  
console.log(counter()); // => 2  
console.log(counter()); // => 3 (count persists – that's the closure!)
```

```
// Each closure gets its OWN private copy:
```

```
const counterA = makeCounter();  
const counterB = makeCounter();  
console.log(counterA()); // => 1  
console.log(counterA()); // => 2  
console.log(counterB()); // => 1 (independent from counterA)
```

```
// — 4. Practical use: data privacy —————
```

```
// Closures let you create "private" variables that can't be touched directly.
```

```
function createBankAccount(initial) {  
  let balance = initial; // private – no outside code can read/write it  
  return {
```

```

    deposit(amount) { balance += amount; return balance; },
    withdraw(amount) {
      if (amount > balance) return 'Insufficient funds';
      balance -= amount;
      return balance;
    },
    getBalance() { return balance; },
  };
}

```

```

const account = createBankAccount(100);
console.log(account.deposit(50)); // => 150
console.log(account.withdraw(30)); // => 120
console.log(account.getBalance()); // => 120
// account.balance is undefined – it's truly private.
console.log(account.balance);      // => undefined

```

```

/* PRACTICE -----
*   1. Write makeAdder(x) that returns a function adding x to its argument.
*     const add5 = makeAdder(5); add5(10) === 15
*   2. Build a once(fn) wrapper that runs fn only the first time it's called.
*   3. Explain in your own words why `count` survives between counter() calls.
* ----- */

```

## 11 • THE `this` KEYWORD

---

```

/* =====
* 11 • THE `this` KEYWORD
* =====
* Run: node lessons/11-this-keyword.js
*
* WHAT YOU'LL LEARN
* • What `this` refers to (it depends on HOW a function is called)
* • Why arrow functions don't have their own `this`
* • call, apply, and bind
*
* BIG IDEA: `this` is NOT decided where a function is written – it's decided
*           by HOW the function is called.
* ===== */

// — 1. `this` inside an object method = the object —————
const user = {
  name: 'Sam',
  greet() {
    return `Hi, I'm ${this.name}`; // `this` = the object the method is called on
  },
};

```

```
console.log(user.greet()); // => Hi, I'm Sam
```

// — 2. The classic gotcha: a regular function inside a method —————

```
const team = {
  name: 'Rockets',
  members: ['Ana', 'Bob'],
  listBroken() {
    // ❌ A normal function gets its OWN `this` (undefined in strict mode),
    //    so `this.name` is lost here.
    this.members.forEach(function (m) {
      // console.log(`${m} is on ${this.name}`); // would throw / be undefined
    });
  },
  listFixed() {
    // ✅ Arrow functions DON'T have their own `this` — they reuse the
    //    surrounding `this` (the team object). This is why arrows shine here.
    this.members.forEach((m) => {
      console.log(`${m} is on ${this.name}`);
    });
  },
};
team.listFixed(); // => Ana is on Rockets / Bob is on Rockets
```

// — 3. Arrow functions inherit `this` —————

```
const counter = {
  count: 0,
  // ❌ Don't use an arrow for the METHOD itself — it would grab the outer
  //    (module/global) `this`, not the object.
  increment() {
    const helper = () => ++this.count; // arrow keeps `this` = counter
    return helper();
  },
};
console.log(counter.increment()); // => 1
console.log(counter.increment()); // => 2
```

// — 4. call / apply / bind — manually set `this` —————

```
function introduce(greeting) {
  return `${greeting}, I'm ${this.name}`;
}
const person = { name: 'Alex' };

// call: run now, pass `this` then arguments one by one
console.log(introduce.call(person, 'Hello')); // => Hello, I'm Alex

// apply: like call, but arguments come as an array
```



```

console.log(introduce.apply(person, ['Hi'])); // => Hi, I'm Alex

// bind: DON'T run now – return a NEW function with `this` permanently fixed
const boundIntro = introduce.bind(person);
console.log(boundIntro('Hey')); // => Hey, I'm Alex

/* QUICK REFERENCE -----
*   obj.method()          → this = obj
*   standalone()         → this = undefined (strict) or globalThis
*   arrow function       → this = whatever it was in the enclosing scope
*   fn.call/apply/bind   → this = the object you provide
* ----- */

/* PRACTICE -----
*   1. Make an object `dog` with a name and a bark() method using `this`.
*   2. Break it by extracting the method: const b = dog.bark; b(); – see why.
*   3. Fix it with .bind(dog).
* ----- */

```

## 12 · ARRAYS — basics

---

```

/* =====
* 12 · ARRAYS – basics
* =====
* Run:  node lessons/12-arrays.js
*
* WHAT YOU'LL LEARN
*   • Creating and accessing arrays
*   • Adding / removing elements (the mutating methods)
*   • Mutating vs non-mutating operations
* (Transform methods like map/filter/reduce get their own lesson – 13.)
* ===== */

// — 1. Creating & accessing -----
const fruits = ['apple', 'banana', 'cherry'];
console.log(fruits[0]);           // => apple   (zero-indexed)
console.log(fruits.at(-1));       // => cherry  (.at supports negative indexes)
console.log(fruits.length);       // => 3

// Arrays can hold any mix of types (but usually keep them consistent):
const mixed = [1, 'two', true, null, { id: 1 }];
console.log(mixed.length);        // => 5

// — 2. Add / remove at the ENDS (these MUTATE the array) -----
const stack = ['a', 'b'];

```

```
stack.push('c');          // add to end      → ['a','b','c']
console.log(stack.pop()); // remove from end → returns 'c', array is ['a','b']
stack.unshift('z');       // add to start    → ['z','a','b']
console.log(stack.shift()); // remove from start → returns 'z', array is ['a','b']
console.log(stack);       // => [ 'a', 'b' ]
```

```
// — 3. splice – remove/insert ANYWHERE (mutates) —————
const nums = [1, 2, 3, 4, 5];
// splice(startIndex, deleteCount, ...itemsToInsert)
nums.splice(1, 2);          // remove 2 items starting at index 1
console.log(nums);          // => [ 1, 4, 5 ]
nums.splice(1, 0, 'x', 'y'); // insert without deleting
console.log(nums);          // => [ 1, 'x', 'y', 4, 5 ]
```

```
// — 4. slice – COPY a portion (does NOT mutate) —————
const letters = ['a', 'b', 'c', 'd'];
console.log(letters.slice(1, 3)); // => [ 'b', 'c' ] (end not included)
console.log(letters.slice(-2));   // => [ 'c', 'd' ]
console.log(letters);             // => unchanged: [ 'a','b','c','d' ]
```

```
// — 5. Searching —————
const colors = ['red', 'green', 'blue'];
console.log(colors.includes('green')); // => true
console.log(colors.indexOf('blue'));   // => 2 (-1 if not found)
```

```
// — 6. Combining & converting —————
console.log([1, 2].concat([3, 4]));    // => [1,2,3,4] (non-mutating join)
console.log([1, 2, 3].join('-'));      // => "1-2-3" (array → string)
console.log([3, 1, 2].sort());         // => [1,2,3] (⚠ MUTATES, sorts as text)
console.log([1, 2, 3].reverse());      // => [3,2,1] (⚠ MUTATES)
```

```
// ⚠ Default sort is alphabetical! For numbers, pass a comparator:
console.log([10, 2, 1].sort());         // => [1, 10, 2] (wrong!)
console.log([10, 2, 1].sort((a, b) => a - b)); // => [1, 2, 10] (correct)
```

```
// — 7. Copying safely (avoid accidental shared references) —————
const source = [1, 2, 3];
const copy1 = [...source];             // spread copy
const copy2 = source.slice();          // slice copy
copy1.push(4);
console.log(source, copy1);            // => [1,2,3] [1,2,3,4] (source untouched)
```

```
/* MUTATING vs NON-MUTATING -----
```

```

*   MUTATES:      push pop shift unshift splice sort reverse fill
*   RETURNS NEW:  slice concat map filter [...spread] join
*   When in doubt, prefer non-mutating – it prevents sneaky bugs.
*   ----- */

/* PRACTICE -----
*   1. Start with [], push 3 items, then remove the first one.
*   2. Sort [42, 7, 100, 1] numerically ascending and descending.
*   3. Copy an array, mutate the copy, and prove the original is unchanged.
*   ----- */

```

## 13 · ARRAY METHODS — map, filter, reduce & friends

---

```

/* =====
* 13 · ARRAY METHODS – map, filter, reduce & friends
* =====
* Run:  node lessons/13-array-methods.js
*
* WHAT YOU'LL LEARN
*   • The "iteration" methods every JS developer uses daily
*   • How they take a CALLBACK function
*   • Chaining them together for clean data transformations
*
* These are NON-MUTATING – they return new arrays/values and leave the
* original alone. This is the heart of modern, readable JavaScript.
* ===== */

const numbers = [1, 2, 3, 4, 5, 6];

// — 1. forEach – do something for each item (returns nothing) —————
numbers.forEach((n) => console.log('item:', n)); // side effects only

// — 2. map – transform EACH item into a new array (same length) —————
const doubled = numbers.map((n) => n * 2);
console.log(doubled); // => [ 2, 4, 6, 8, 10, 12 ]

const names = ['ana', 'bob'];
console.log(names.map((name) => name.toUpperCase())); // => [ 'ANA', 'BOB' ]

// — 3. filter – keep only items that pass a test (length may shrink) —————
const evens = numbers.filter((n) => n % 2 === 0);
console.log(evens); // => [ 2, 4, 6 ]

// — 4. reduce – boil an array down to a SINGLE value —————

```

```

// reduce((accumulator, current) => ..., startingValue)
const total = numbers.reduce((sum, n) => sum + n, 0);
console.log(total); // => 21

// reduce is powerful – e.g. find the max, or count occurrences:
const max = numbers.reduce((biggest, n) => (n > biggest ? n : biggest));
console.log(max); // => 6

const letters = ['a', 'b', 'a', 'c', 'b', 'a'];
const counts = letters.reduce((tally, letter) => {
  tally[letter] = (tally[letter] || 0) + 1;
  return tally;
}, {});
console.log(counts); // => { a: 3, b: 2, c: 1 }

// — 5. find / findIndex – get the FIRST match —————
const users = [
  { id: 1, name: 'Ana' },
  { id: 2, name: 'Bob' },
];
console.log(users.find((u) => u.id === 2)); // => { id: 2, name: 'Bob' }
console.log(users.findIndex((u) => u.id === 2)); // => 1
// (find returns undefined and findIndex returns -1 when nothing matches)

// — 6. some / every – boolean tests across the array —————
console.log(numbers.some((n) => n > 5)); // => true (at least one matches)
console.log(numbers.every((n) => n > 0)); // => true (all match)

// — 7. sort (non-mutating with a copy) —————
const scores = [42, 7, 100, 1];
const sorted = [...scores].sort((a, b) => a - b); // copy first to avoid mutation
console.log(sorted); // => [ 1, 7, 42, 100 ]
console.log(scores); // => [ 42, 7, 100, 1 ] (original preserved)

// — 8. flat / flatMap —————
console.log([1, [2, [3]]].flat()); // => [ 1, 2, [3] ] (one level)
console.log([1, [2, [3]]].flat(Infinity)); // => [ 1, 2, 3 ] (all levels)

// — 9. Chaining – the real power —————
// "From these orders, get the total of completed orders over $50."
const orders = [
  { item: 'pen', price: 5, done: true },
  { item: 'laptop', price: 900, done: true },
  { item: 'mouse', price: 60, done: false },

```

```

    { item: 'monitor', price: 200, done: true },
  ];

const revenue = orders
  .filter((o) => o.done)           // keep completed
  .filter((o) => o.price > 50)     // keep > $50
  .map((o) => o.price)            // get just the prices
  .reduce((sum, p) => sum + p, 0); // add them up
console.log(revenue); // => 1100

/* PRACTICE -----
* 1. From [1..10], map to squares, then filter to keep only even squares.
* 2. Use reduce to find the longest word in ['hi', 'hello', 'hey', 'howdy'].
* 3. From a list of {name, age} people, get the names of everyone 18+.
* ----- */

```

## 14 • OBJECTS

---

```

/* =====
* 14 • OBJECTS
* =====
* Run:  node lessons/14-objects.js
*
* WHAT YOU'LL LEARN
* • Creating objects, reading/writing/deleting properties
* • Methods and shorthand syntax
* • Iterating with Object.keys / values / entries
* • Copying and merging objects
* • Getters/setters, property descriptors, and freezing
* • JSON – turning objects into text and back
* ===== */

// — 1. Creating & accessing -----
const user = {
  name: 'Sam',
  age: 25,
  'favorite color': 'blue', // keys with spaces need quotes
};

console.log(user.name);           // => Sam   (dot notation – preferred)
console.log(user['age']);         // => 25    (bracket notation)
console.log(user['favorite color']); // => blue (brackets required for odd keys)

// Bracket notation is essential when the key is in a variable:
const key = 'name';
console.log(user[key]);           // => Sam

```

```
// — 2. Adding, updating, deleting —————  
user.email = 'sam@example.com'; // add  
user.age = 26; // update  
delete user['favorite color']; // delete  
console.log(user); // => { name: 'Sam', age: 26, email: 'sam@example.com' }
```

```
// Check if a key exists:  
console.log('email' in user); // => true  
console.log(user.phone === undefined); // => true (phone doesn't exist)
```

```
// — 3. Methods & `this` —————  
const account = {  
  owner: 'Alex',  
  balance: 100,  
  // method shorthand (no `function` keyword needed):  
  describe() {  
    return `${this.owner} has ${this.balance}`;  
  },  
};  
console.log(account.describe()); // => Alex has $100
```

```
// — 4. Shorthand property names —————  
const name = 'Dana';  
const age = 30;  
// Instead of { name: name, age: age }:  
const person = { name, age };  
console.log(person); // => { name: 'Dana', age: 30 }
```

```
// — 5. Computed property names —————  
const field = 'score';  
const result = { [field]: 95 }; // the key comes from a variable  
console.log(result); // => { score: 95 }
```

```
// — 6. Iterating over an object —————  
const car = { make: 'Toyota', model: 'Corolla', year: 2020 };  
  
console.log(Object.keys(car)); // => [ 'make', 'model', 'year' ]  
console.log(Object.values(car)); // => [ 'Toyota', 'Corolla', 2020 ]  
console.log(Object.entries(car)); // => [ ['make','Toyota'], ['model','Corolla'], ['year',2020] ]  
  
// entries() pairs nicely with for...of and destructuring:  
for (const [k, v] of Object.entries(car)) {  
  console.log(`${k} = ${v}`); // => make = Toyota, etc.
```

```
}
```

```
// — 7. Copying & merging (shallow) —————
```

```
const defaults = { theme: 'light', fontSize: 14 };
const overrides = { fontSize: 18 };
```

```
const settings = { ...defaults, ...overrides }; // later keys win
console.log(settings); // => { theme: 'light', fontSize: 18 }
```

```
// Object.assign does the same (mutates the first argument):
const merged = Object.assign({}, defaults, overrides);
console.log(merged); // => { theme: 'light', fontSize: 18 }
```

```
// ⚠ Spread is a SHALLOW copy – nested objects are still shared:
```

```
const original = { nested: { a: 1 } };
const shallow = { ...original };
shallow.nested.a = 99;
console.log(original.nested.a); // => 99 (the nested object is shared!)
// For a deep copy of plain data: structuredClone(original)
const deep = structuredClone(original);
deep.nested.a = 1;
console.log(original.nested.a, deep.nested.a); // => 99 1
```

```
// — 8. Nested objects & optional chaining —————
```

```
const company = { ceo: { name: 'Lee', contact: { city: 'Pune' } } };
console.log(company.ceo.contact.city); // => Pune
console.log(company.cfo?.contact?.city); // => undefined (no crash)
```

```
// — 9. Getters & setters on plain objects —————
```

```
// A getter/setter looks like a normal property from the outside, but runs a
// function. Great for computed/derived values. (Classes do this too – lesson 17.)
const temperature = {
  celsius: 25,
  get fahrenheit() { // read like a property: temperature.fahrenheit
    return this.celsius * 1.8 + 32;
  },
  set fahrenheit(f) { // write like a property: temperature.fahrenheit = 212
    this.celsius = (f - 32) / 1.8;
  },
};
console.log(temperature.fahrenheit); // => 77 (computed, no parentheses)
temperature.fahrenheit = 212;
console.log(temperature.celsius); // => 100 (the setter ran)
```

```
// — 10. Property descriptors & Object.defineProperty —————
```

```

// Every property secretly has flags: writable, enumerable, configurable.
// Normal assignment makes all three true. defineProperty lets you control them.
const config = {};
Object.defineProperty(config, 'VERSION', {
  value: '1.0',
  writable: false,    // can't be reassigned
  enumerable: false,  // hidden from Object.keys / for...in / JSON
  configurable: false, // can't be deleted or redefined
});
config.VERSION = '2.0';           // silently ignored (throws in strict mode)
console.log(config.VERSION);       // => 1.0
console.log(Object.keys(config));  // => [] (it's non-enumerable)
console.log(Object.getOwnPropertyDescriptor(config, 'VERSION'));
// => { value: '1.0', writable: false, enumerable: false, configurable: false }

// — 11. Locking objects: freeze & seal —————
// freeze: no add, no remove, no change – fully read-only (shallow).
const settingsFrozen = Object.freeze({ theme: 'dark', fontSize: 14 });
settingsFrozen.fontSize = 99;      // ignored (throws in strict mode)
settingsFrozen.newKey = 1;         // ignored – can't add either
console.log(settingsFrozen, Object.isFrozen(settingsFrozen)); // => {...unchanged}, true

// seal: can CHANGE existing values, but can't ADD or REMOVE keys.
const sealed = Object.seal({ count: 0 });
sealed.count = 5;                  // ✅ allowed (existing key)
sealed.extra = 1;                  // ❌ ignored (can't add)
console.log(sealed, Object.isSealed(sealed)); // => { count: 5 }, true

// ⚠ freeze is SHALLOW – nested objects can still change.
// For deep immutability, freeze recursively or keep data flat.

// — 12. JSON – objects as text —————
// JSON (JavaScript Object Notation) is how objects travel: saved to a file,
// stored in localStorage, or sent over the network. It is plain TEXT, not an
// object. Two functions convert back and forth:
const profile = { name: 'Sam', age: 25, tags: ['js', 'css'], active: true };

const text = JSON.stringify(profile); // object → JSON string
console.log(text); // => {"name":"Sam","age":25,"tags":["js","css"],"active":true}

const back = JSON.parse(text);        // JSON string → object (a NEW object)
console.log(back.name, back.tags[0]); // => Sam js

// Pretty-print: pass a "space" argument (great for logs and files):
console.log(JSON.stringify(profile, null, 2)); // 2-space indented, multi-line

// Pick/transform fields with a replacer (array = allowlist, or a function):

```



```

console.log(JSON.stringify(profile, ['name', 'age'])); // => {"name":"Sam","age":25}

// Revive values while parsing with a reviver (e.g. rebuild Dates):
const order = JSON.parse('{ "id":1,"when":"2026-06-05T00:00:00.000Z"}', (k, v) =>
  k === 'when' ? new Date(v) : v
);
console.log(order.when instanceof Date); // => true

// △ WHAT JSON DROPS – it only understands plain data:
//   • functions, undefined, and Symbols are OMITTED from objects
//   • Date becomes a STRING (use a reviver to turn it back, as above)
//   • Map/Set become {} ; NaN and Infinity become null
const messy = { fn() {}, skip: undefined, when: new Date(0), n: NaN };
console.log(JSON.stringify(messy)); // => {"when":"1970-01-01T00:00:00.000Z","n":null}

// △ ALWAYS guard JSON.parse – bad text throws:
try {
  JSON.parse('{ not valid }');
} catch (err) {
  console.log('parse failed:', err.name); // => SyntaxError
}

// The "JSON deep-copy trick" – quick, but only for plain JSON-safe data
// (loses the same things listed above). Prefer structuredClone for general use.
const copy = JSON.parse(JSON.stringify(profile));
copy.tags.push('html');
console.log(profile.tags.length, copy.tags.length); // => 2 3 (independent)

/* PRACTICE -----
*   1. Build a `book` object with title, author, and a summary() method.
*   2. Loop over its entries and print "key: value" lines.
*   3. Merge two settings objects so the second overrides the first.
*   4. JSON.stringify the book, then JSON.parse it back – notice the
*       summary() method is gone. Why? (JSON only stores plain data.)
* ----- */

```

## 15 • DESTRUCTURING & SPREAD / REST

---

```

/* =====
* 15 • DESTRUCTURING & SPREAD / REST
* =====
* Run: node lessons/15-destructuring-spread.js
*
* WHAT YOU'LL LEARN
*   • Pulling values out of arrays and objects in one line (destructuring)
*   • Spreading (...) to expand arrays/objects

```

```

*   • Rest (...) to collect leftovers
* These show up EVERYWHERE in modern JS – learn them well.
* ===== */

// — 1. Array destructuring —————
const rgb = [255, 128, 0];
const [red, green, blue] = rgb; // unpack by position
console.log(red, green, blue); // => 255 128 0

// Skip items with empty commas:
const [first, , third] = ['a', 'b', 'c'];
console.log(first, third); // => a c

// Default values for missing items:
const [x = 0, y = 0, z = 0] = [10, 20];
console.log(x, y, z); // => 10 20 0

// Swap variables without a temp variable:
let a = 1, b = 2;
[a, b] = [b, a];
console.log(a, b); // => 2 1

// — 2. Object destructuring —————
const user = { name: 'Sam', age: 25, city: 'Pune' };
const { name, age } = user; // unpack by KEY name (order doesn't matter)
console.log(name, age); // => Sam 25

// Rename while unpacking:
const { name: fullName } = user;
console.log(fullName); // => Sam

// Defaults for missing keys:
const { country = 'India' } = user;
console.log(country); // => India

// Nested destructuring:
const order = { id: 7, customer: { email: 'a@b.com' } };
const { customer: { email } } = order;
console.log(email); // => a@b.com

// — 3. Destructuring in function parameters (very common) —————
// Instead of accepting an object and reading props inside:
function greet({ name, greeting = 'Hello' }) {
  return `${greeting}, ${name}!`;
}

console.log(greet({ name: 'Ana' })); // => Hello, Ana!
console.log(greet({ name: 'Bob', greeting: 'Hey' })); // => Hey, Bob!

```

// — 4. Spread (...) – expand an iterable into individual elements —————

```
// Combine arrays:
const lows = [1, 2];
const highs = [3, 4];
console.log([...lows, ...highs]); // => [ 1, 2, 3, 4 ]
```

```
// Copy an array (shallow):
const copy = [...lows];
```

```
// Pass array items as separate arguments:
console.log(Math.max(...[5, 9, 2])); // => 9
```

```
// Merge / clone objects (later keys win):
const base = { theme: 'light', size: 14 };
const custom = { ...base, size: 18 }; // => { theme: 'light', size: 18 }
console.log(custom);
```

```
// Spread a string into characters:
console.log([...'abc']); // => [ 'a', 'b', 'c' ]
```

// — 5. Rest (...) – collect "the rest" into an array/object —————

```
// In array destructuring:
const [winner, ...runnersUp] = ['gold', 'silver', 'bronze'];
console.log(winner); // => gold
console.log(runnersUp); // => [ 'silver', 'bronze' ]
```

```
// In object destructuring:
const { id, ...details } = { id: 1, name: 'Sam', age: 25 };
console.log(id); // => 1
console.log(details); // => { name: 'Sam', age: 25 }
```

```
// In function parameters (gather all arguments):
function sum(...nums) {
  return nums.reduce((t, n) => t + n, 0);
}
console.log(sum(1, 2, 3, 4)); // => 10
```

```
// SPREAD vs REST: same `...` symbol, opposite jobs.
// Spread → EXPANDS something out (used in calls/literals).
// Rest → COLLECTS many into one (used in parameters/patterns).
```

```
/* PRACTICE -----
* 1. Destructure { title, year } from a movie object, defaulting year to 2024.
* 2. Merge two arrays and dedupe them with [...new Set([...a, ...b])].
* 3. Write a function printFirstAndRest(first, ...rest) and test it.
```

```
* ----- */
```

## 16 · SETS & MAPS

---

```
/* =====
 * 16 · SETS & MAPS
 * =====
 * Run:  node lessons/16-sets-maps.js
 *
 * WHAT YOU'LL LEARN
 *   • Set – a collection of UNIQUE values
 *   • Map – key/value pairs where keys can be ANY type
 *   • When to reach for these instead of arrays/objects
 * ===== */

// — 1. Set – unique values, fast membership checks —————
const ids = new Set();
ids.add(1);
ids.add(2);
ids.add(2);  // ignored – already present
ids.add(3);
console.log(ids);      // => Set(3) { 1, 2, 3 }
console.log(ids.size); // => 3
console.log(ids.has(2)); // => true  (fast lookup, unlike array.includes)
ids.delete(1);
console.log([...ids]);  // => [ 2, 3 ] (spread to convert back to an array)

// ★ Most common use: remove duplicates from an array in one line.
const withDups = [1, 2, 2, 3, 3, 3, 4];
const unique = [...new Set(withDups)];
console.log(unique); // => [ 1, 2, 3, 4 ]

// Iterating a Set:
for (const id of ids) console.log('id:', id);

// — 2. Map – key/value pairs with ANY key type —————
// Unlike objects (whose keys are always strings), Map keys can be numbers,
// objects, functions – anything. And it remembers insertion order.
const scores = new Map();
scores.set('Ana', 90);
scores.set('Bob', 75);
console.log(scores.get('Ana')); // => 90
console.log(scores.has('Bob')); // => true
console.log(scores.size);       // => 2
scores.delete('Bob');
```

```
// Object keys (impossible with a plain object):
const userA = { id: 1 };
const lastSeen = new Map();
lastSeen.set(userA, '2026-06-04');
console.log(lastSeen.get(userA)); // => 2026-06-04

// Iterating a Map (entries are [key, value] pairs):
const colors = new Map([
  ['red', '#f00'],
  ['green', '#0f0'],
]);
for (const [name, hex] of colors) {
  console.log(`${name} → ${hex}`); // => red → #f00, green → #0f0
}
console.log([...colors.keys()]); // => [ 'red', 'green' ]
console.log([...colors.values()]); // => [ '#f00', '#0f0' ]

// — 3. Object vs Map – which to use? —————
//   Use a plain OBJECT when:
//     • keys are fixed, known strings ("structured record")
//     • you want JSON serialization (JSON.stringify works on objects)
//   Use a MAP when:
//     • keys are dynamic, or not strings
//     • you frequently add/remove entries
//     • you need a reliable .size and guaranteed insertion order

// Convert between them:
const obj = Object.fromEntries(colors); // Map → object
console.log(obj); // => { red: '#f00', green: '#0f0' }
const backToMap = new Map(Object.entries(obj)); // object → Map
console.log(backToMap.size); // => 2

/* PRACTICE -----
*   1. Count unique words in "the cat the dog the bird" using a Set.
*   2. Build a Map of country → capital and print each pair.
*   3. Use a Map to tally how many times each letter appears in "banana".
* ----- */
```

## 17 · CLASSES & OBJECT-ORIENTED PROGRAMMING

---

```
/* =====
* 17 · CLASSES & OBJECT-ORIENTED PROGRAMMING
* =====
* Run: node lessons/17-classes-oop.js
*
```

```

* WHAT YOU'LL LEARN
*   • Defining a class with a constructor and methods
*   • Creating instances with `new`
*   • Inheritance with `extends` and `super`
*   • Getters/setters, static members, and private fields (#)
* ===== */

// — 1. A basic class —————
class Animal {
  // The constructor runs when you do `new Animal(...)`
  constructor(name, sound) {
    this.name = name;    // instance properties
    this.sound = sound;
  }

  // A method (shared by all instances):
  speak() {
    return `${this.name} says ${this.sound}`;
  }
}

const dog = new Animal('Rex', 'Woof');
console.log(dog.speak());    // => Rex says Woof
console.log(dog instanceof Animal); // => true

// — 2. Inheritance with extends / super —————
class Dog extends Animal {
  constructor(name, breed) {
    super(name, 'Woof');    // call the parent constructor FIRST
    this.breed = breed;
  }

  // Add new behavior:
  fetch() {
    return `${this.name} fetches the ball`;
  }

  // Override a parent method (and optionally reuse it via super):
  speak() {
    return `${super.speak()} (a ${this.breed})`;
  }
}

const buddy = new Dog('Buddy', 'Labrador');
console.log(buddy.speak()); // => Buddy says Woof (a Labrador)
console.log(buddy.fetch()); // => Buddy fetches the ball

```

```
// — 3. Getters & setters – computed/guarded properties —————
class Temperature {
  constructor(celsius) {
    this.celsius = celsius;
  }
  // Accessed like a property (no parentheses): temp.fahrenheit
  get fahrenheit() {
    return this.celsius * 1.8 + 32;
  }
  set fahrenheit(value) {
    this.celsius = (value - 32) / 1.8;
  }
}

const temp = new Temperature(25);
console.log(temp.fahrenheit); // => 77 (getter – note: no () )
temp.fahrenheit = 212;        // setter
console.log(temp.celsius);    // => 100

// — 4. Static members – belong to the class, not instances —————
class MathUtils {
  static PI = 3.14159;
  static double(n) {
    return n * 2;
  }
}
console.log(MathUtils.PI);      // => 3.14159 (called on the class itself)
console.log(MathUtils.double(8)); // => 16
// const m = new MathUtils(); m.double(8) // ❌ static methods aren't on instances

// — 5. Private fields (#) – true encapsulation —————
class BankAccount {
  #balance = 0;                // private – only accessible inside this class

  constructor(initial) {
    this.#balance = initial;
  }
  deposit(amount) {
    this.#balance += amount;
    return this.#balance;
  }
  get balance() {
    return this.#balance;
  }
}

const acct = new BankAccount(100);
console.log(acct.deposit(50)); // => 150
```

```

console.log(acct.balance);    // => 150
// console.log(acct.#balance); // ❌ SyntaxError – private outside the class

/* OOP VOCAB -----
 * class      → a blueprint for objects
 * instance   → an object made from a class (via `new`)
 * constructor → sets up a new instance
 * inheritance → a class building on another (extends/super)
 * encapsulation → hiding internal state (#private fields)
 * ----- */

/* PRACTICE -----
 * 1. Make a class Rectangle(width, height) with an area() method.
 * 2. Extend it as Square(side) using super.
 * 3. Add a static method Rectangle.fromObject({width, height}).
 * ----- */

```

## 18 · PROTOTYPES — how inheritance REALLY works

---

```

/* =====
 * 18 · PROTOTYPES – how inheritance REALLY works
 * =====
 * Run:  node lessons/18-prototypes.js
 *
 * WHAT YOU'LL LEARN
 * • The prototype chain – JS's actual inheritance mechanism
 * • That `class` (lesson 17) is "syntactic sugar" over prototypes
 * • How property lookup walks up the chain
 *
 * This is more advanced/theoretical. You can write JS for a long time using
 * only `class`, but understanding prototypes explains WHY things work.
 * ===== */

// — 1. Every object has a hidden link to a "prototype" —————
// When you read obj.something, JS looks on obj first; if not found, it follows
// the link to obj's prototype, then ITS prototype, and so on – the "chain".

const animal = {
  eats: true,
  walk() {
    return 'walking...';
  },
};

// Create `rabbit` whose prototype is `animal`:
const rabbit = Object.create(animal);

```



```

rabbit.jumps = true;

console.log(rabbit.jumps); // => true (own property)
console.log(rabbit.eats); // => true (inherited from animal via the chain)
console.log(rabbit.walk()); // => walking... (method found on the prototype)

// Where did each property come from?
console.log(rabbit.hasOwnProperty('jumps')); // => true (its own)
console.log(rabbit.hasOwnProperty('eats')); // => false (inherited)

// — 2. The chain ends at null —————
// rabbit → animal → Object.prototype → null
console.log(Object.getPrototypeOf(rabbit) === animal); // => true
console.log(Object.getPrototypeOf(animal) === Object.prototype); // => true
console.log(Object.getPrototypeOf(Object.prototype)); // => null

// — 3. Built-in methods live on prototypes —————
// When you call [1,2,3].map(...), `map` isn't on your array – it's on
// Array.prototype, found by walking the chain.
const arr = [1, 2, 3];
console.log(Object.getPrototypeOf(arr) === Array.prototype); // => true
console.log(arr.map === Array.prototype.map); // => true

// — 4. Constructor functions (the pre-`class` style) —————
// Before ES6 classes, this is how "classes" were written. Methods go on
// the constructor's .prototype so all instances SHARE one copy (memory-efficient).
function Person(name) {
  this.name = name; // instance-specific data
}
Person.prototype.greet = function () { // shared method
  return `Hi, I'm ${this.name}`;
};

const sam = new Person('Sam');
console.log(sam.greet()); // => Hi, I'm Sam
console.log(sam.greet === new Person('X').greet); // => true (same shared function)

// — 5. `class` is sugar over all of this —————
// These two are essentially equivalent:
class PersonClass {
  constructor(name) { this.name = name; }
  greet() { return `Hi, I'm ${this.name}`; }
}
const sam2 = new PersonClass('Sam');
console.log(typeof PersonClass); // => function (!)

```

```
console.log(PersonClass.prototype.greet === Object.getPrototypeOf(sam2).greet); // => true
// The class syntax just put `greet` on the prototype for you, like §4.
```

```
/* MENTAL MODEL -----
 *   • Reading a property → JS walks UP the prototype chain until it finds it.
 *   • Methods are stored ONCE on the prototype and shared by all instances.
 *   • `class` is friendlier syntax; under the hood it's prototypes all the way.
 * ----- */

/* PRACTICE -----
 *   1. Use Object.create to make an object that inherits a `describe` method.
 *   2. Check hasOwnProperty for an own vs inherited property.
 *   3. Log the prototype chain of [] until you reach null.
 * ----- */
```

## 19 · CALLBACKS & PROMISES (asynchronous JavaScript)

---

```
/* =====
 * 19 · CALLBACKS & PROMISES (asynchronous JavaScript)
 * =====
 * Run:  node lessons/19-callbacks-promises.js
 *
 * WHAT YOU'LL LEARN
 *   • Why JS needs async (it's single-threaded)
 *   • Callbacks and "callback hell"
 *   • Promises: then / catch / finally
 *   • Promise.all / race / allSettled
 *
 * Watch the OUTPUT ORDER when you run this – it teaches the event loop.
 * ===== */

// — 1. JS is single-threaded – it can't "wait" and block —————
// Slow things (timers, network, files) are handed off and finish LATER.
console.log('1: start');

setTimeout(() => console.log('3: timeout finished (later)'), 0);

console.log('2: end');
// Output order: 1, 2, THEN 3 – even though the timeout was 0ms. The synchronous
// code runs first; the callback waits in a queue until the stack is clear.

// — 2. Callbacks – "call this function when you're done" —————
function fetchUser(id, callback) {
  setTimeout(() => {
    callback({ id, name: 'Sam' }); // simulate a 200ms network delay
```

```

    }, 200);
  }
  fetchUser(1, (user) => console.log('callback got:', user.name)); // => Sam (after 200ms)

// ▲ CALLBACK HELL: nesting callbacks for sequential async steps gets ugly:
//   fetchUser(1, (user) => {
//     fetchPosts(user, (posts) => {
//       fetchComments(posts[0], (comments) => { ... }); // deeply nested 😞
//     });
//   });
// Promises were invented to flatten this.

// — 3. Promises – an object representing a future value —————
// A Promise is in one of three states: pending → fulfilled (resolve) or rejected.
const promise = new Promise((resolve, reject) => {
  const success = true;
  setTimeout(() => {
    if (success) resolve('✅ data loaded');
    else reject(new Error('❌ failed'));
  }, 100);
});

promise
  .then((result) => console.log('then:', result)) // runs on resolve
  .catch((err) => console.log('catch:', err.message)) // runs on reject
  .finally(() => console.log('finally: always runs'));

// — 4. Chaining – each .then returns a new promise (no more nesting!) —————
function getUser(id) {
  return new Promise((resolve) =>
    setTimeout(() => resolve({ id, name: 'Ana' }), 50)
  );
}

function getPosts(user) {
  return new Promise((resolve) =>
    setTimeout(() => resolve(`${user.name}'s post`), 50)
  );
}

getUser(1)
  .then((user) => getPosts(user)) // return a promise → the chain waits for it
  .then((posts) => console.log('chained:', posts)) // => [ "Ana's post" ]
  .catch((err) => console.log('error:', err));

// — 5. Running promises together —————
const p1 = Promise.resolve('a');
```

```

const p2 = new Promise((r) => setTimeout(() => r('b'), 80));
const p3 = Promise.resolve('c');

// Promise.all – wait for ALL to finish (rejects if ANY one rejects):
Promise.all([p1, p2, p3]).then((values) => console.log('all:', values)); // => ['a','b','c']

// Promise.race – settle as soon as the FIRST one settles:
Promise.race([p2, Promise.resolve('fast')]).then((v) => console.log('race:', v)); // => fast

// Promise.allSettled – wait for all, never rejects; reports each outcome:
Promise.allSettled([Promise.resolve('ok'), Promise.reject('bad')]).then((results) =>
  console.log('allSettled:', results)
); // => [{status:'fulfilled',value:'ok'}, {status:'rejected',reason:'bad'}]

/* KEY TAKEAWAYS -----
*   • Synchronous code always runs before any async callback/promise.
*   • .then chains replace nested callbacks.
*   • Always end a chain with .catch to handle errors.
*   • Next lesson (async/await) makes promises read like normal code.
* ----- */

/* PRACTICE -----
*   1. Wrap setTimeout in a `delay(ms)` function that returns a Promise.
*   2. Chain delay(100) → log "done after 100ms".
*   3. Use Promise.all to wait for delay(50) and delay(150) together.
* ----- */

```

## 20 • ASYNC / AWAIT

---

```

/* =====
* 20 • ASYNC / AWAIT
* =====
* Run:  node lessons/20-async-await.js
*
* WHAT YOU'LL LEARN
*   • async/await – write asynchronous code that READS like synchronous code
*   • Error handling with try/catch
*   • Running tasks in sequence vs in parallel (a common performance trap)
*
* async/await is just nicer syntax over Promises (lesson 19). Under the hood,
* an async function always returns a Promise.
* ===== */

// A tiny helper used throughout: resolves after `ms` milliseconds.
function delay(ms, value) {
  return new Promise((resolve) => setTimeout(() => resolve(value), ms));
}

```

```
}
```

```
// — 1. The basics: async marks a function, await pauses for a Promise —————
```

```
async function loadData() {  
  console.log('loading...');  
  const result = await delay(100, '✅ data'); // pause here until it resolves  
  console.log('got:', result);                // runs AFTER the await  
  return result;                             // async functions return a Promise  
}
```

```
// Compare to the promise version – same thing, fewer .then() callbacks:
```

```
//   loadData() === getData().then(result => ...)
```

```
// — 2. Error handling with try/catch (clean!) —————
```

```
async function riskyLoad() {  
  try {  
    const data = await Promise.reject(new Error('network down'));  
    console.log(data); // skipped – the await threw  
  } catch (err) {  
    console.log('caught:', err.message); // => caught: network down  
  } finally {  
    console.log('cleanup runs either way');  
  }  
}
```

```
// — 3. Sequential vs parallel – a critical distinction —————
```

```
async function sequential() {  
  // ❌ SLOW: each await waits for the previous one. Total ≈ 100+100+100 = 300ms.  
  const a = await delay(100, 'a');  
  const b = await delay(100, 'b');  
  const c = await delay(100, 'c');  
  return [a, b, c];  
}
```

```
async function parallel() {  
  // ✅ FAST: start all three FIRST, then await them together. Total ≈ 100ms.  
  const pA = delay(100, 'a'); // started now  
  const pB = delay(100, 'b'); // started now  
  const pC = delay(100, 'c'); // started now  
  return Promise.all([pA, pB, pC]); // wait for all in parallel  
}  
// Rule: if tasks DON'T depend on each other, run them in parallel.
```

```
// — 4. await in a loop —————
```

```
async function processInOrder(items) {
```

```

const results = [];
for (const item of items) {
  const r = await delay(30, item.toUpperCase()); // one at a time, in order
  results.push(r);
}
return results;
}
// For independent items, prefer: await Promise.all(items.map(processOne))

// — 5. Run everything (top-level) —————
// We use an async IIFE so we can `await` at the top level in any environment.
(async () => {
  await loadData(); // => loading... / got:  data
  await riskyLoad(); // => caught: network down / cleanup...
  console.log('sequential:', await sequential()); // => [ 'a', 'b', 'c' ] (~300ms)
  console.log('parallel:', await parallel()); // => [ 'a', 'b', 'c' ] (~100ms)
  console.log('inOrder:', await processInOrder(['x', 'y', 'z'])); // => ['X','Y','Z']
})();

/* MENTAL MODEL -----
 * • `await x` means "pause this function until promise x settles, then
 *   give me its value (or throw its error)".
 * • The rest of your program keeps running while an async function awaits.
 * • await ONLY works inside an async function.
 * ----- */

/* PRACTICE -----
 * 1. Rewrite a .then() chain from lesson 19 using async/await.
 * 2. Fetch two things in parallel with Promise.all and time the difference.
 * 3. Wrap a failing await in try/catch and log a friendly error message.
 * ----- */

```

## 21 • ERROR HANDLING

---

```

/* =====
 * 21 • ERROR HANDLING
 * =====
 * Run:  node lessons/21-error-handling.js
 *
 * WHAT YOU'LL LEARN
 * • try / catch / finally
 * • throw and the Error object
 * • Custom error classes
 * • Handling errors in async code
 * • Global "last resort" handlers (browser & Node)

```

```

* ===== */

// — 1. try / catch / finally —————
try {
  console.log('trying...');
  const data = JSON.parse('{ invalid json }'); // this throws
  console.log(data); // skipped
} catch (err) {
  console.log('caught:', err.message); // => caught: ... (parse error)
} finally {
  console.log('finally always runs (cleanup, closing files, etc.)');
}

// — 2. Throwing your own errors —————
// `throw` stops the function and jumps to the nearest catch.
function divide(a, b) {
  if (b === 0) {
    throw new Error('Cannot divide by zero');
  }
  return a / b;
}

try {
  console.log(divide(10, 2)); // => 5
  console.log(divide(10, 0)); // throws
} catch (err) {
  console.log('error:', err.message); // => error: Cannot divide by zero
}
// Always throw an Error OBJECT (not a string) – it carries a message + stack.

// — 3. The Error object & built-in error types —————
const err = new Error('something broke');
console.log(err.name); // => Error
console.log(err.message); // => something broke
// err.stack contains the file/line trace (useful when debugging/logging).

// Built-in subtypes JS throws automatically:
//   TypeError      → wrong type (e.g. null.foo)
//   ReferenceError → using an undeclared variable
//   SyntaxError    → invalid code / bad JSON
//   RangeError     → value out of range (e.g. new Array(-1))
try {
  null.foo;
} catch (e) {
  console.log(e.name); // => TypeError
}

```

// — 4. Custom error classes – meaningful, catchable error types —————

```
class ValidationError extends Error {
  constructor(message, field) {
    super(message);
    this.name = 'ValidationError';
    this.field = field; // attach extra context
  }
}

function createUser(name) {
  if (!name) {
    throw new ValidationError('Name is required', 'name');
  }
  return { name };
}

try {
  createUser('');
} catch (err) {
  if (err instanceof ValidationError) {
    console.log(`[${err.field}] ${err.message}`); // => [name] Name is required
  } else {
    throw err; // not ours – re-throw so it isn't silently swallowed
  }
}
```

// — 5. Errors in async code —————

```
async function fetchThing() {
  // In async functions, use plain try/catch around awaits:
  try {
    await Promise.reject(new Error('timeout'));
  } catch (err) {
    console.log('async caught:', err.message); // => async caught: timeout
  }
}

fetchThing();

// For raw promises, use .catch():
Promise.reject(new Error('boom')).catch((e) => console.log('promise caught:', e.message));
```

// — 6. Global "last resort" handlers —————

```
// try/catch handles errors you EXPECT. But something always slips through – an
// unexpected throw, a promise with no .catch(). A global handler is your safety
// net: log it (to a monitoring service – lesson 47) so you find out in prod.
//
// IN THE BROWSER:
```



```
// window.addEventListener('error', (e) => {
//   report({ msg: e.message, file: e.filename, line: e.lineno }); // log it
// });
// // A promise that rejects with nobody listening fires THIS instead:
// window.addEventListener('unhandledrejection', (e) => {
//   report({ msg: 'unhandled rejection', reason: e.reason });
//   e.preventDefault(); // stops the default console error (optional)
// });
//
// IN NODE (this part runs):
process.on('unhandledRejection', (reason) => {
  console.log('global: unhandled rejection ->', reason?.message ?? reason);
});
process.on('uncaughtException', (err) => {
  console.log('global: uncaught exception ->', err.message);
  // A truly uncaught error left the app in an UNKNOWN state. Best practice:
  // log, then exit and let a process manager restart you cleanly:
  // process.exit(1);
});
// Trigger the rejection handler on purpose (no .catch on this one):
Promise.reject(new Error('nobody caught me'));
// △ Global handlers are a NET, not a strategy – handle errors locally where you
// can actually recover; let the global handler only LOG the ones that escape.

/* BEST PRACTICES -----
*   • Throw Error objects, not strings.
*   • Only catch errors you can meaningfully handle; re-throw the rest.
*   • Don't leave an empty catch {} – at minimum, log it.
*   • Use custom error classes so callers can react to specific failures.
* ----- */

/* PRACTICE -----
*   1. Write parseAge(str) that throws a RangeError if the age is negative.
*   2. Create a NotFoundError class and throw it from a lookup function.
*   3. Wrap a JSON.parse in try/catch and return a default object on failure.
*   4. Add a process.on('unhandledRejection', ...) and trigger it with a
*       Promise.reject that has no .catch – confirm your handler logs it.
* ----- */
```

## 22 • MODULES — import / export

---

```
/* =====
* 22 • MODULES – import / export
* =====
* Run: node lessons/22-modules.js (the runnable demo at the bottom)
*
* WHAT YOU'LL LEARN
```

```

*   • Why we split code into modules (files)
*   • ES Modules (import/export) – the modern standard
*   • CommonJS (require/module.exports) – Node's older system
*   • Dynamic import() – load a module on demand (code splitting)
*
* Modules need MULTIPLE files to truly demonstrate, so the import/export
* examples below are shown as comments – but there is a REAL, runnable pair
* of files in lessons/modules-demo/ that you can run right now:
*
*   node lessons/modules-demo/app.mjs    # ES Modules (import/export)
*   node lessons/modules-demo/app.cjs    # CommonJS (require/exports)
*
* Open those four files (math.mjs/app.mjs and math.cjs/app.cjs) alongside this
* lesson. The self-contained demo at the bottom of THIS file just lets it run
* on its own with `node lessons/22-modules.js`.
* ===== */

/* — 1. WHY modules? —————
* One giant file is hard to navigate and reuse. Modules let each file own one
* job and explicitly share (export) only what others need (import). Anything
* not exported stays private to the file.
*/

/* — 2. ES MODULES (ESM) – the modern standard —————
* Enable in Node by adding "type": "module" to package.json,
* or by naming files with the .mjs extension. In the browser, use
* <script type="module" src="app.js"></script>.
*
* ---- file: math.js ----
* // Named exports – export many values by name:
* export const PI = 3.14159;
* export function add(a, b) { return a + b; }
*
* // Default export – ONE main value per file:
* export default function multiply(a, b) { return a * b; }
*
* ---- file: app.js ----
* import multiply, { PI, add } from './math.js';
* //    ^default    ^named (names must match, braces required)
*
* import { add as sum } from './math.js'; // rename on import
* import * as math from './math.js';     // grab everything as a namespace
* console.log(math.PI, math.add(2, 3));
*
* Notes:
*   • import lines are hoisted and run first, before other code.
*   • Paths are relative ('./') and usually need the .js extension in Node ESM.
*/

```

```

/* — 3. COMMONJS (CJS) – Node's original system —————
* The default in Node when there's no "type": "module". You'll see this a lot
* in existing Node code.
*
* ---- file: math.cjs ----
*   const PI = 3.14159;
*   function add(a, b) { return a + b; }
*   module.exports = { PI, add };    // export an object
*
* ---- file: app.cjs ----
*   const { PI, add } = require('./math.cjs');
*   const math = require('./math.cjs'); // or grab the whole object
*
* ESM vs CJS quick map:
*   export default x    ↔   module.exports = x
*   export { a, b }     ↔   module.exports = { a, b }
*   import x from '...' ↔   const x = require('...')
*/

/* — 3b. DYNAMIC import() – load a module ON DEMAND —————
* The `import ... from` form is STATIC: it runs at the top, always, before
* anything else. `import()` is a FUNCTION that returns a Promise – so you can
* load a module later, conditionally, or only when it's actually needed.
*
*   button.addEventListener('click', async () => {
*     const { renderChart } = await import('./chart.js'); // fetched only on click
*     renderChart(data);
*   });
*
* WHY IT MATTERS – CODE SPLITTING (lesson 41):
* Bundlers (lesson 44) split each dynamic import into its own file, so the
* initial page ships LESS JavaScript and the heavy bits load on demand.
*
* Other uses:
*   • Conditional/lazy loading: if (isAdmin) await import('./admin-panel.js');
*   • Load a CommonJS module from ESM: const cjs = await import('./old.cjs');
*   • import() works even in plain <script> (no type="module" required).
*
* It returns the module's NAMESPACE object (default export is `.default`):
*   const mod = await import('./math.js');
*   mod.add(2, 3);    mod.default(4, 5);    // named + default
*/

// Runnable taste of it: import a built-in Node module dynamically.
import('node:os').then((os) => {
  console.log('dynamic import → platform:', os.platform()); // e.g. 'linux'
});

```

```

/* — 3c. TOP-LEVEL await – `await` at module scope (ES2022) —————
* Inside an ES module you can use `await` at the TOP LEVEL – no `async`
* function wrapper needed. The module's own evaluation pauses until it resolves:
*
* // config.mjs
* const res = await fetch('https://api.example.com/config');
* export const config = await res.json(); // ← awaited at top level
*
* // app.mjs – importing waits until config.mjs has finished awaiting:
* import { config } from './config.mjs'; // `config` is already resolved
*
* WHY IT MATTERS:
* • Initialise things that need async work (config, DB connection, a WASM
*   module) before the module's exports are usable – no half-ready exports.
* • Combine with dynamic import() to pick a module at load time:
*   const { strings } = await import(`./il8n/${navigator.language}.mjs`);
*
* CAVEATS:
* • ESM ONLY. It does NOT work in CommonJS (.cjs) or in classic <script>;
*   needs type="module" / a .mjs file / "type":"module" in package.json.
* • A slow top-level await DELAYS every module that imports this one, so keep
*   it for genuine startup work – not per-request logic.
*/

// Runnable taste of it: this file is CommonJS, so we show it via a dynamic
// import of an ESM data: URL (the imported module uses top-level await itself).
import('data:text/javascript,export default await Promise.resolve(42)')
  .then((m) => console.log('top-level await → default export:', m.default)); // => 42


// — 4. Runnable demo (no real imports needed) —————
// This simulates a "module" as a self-contained object so the file still runs.
const mathModule = (() => {
  const PI = 3.14159; // "private" until exported
  const add = (a, b) => a + b;
  const multiply = (a, b) => a * b;
  return { PI, add, multiply }; // the file's "exports"
})();

console.log('PI:', mathModule.PI); // => 3.14159
console.log('add:', mathModule.add(2, 3)); // => 5
console.log('multiply:', mathModule.multiply(4, 5)); // => 20


/* PRACTICE -----
* First, RUN the real demo and read its files:
*   node lessons/modules-demo/app.mjs
*   node lessons/modules-demo/app.cjs
* Then try these:
* 1. Add a `subtract` named export to modules-demo/math.mjs and import it
*   in app.mjs. Re-run: node lessons/modules-demo/app.mjs

```

```

* 2. Change which function is the `default` export and update app.mjs.
* 3. Add a `multiply` export to math.cjs and use it from app.cjs.
* 4. In app.mjs, load math.mjs with a dynamic `await import('./math.mjs')`
*    instead of a top-of-file import – log the returned namespace object.
* 5. Add a top-level `await` to math.mjs (e.g. `export const ready =
*    await Promise.resolve(true);`), import `ready` in app.mjs, and confirm
*    it's already resolved when app.mjs runs.
* ----- */

```

## 23 · THE DOM — Document Object Model ⚠ BROWSER ONLY

```

/* =====
* 23 · THE DOM — Document Object Model ⚠ BROWSER ONLY
* =====
* The DOM is how JS reads and changes the web page. It DOES NOT exist in Node
* (there's no page), so running this with `node` just prints a reminder.
*
* HOW TO RUN THIS LESSON:
* 1. Open index.html in the JS-learn folder.
* 2. Change its script tag to: <script src="lessons/23-dom.js"></script>
* 3. Open the browser, then open DevTools (F12) → Console tab.
*
* WHAT YOU'LL LEARN
* • Selecting elements
* • Reading/changing text, HTML, attributes, classes, and styles
* • Creating and removing elements
* ===== */

// Guard so this file is harmless when run in Node:
if (typeof document === 'undefined') {
  console.log('⚠ This is a BROWSER lesson. Open it via index.html – see the header.');
```

```

} else {
  // — 1. Selecting elements —————
  // querySelector uses CSS selectors and returns the FIRST match (or null).
  const title = document.querySelector('h1');           // by tag
  const box = document.querySelector('#box');           // by id (#)
  const items = document.querySelectorAll('.item');     // ALL matches (a NodeList)

  // Older, still-valid alternatives:
  // document.getElementById('box');
  // document.getElementsByClassName('item');

  // — 2. Reading & changing content —————
  if (title) {
    console.log(title.textContent);                      // read the text
    title.textContent = 'Updated by JS!';               // change the text (safe – no HTML)
    title.innerHTML = 'Now <em>italic</em>';             // parses HTML (⚠ never with user input – XSS risk)
  }
}

```

```
// — 3. Attributes —————
const link = document.querySelector('a');
if (link) {
  link.getAttribute('href');
  link.setAttribute('target', '_blank');
  link.href = 'https://example.com'; // many attributes are also properties
}

// — 4. Classes & styles —————
if (box) {
  box.classList.add('active');
  box.classList.remove('hidden');
  box.classList.toggle('open'); // add if absent, remove if present
  box.classList.contains('active'); // => true/false

  box.style.color = 'white';
  box.style.backgroundColor = 'navy'; // note: camelCase (background-color)
}

// — 5. Creating & inserting elements —————
const list = document.querySelector('ul');
if (list) {
  const li = document.createElement('li'); // create (not yet on the page)
  li.textContent = 'New item';
  li.classList.add('item');
  list.appendChild(li); // add as the last child
  // list.prepend(li); // add as the first child
  // li.remove(); // remove an element
}

// — 6. Looping over a NodeList —————
items.forEach((item, i) => {
  item.textContent = `Item ${i + 1}`;
});
}

/* PRACTICE (in the browser) -----
* 1. Add a <ul id="list"></ul> to index.html, then append 3 <li>s with JS.
* 2. Select the <h1> and change its color and text.
* 3. Toggle a "dark" class on the <body> from the console.
* ----- */
```

## 24 • EVENTS BROWSER ONLY

```
/* =====
* 24 • EVENTS  ⚠ BROWSER ONLY
```

```

* =====
* Events are how your page REACTS to the user: clicks, typing, submitting forms,
* scrolling, etc. Like the DOM lesson, this runs in a browser, not Node.
*
* HOW TO RUN: see lesson 23's header (open via index.html, check the console).
*
* WHAT YOU'LL LEARN
*   • addEventListener
*   • The event object (event.target, preventDefault)
*   • Event bubbling and delegation
*   • Common events (click, input, submit, keydown)
* ===== */

if (typeof document === 'undefined') {
  console.log('⚠ This is a BROWSER lesson. Open it via index.html – see lesson 23.');
```

```

} else {
  // — 1. addEventListener – the standard way to respond to events —————
  const button = document.querySelector('#myButton');
  if (button) {
    button.addEventListener('click', (event) => {
      console.log('Button clicked!');
      console.log('clicked element:', event.target); // the element that fired it
    });
    // You can attach MANY listeners to the same element – they all run.
    // Remove one later with button.removeEventListener('click', namedHandler).
  }

  // — 2. The event object —————
  const input = document.querySelector('#nameInput');
  if (input) {
    // 'input' fires on every keystroke; event.target.value is the current text.
    input.addEventListener('input', (event) => {
      console.log('typing:', event.target.value);
    });
  }

  // — 3. Forms – preventDefault stops the page reload —————
  const form = document.querySelector('#myForm');
  if (form) {
    form.addEventListener('submit', (event) => {
      event.preventDefault(); // stop the browser's default submit/reload
      const data = new FormData(form);
      console.log('form name field:', data.get('name'));
    });
  }

  // — 4. Keyboard events —————
  document.addEventListener('keydown', (event) => {
    console.log('key pressed:', event.key); // e.g. "Enter", "a", "Escape"
    if (event.key === 'Escape') console.log('Escape pressed!');
  });
}

```

```

});

// — 5. Bubbling & delegation —————
// When you click a child, the event "bubbles" UP through its ancestors.
// DELEGATION uses this: attach ONE listener to a parent instead of many to
// each child – great for lists that change over time.
const list = document.querySelector('#todoList');
if (list) {
  list.addEventListener('click', (event) => {
    // Did the click land on an <li>? (closest walks up to find a match)
    const li = event.target.closest('li');
    if (li) {
      li.classList.toggle('done');
      console.log('toggled:', li.textContent);
    }
  });
  // Now ANY <li> inside #todoList responds – even ones added later.
}
}

/* COMMON EVENTS -----
*   click, dblclick    → mouse clicks
*   input, change      → form field value changed
*   submit             → form submitted (use preventDefault)
*   keydown, keyup     → keyboard
*   mouseover, mouseout → hover
*   load, DOMContentLoaded → page/resources ready
* ----- */

/* PRACTICE (in the browser) -----
*   1. Add a button that increments a counter shown in a <span>.
*   2. Add an input that live-updates an <h2> as you type.
*   3. Use delegation: one listener on a <ul> that strikes through any <li>
*       you click.
* ----- */

```

## 25 · FETCH & APIs — talking to the internet

---

```

/* =====
* 25 · FETCH & APIs – talking to the internet
* =====
* Run:  node lessons/25-fetch-apis.js    (needs internet; Node 18+ has fetch)
*
* WHAT YOU'LL LEARN
* • HTTP fundamentals: methods, status codes, headers, REST
* • fetch() – making HTTP requests (GET and POST)
* • Working with JSON
* • Handling errors and HTTP status codes

```



```

*   • This ties together Promises + async/await from lessons 19–20
*
* `fetch` is built into modern browsers AND Node 18+. It returns a Promise.
* ===== */

/* — 0. HTTP IN 60 SECONDS – what fetch is actually doing —————
* Every request is: a METHOD + a URL + HEADERS (+ an optional BODY). The server
* replies with a STATUS CODE + headers + a body. Know these and APIs stop being
* mysterious.
*
* METHODS (the "verb" – what you want to do):
*   GET      read data (no body)          POST   create something (has a body)
*   PUT      replace a whole resource     PATCH   update part of a resource
*   DELETE   remove a resource
*
* STATUS CODES (grouped by first digit):
*   2xx success → 200 OK, 201 Created, 204 No Content
*   3xx redirect → 301 Moved, 304 Not Modified (cached)
*   4xx YOUR fault → 400 Bad Request, 401 Unauthorized, 403 Forbidden,
*                   404 Not Found, 429 Too Many Requests
*   5xx SERVER fault → 500 Internal Error, 502/503 upstream/unavailable
*
* COMMON HEADERS:
*   Content-Type: application/json → what the body IS
*   Accept: application/json → what you WANT back
*   Authorization: Bearer <token> → who you are (lesson 39)
*
* REST = a convention for designing URLs around "resources" (nouns) + methods:
*   GET    /users      list users      POST   /users      create a user
*   GET    /users/42   read user 42   PATCH  /users/42   update user 42
*   DELETE /users/42   delete user 42
* The same idea powers almost every API you'll call.
*/

// — 1. JSON – the data format of the web —————
// APIs send/receive text in JSON format. Convert between JS and JSON like so:
const user = { name: 'Sam', age: 25, hobbies: ['code', 'chess'] };

const json = JSON.stringify(user);          // JS object → JSON string
console.log(json); // => {"name":"Sam","age":25,"hobbies":["code","chess"]}

const parsed = JSON.parse(json);            // JSON string → JS object
console.log(parsed.name); // => Sam

// Pretty-print with indentation (handy for logging/debugging):
console.log(JSON.stringify(user, null, 2));

```

```
// — 1b. Building URLs with query strings (URLSearchParams) —————  
// Many GET requests need a "?key=value&key=value" query string. Don't glue it  
// by hand — URLSearchParams encodes spaces and special characters correctly.  
const params = new URLSearchParams({ q: 'iced coffee', page: 2, sort: 'new' });  
console.log(params.toString()); // => q=iced+coffee&page=2&sort=new  
  
const searchUrl = `https://api.example.com/search?${params}`;  
console.log(searchUrl); // => https://api.example.com/search?q=iced+coffee&page=2&sort=new  
// You'd then: await fetch(searchUrl)  
// (URLSearchParams also PARSES query strings — see lesson 34.)
```

```
// — 2. A GET request with async/await —————  
async function getPost(id) {  
  // fetch returns a Response. .json() reads & parses the body (also a Promise).  
  const response = await fetch(`https://jsonplaceholder.typicode.com/posts/${id}`);  
  
  // ⚠ fetch does NOT throw on 404/500 — you must check response.ok yourself.  
  if (!response.ok) {  
    throw new Error(`HTTP ${response.status} ${response.statusText}`);  
  }  
  
  const data = await response.json();  
  return data;  
}
```

```
// — 3. A POST request — sending data —————  
async function createPost(post) {  
  const response = await fetch('https://jsonplaceholder.typicode.com/posts', {  
    method: 'POST',  
    headers: { 'Content-Type': 'application/json' },  
    body: JSON.stringify(post), // body must be a string  
  });  
  return response.json();  
}
```

```
// — 4. Fetching several things in parallel —————  
async function getManyPosts(ids) {  
  // Start all requests at once, then await them together (fast — see lesson 20).  
  const requests = ids.map((id) => getPost(id));  
  return Promise.all(requests);  
}
```

```
// — 5. Run it (with proper error handling) —————  
(async () => {  
  try {
```

```

const post = await getPost(1);
console.log('GET post 1 title:', post.title);

const created = await createPost({ title: 'Hello', body: 'World', userId: 1 });
console.log('POST created id:', created.id); // => 101 (this fake API echoes back)

const many = await getManyPosts([1, 2, 3]);
console.log('parallel fetched titles:', many.map((p) => p.title.slice(0, 15)));
} catch (err) {
  // Network failure (e.g. offline) OR a thrown HTTP error lands here.
  console.log('Request failed:', err.message);
  console.log('(If you are offline, that is expected – the code is correct.)');
}
})();

/* THE PATTERN TO REMEMBER -----
*   const res = await fetch(url, options);
*   if (!res.ok) throw new Error(res.status); // fetch won't throw for you
*   const data = await res.json();           // parse the body
*   ... wrap the whole thing in try/catch.
* ----- */

/* PRACTICE -----
*   1. Fetch https://jsonplaceholder.typicode.com/users and log every name.
*   2. Fetch a single todo and log whether it's completed.
*   3. Add a check that logs a friendly message when response.ok is false
*       (try a URL ending in /posts/99999999).
*   4. Log response.status and response.headers.get('content-type') for a GET.
*   5. Send a DELETE to /posts/1 and log the status code you get back.
* ----- */

```

## 26 · REGULAR EXPRESSIONS (Regex)

---

```

/* =====
* 26 · REGULAR EXPRESSIONS (Regex)
* =====
* Run:  node lessons/26-regular-expressions.js
*
* WHAT YOU'LL LEARN
*   • Creating patterns and the common flags
*   • Character classes, quantifiers, anchors, groups
*   • Lookahead & lookbehind (match based on what's around)
*   • The modern flags: u (unicode), s (dotAll), y (sticky), v (unicodeSets)
*   • test / match / matchAll / replace / split with regex
*
* A regex is a tiny pattern language for finding/validating/replacing text.

```

```

* Read each pattern's comment slowly – that's how regex "clicks".
* ===== */

// — 1. Creating a regex —————
const literal = /cat/; // literal notation (preferred)
const fromString = new RegExp('cat'); // when the pattern is dynamic/in a variable
console.log(literal.test('a cat')); // => true (.test → does it match? boolean)
console.log(fromString.test('dog')); // => false

// — 2. Flags (after the closing slash) —————
// g → global (find ALL matches, not just the first)
// i → case-insensitive
// m → multiline (^ and $ match per line)
console.log('Cat cat CAT'.match(/cat/gi)); // => [ 'Cat', 'cat', 'CAT' ]

// — 3. Character classes – what kind of character —————
// \d digit \w word char (a-z A-Z 0-9 _) \s whitespace
// \D not-digit \W not-word \S not-space
// . any char (except newline)
// [abc] one of a/b/c [a-z] a range [^abc] NOT a/b/c
console.log('Order #42 ok'.match(/#\d+/)); // => [ '42', ... ] (\d+ = one or more digits)
console.log('a1 b2 c3'.match(/[a-c]\d/g)); // => [ 'a1', 'b2', 'c3' ]

// — 4. Quantifiers – how many —————
// * zero or more + one or more ? zero or one (optional)
// {3} exactly 3 {2,4} between 2-4 {2,} 2 or more
console.log(/colou?r/.test('color')); // => true (the u is optional)
console.log(/colou?r/.test('colour')); // => true
console.log('aaa'.match(/a{2}/)); // => [ 'aa', ... ]

// — 5. Anchors – position —————
// ^ start of string $ end of string \b word boundary
console.log(/^hello/.test('hello world')); // => true (starts with hello)
console.log(/world$/.test('hello world')); // => true (ends with world)
console.log(/\bcat\b/.test('the cat sat')); // => true (whole word "cat")
console.log(/\bcat\b/.test('category')); // => false (cat is part of a word)

// — 6. Groups & capturing —————
// ( ) captures part of the match so you can reuse it.
const date = '2026-06-04';
const match = date.match(/(\d{4})-(\d{2})-(\d{2})/);
console.log(match[0]); // => 2026-06-04 (whole match)
console.log(match[1]); // => 2026 (first group)
console.log(match[2]); // => 06 (second group)

```

```

// Named groups are clearer:
const named = date.match(/(<year>\d{4})-(?<month>\d{2})-(?<day>\d{2})/);
console.log(named.groups.year, named.groups.month); // => 2026 06

// Non-capturing group (?:...) – group for structure WITHOUT creating a $1 slot:
console.log('abcabc'.match(/(?:abc)+/)[0]); // => abcabc (grouped, not captured)

// — 6b. Lookahead & lookbehind – match by CONTEXT (not consumed) —————
// These assert what comes BEFORE/AFTER without including it in the match.
// (?:=...) positive lookahead – followed by ...
// (?!...) negative lookahead – NOT followed by ...
// (?<=...) positive lookbehind – preceded by ...
// (?<!)... negative lookbehind – NOT preceded by ...

// Get the number, but only when it's followed by "px" (the px isn't captured):
console.log('width: 42px'.match(/\d+(?=px)/)[0]); // => 42

// Get the amount AFTER a "$" (the $ isn't part of the match):
console.log('Total $59 now'.match(/(?<=\$)\d+/)[0]); // => 59

// Negative lookahead – a word NOT followed by "ing":
console.log('run runs running'.match(/run(?!ing)/g)); // => [ 'run', 'run' ]

// Classic use: add thousands separators by inserting commas at the right spots.
console.log('1234567'.replace(/\B(?=(\d{3})+(?!\d))/g, ',')); // => 1,234,567

// — 6c. Modern flags: u (unicode), s (dotAll), y (sticky) —————
// u → full Unicode mode. Required for \p{...} property escapes & astral chars.
console.log(/\p{Emoji}/u.test('hi 🙌')); // => true (match by Unicode property)
console.log([... 'a🍌b'].length); // => 3 (spread is unicode-aware)
// s → "dotAll": let . match newlines too (by default it doesn't).
console.log(/a.b/s.test('a\nb')); // => true (without /s → false)
// y → "sticky": match ONLY at regex.lastIndex, then advance it. Great for
// tokenizers/parsers that scan a string piece by piece.
const sticky = /\d+/y;
sticky.lastIndex = 5;
console.log(sticky.exec('abc 123')[0]); // => 123 (matches starting at idx 5)
// v → "unicodeSets" (ES2024): a smarter, stricter superset of u. It adds set
// operations inside character classes and matches multi-codepoint graphemes.
// • Intersection [A&&B] – match chars in BOTH sets:
console.log('café'.replace(/[\p{L}&&\p{ASCII}]/gu, '*')); // => "*****" (u: can't AND; && is
literal → letters OR ascii)
console.log('café'.replace(/[\p{L}&&\p{ASCII}]/gv, '*')); // => "****é" (v: letters AND ascii → é
kept, it's non-ASCII)
// • Subtraction [A--B] – match chars in A but NOT B:
console.log('café 123!'.replace(/[\p{L}--[aeiou]]/gv, '*')); // => "*a** 123!" (letters except

```

```

vowels)
// • Match a whole multi-codepoint grapheme via \p{} string literals:
console.log(/\p{👍}/v.test('👍')); // => true (one class entry = a string)
// Rule of thumb: prefer /v over /u in new code when you use \p{...} or classes.

// — 7. The string methods that take a regex —————
// match – first match (or with /g, all matches as an array)
console.log('cat bat hat'.match(/[cbh]at/g)); // => [ 'cat', 'bat', 'hat' ]

// matchAll – all matches WITH their groups (returns an iterator)
for (const m of 'a1 b2'.matchAll(/(\w)(\d)/g)) {
  console.log(m[1], m[2]); // => a 1 / b 2
}

// replace – find & replace ($1 refers to a captured group)
console.log('2026-06-04'.replace(/(\d{4})-(\d{2})-(\d{2})/, '$3/$2/$1')); // => 04/06/2026
console.log('a b c d'.replace(/\s+/g, ' ')); // => "a b c d" (collapse spaces)

// split – split on a pattern
console.log('a, b ,c , d'.split(/\s*,\s*/)); // => [ 'a', 'b', 'c', 'd' ]

// — 8. Practical validators —————
const isEmail = (s) => /^[^\s@]+@[^\s@]+\.[^\s@]+$/i.test(s);
console.log(isEmail('sam@example.com')); // => true
console.log(isEmail('not-an-email')); // => false

// ⚠ Don't over-trust regex for emails/URLs in production – but it's great for
// quick format checks and pulling data out of text.

/* CHEAT SHEET -----
* \d \w \s digit / word / space ^ $ \b start / end / boundary
* * + ? many / 1+ / optional {n} {n,m} exact / range counts
* [abc] [^abc] one-of / none-of ( ) capture (?<name> ) named
* (?:..) non-capturing group (?:) (?!) lookahead pos/neg
* (?<=) (?<!) lookbehind pos/neg \p{...} unicode property (needs u)
* flags: g (all) i (ignore case) m (multiline) s (dot matches \n)
* u (unicode) y (sticky – match at lastIndex only) d (match indices)
* v (unicodeSets – superset of u: set ops [A&B]/[A--B] & \p{} strings)
* Tip: build & test patterns at regex101.com – it explains every token.
* ----- */

/* PRACTICE -----
* 1. Write a regex that matches a 10-digit phone number.
* 2. Extract all hashtags from "love #js and #coding" using matchAll.
* 3. Replace every vowel in "hello world" with "*".
* 4. Use a lookbehind to grab the number after "id=" in "?id=42&x=1".

```

```
* 5. Use a negative lookahead to match "cat" only when NOT followed by "alog".
* ----- */
```

## 27 • GENERATORS & ITERATORS

---

```
/* =====
* 27 • GENERATORS & ITERATORS
* =====
* Run: node lessons/27-generators-iterators.js
*
* WHAT YOU'LL LEARN
* • The iterator protocol (what makes something "for...of"-able)
* • Generator functions (function*) and yield
* • Lazy / infinite sequences
* • Making your own objects iterable with Symbol.iterator
* ===== */

// — 1. The iterator protocol —————
// An "iterator" is any object with a .next() that returns { value, done }.
// Arrays, strings, Sets, Maps are "iterable" because they can produce one.
const arr = [10, 20];
const it = arr[Symbol.iterator](); // get the array's iterator
console.log(it.next()); // => { value: 10, done: false }
console.log(it.next()); // => { value: 20, done: false }
console.log(it.next()); // => { value: undefined, done: true }

// — 2. Generator functions – the easy way to make iterators —————
// function* + yield. Calling it does NOT run the body – it returns a generator.
// Each .next() runs UNTIL the next yield, then PAUSES there.
function* countToThree() {
  yield 1;
  yield 2;
  yield 3;
}

const gen = countToThree();
console.log(gen.next()); // => { value: 1, done: false }
console.log(gen.next()); // => { value: 2, done: false }
console.log(gen.next()); // => { value: 3, done: false }
console.log(gen.next()); // => { value: undefined, done: true }

// Generators are iterable, so for...of and spread just work:
console.log([...countToThree()]); // => [ 1, 2, 3 ]
for (const n of countToThree()) console.log('got', n); // => 1, 2, 3
```

```
// — 3. Generators keep state between yields —————
function* idMaker() {
  let id = 1;
  while (true) {          // an "infinite" generator – safe because it's LAZY
    yield id++;
  }
}
const ids = idMaker();
console.log(ids.next().value); // => 1
console.log(ids.next().value); // => 2
console.log(ids.next().value); // => 3
// It only computes the next value when you ask – never runs forever.

// — 4. Lazy sequences – compute only what you need —————
function* take(iterable, n) {
  let count = 0;
  for (const item of iterable) {
    if (count++ >= n) return;
    yield item;
  }
}
function* naturals() {
  let i = 1;
  while (true) yield i++;
}
console.log([...take(naturals(), 5)]); // => [ 1, 2, 3, 4, 5 ] (from an infinite source!)

// — 5. yield* – delegate to another generator/iterable —————
function* letters() { yield 'a'; yield 'b'; }
function* combined() {
  yield* letters();    // yield everything from letters()
  yield* [1, 2];       // yield* works on any iterable
}
console.log([...combined()]); // => [ 'a', 'b', 1, 2 ]

// — 6. Make your OWN object iterable —————
// Add a [Symbol.iterator] method (a generator is the simplest way).
const range = {
  start: 1,
  end: 5,
  *[Symbol.iterator]() {
    for (let i = this.start; i <= this.end; i++) yield i;
  },
};
console.log([...range]);           // => [ 1, 2, 3, 4, 5 ]
for (const n of range) process.stdout.write(n + ' '); // => 1 2 3 4 5
console.log();
```



```

/* WHY USE GENERATORS? -----
 *   • Represent infinite or huge sequences without storing them all in memory.
 *   • Produce values lazily, on demand.
 *   • Write custom iteration for your own data structures cleanly.
 * ----- */

/* PRACTICE -----
 *   1. Write a generator fibonacci() and take the first 10 numbers.
 *   2. Make an object { from, to } iterable so [...it] counts from→to.
 *   3. Write a generator that yields only the even numbers of an array.
 * ----- */

```

## 28 · SYMBOLS, PROXY & REFLECT (metaprogramming)

---

```

/* =====
 * 28 · SYMBOLS, PROXY & REFLECT (metaprogramming)
 * =====
 * Run:  node lessons/28-symbols-proxy-reflect.js
 *
 * WHAT YOU'LL LEARN
 *   • Symbol – unique, hidden-ish keys
 *   • Proxy – intercept operations on an object (get/set/has/delete...)
 *   • Reflect – the standard helpers Proxy traps delegate to
 *
 * This is "metaprogramming": code that customizes how objects behave.
 * You won't use it daily, but libraries (Vue, MobX) are built on it.
 * ===== */

// — 1. Symbols – guaranteed-unique values -----
const a = Symbol('id'); // the 'id' is just a description (for debugging)
const b = Symbol('id');
console.log(a === b);    // => false  (every Symbol is unique, even same desc)
console.log(a.toString()); // => Symbol(id)

// Use as object keys that won't collide with normal string keys:
const ID = Symbol('id');
const user = { name: 'Sam', [ID]: 12345 };
console.log(user[ID]);    // => 12345
console.log(Object.keys(user)); // => [ 'name' ] (symbol keys are skipped here)
console.log(user.name);    // normal keys still work as usual

// Well-known symbols let you hook into language behavior (lesson 27 used
// Symbol.iterator). Another example – customize toString tagging:
const widget = { [Symbol.toStringTag]: 'Widget' };
console.log(Object.prototype.toString.call(widget)); // => [object Widget]

```

// — 2. Proxy – wrap an object and intercept operations —————

// A Proxy sits in front of a "target" and runs your "traps" on access.

```
const target = { message: 'hello' };
```

```
const logged = new Proxy(target, {
  get(obj, prop) {
    console.log(`(read ${String(prop)})`);
    return obj[prop];
  },
  set(obj, prop, value) {
    console.log(`(write ${String(prop)} = ${value})`);
    obj[prop] = value;
    return true; // set traps MUST return true on success
  },
});
```

```
logged.message;           // => (read message)
```

```
logged.message = 'hi';    // => (write message = hi)
```

```
console.log(target.message); // => hi    (the real object was updated)
```

// — 3. Proxy use-case: validation —————

```
function createValidatedUser() {
```

```
  return new Proxy({}, {
```

```
    set(obj, prop, value) {
```

```
      if (prop === 'age' && (typeof value !== 'number' || value < 0)) {
```

```
        throw new TypeError('age must be a non-negative number');
```

```
      }
```

```
      obj[prop] = value;
```

```
      return true;
```

```
    },
```

```
  });
```

```
}
```

```
const person = createValidatedUser();
```

```
person.age = 30;
```

```
console.log(person.age); // => 30
```

```
try {
```

```
  person.age = -5;           // blocked by the trap
```

```
} catch (e) {
```

```
  console.log('blocked:', e.message); // => blocked: age must be a non-negative number
```

```
}
```

// — 4. Proxy use-case: default values for missing keys —————

```
const withDefault = new Proxy({}, {
```

```
  get: (obj, prop) => (prop in obj ? obj[prop] : `<no ${String(prop)}>`),
```

```
});
```

```

withDefault.name = 'Ana';
console.log(withDefault.name); // => Ana
console.log(withDefault.email); // => <no email> (instead of undefined)

// — 5. Reflect – the "default behaviors" as functions —————
// Reflect mirrors the proxy traps. Inside a trap, delegate to Reflect so you
// keep the standard behavior while adding your own logic on top.
const obj = { x: 1 };
console.log(Reflect.get(obj, 'x')); // => 1 (same as obj.x)
Reflect.set(obj, 'y', 2); // same as obj.y = 2
console.log(Reflect.has(obj, 'y')); // => true (same as 'y' in obj)
console.log(Reflect.ownKeys(obj)); // => [ 'x', 'y' ]

// Idiomatic proxy using Reflect (forwards everything, logs reads):
const clean = new Proxy({ score: 9 }, {
  get(t, p, r) {
    console.log(`accessed ${String(p)}`);
    return Reflect.get(t, p, r); // default get, no manual t[p]
  },
});
console.log(clean.score); // => accessed score / 9

/* WHEN DO YOU USE THIS? -----
 * • Symbol: unique keys, hooking into iteration/string-tagging.
 * • Proxy + Reflect: reactive state (Vue 3), observable objects, validation,
 *   logging, virtualized/lazy objects, API mocking.
 * • For everyday app code you rarely need these – but knowing they exist
 *   explains how "magic" libraries work.
 * ----- */

/* PRACTICE -----
 * 1. Create two Symbols with the same description and prove they differ.
 * 2. Build a Proxy that makes an object read-only (set throws).
 * 3. Use Reflect.ownKeys to list every key (including symbols) of an object.
 * ----- */

```

## 29 • ADVANCED ASYNC

---

```

/* =====
 * 29 • ADVANCED ASYNC
 * =====
 * Run: node lessons/29-advanced-async.js
 *
 * WHAT YOU'LL LEARN
 * • Microtasks vs macrotasks (the real event-loop order)

```

```

*   • async iteration: for await...of and async generators
*   • AbortController – cancelling async work
*   • debounce & throttle – controlling how often a function runs
*   • retry with backoff, and concurrency limiting (a "pool")
*
* Builds on lessons 19–20. These are the patterns senior devs reach for.
* ===== */

```

```

const delay = (ms, v) => new Promise((r) => setTimeout(() => r(v), ms));

```

```

// — 1. Microtasks vs macrotasks (event-loop ordering) —————
// After the current synchronous code finishes, the engine drains ALL
// microtasks (Promise callbacks) BEFORE the next macrotask (setTimeout).
console.log('A: sync');
setTimeout(() => console.log('D: setTimeout (macrotask)'), 0);
Promise.resolve().then(() => console.log('C: promise (microtask)'));
console.log('B: sync');
// Order: A, B, C, D → promises always jump ahead of setTimeout.

```

```

// — 2. Async generators + for await...of —————
// A generator that yields PROMISES. for await...of awaits each one in turn.
async function* fetchPages() {
  for (let page = 1; page <= 3; page++) {
    const data = await delay(20, `page ${page} data`); // pretend network call
    yield data;
  }
}

async function readPages() {
  for await (const page of fetchPages()) {
    console.log('received:', page); // => received: page 1 data, ... 2, ... 3
  }
}

```

```

// — 3. AbortController – cancel async work —————
// The standard way to cancel fetches, timers, or any awaitable.
function cancellableDelay(ms, signal) {
  return new Promise((resolve, reject) => {
    const id = setTimeout(resolve, ms);
    signal.addEventListener('abort', () => {
      clearTimeout(id);
      reject(new Error('aborted'));
    });
  });
}

```

```

async function abortDemo() {
  const controller = new AbortController();
  setTimeout(() => controller.abort(), 30); // cancel after 30ms
  try {
    await cancellableDelay(1000, controller.signal); // would take 1s
    console.log('completed'); // never reached
  } catch (err) {
    console.log('abort demo:', err.message); // => abort demo: aborted
  }
  // Real fetch supports this directly: fetch(url, { signal: controller.signal })
}

```

```

// — 4. debounce – run only AFTER activity stops —————
// "Wait until the user stops typing for 300ms, THEN search." Resets on each call.
function debounce(fn, wait) {
  let timer;
  return function (...args) {
    clearTimeout(timer);
    timer = setTimeout(() => fn.apply(this, args), wait);
  };
}

```

```

// — 5. throttle – run at most ONCE per interval —————
// "No matter how often this fires (scroll/resize), run at most once per 200ms."
function throttle(fn, interval) {
  let last = 0;
  let lastArgs = null;
  return function (...args) {
    const now = Date.now();
    if (now - last >= interval) {
      last = now;
      fn.apply(this, args);
    } else {
      lastArgs = args; // remember the most recent call (optional trailing run)
    }
  };
}
// debounce → "settle then act" (search box). throttle → "steady rate" (scroll).

```

```

// — 6. Retry with backoff – a real-world async pattern —————
async function withRetry(task, attempts = 3) {
  for (let i = 1; i <= attempts; i++) {
    try {
      return await task();
    } catch (err) {
      if (i === attempts) throw err; // out of tries → give up
      await delay(10 * i); // wait longer each time (backoff)
    }
  }
}

```

```

    console.log(`retry ${i} after failure: ${err.message}`);
  }
}
}

```

// — 7. Concurrency limiting – a "pool" of N at a time —————

// Promise.all (lesson 20) starts EVERYTHING at once. With 1,000 URLs that can  
 // overwhelm the server (or your rate limit). A pool runs at most `limit` tasks  
 // concurrently: as each finishes, the next starts. Keeps results in order.

```

async function mapLimit(items, limit, task) {
  const results = new Array(items.length);
  let next = 0;
  async function worker() {
    while (next < items.length) {
      const i = next++; // claim the next index
      results[i] = await task(items[i], i);
    }
  }
  // start `limit` workers; they pull from the shared queue until it's empty
  const pool = Array.from({ length: Math.min(limit, items.length) }, worker);
  await Promise.all(pool);
  return results;
}

```

// — 8. Run the demos —————

```

(async () => {
  await readPages();
  await abortDemo();

  let flaky = 0;
  const result = await withRetry(async () => {
    flaky++;
    if (flaky < 3) throw new Error(`fail #${flaky}`);
    return 'success on attempt 3';
  });
  console.log(result); // => success on attempt 3

  // concurrency demo: 6 tasks, only 2 run at a time → finishes in ~3 "waves".
  let active = 0, peak = 0;
  const out = await mapLimit([1, 2, 3, 4, 5, 6], 2, async (n) => {
    active++; peak = Math.max(peak, active);
    await delay(20);
    active--;
    return n * 10;
  });
  console.log('mapLimit result:', out); // => [10,20,30,40,50,60] (in order)
  console.log('mapLimit peak concurrency:', peak); // => 2 (never more than the limit)
}
)

```

```
// debounce demo: 5 rapid calls → fn runs once, with the LAST argument.
const search = debounce((q) => console.log('searching for:', q), 50);
['a', 'ap', 'app', 'appl', 'apple'].forEach((q) => search(q));
// => searching for: apple    (only once, after the burst settles)
})();

/* PRACTICE -----
* 1. Predict the log order of a sync log, a setTimeout(0), and a Promise.then.
* 2. Write an async generator that yields 1..n with a small delay each.
* 3. Wrap a debounce around a function and call it rapidly – watch it fire once.
* 4. Use mapLimit to "fetch" 10 fake URLs 3 at a time; log when each starts.
* ----- */
```

## 30 • FUNCTIONAL PROGRAMMING

---

```
/* =====
* 30 • FUNCTIONAL PROGRAMMING
* =====
* Run:  node lessons/30-functional-programming.js
*
* WHAT YOU'LL LEARN
* • Pure functions & immutability
* • Higher-order functions
* • Currying & partial application
* • Function composition (pipe / compose)
*
* FP is a STYLE: build programs from small, predictable functions that don't
* change shared state. It makes code easier to test and reason about.
* ===== */

// — 1. Pure functions —————
// A pure function: (1) same input → same output, (2) no side effects.
const addPure = (a, b) => a + b;      // ✅ pure
console.log(addPure(2, 3));          // => 5 (always)

let total = 0;
const addImpure = (n) => (total += n); // ❌ impure – mutates outside state
addImpure(5);
console.log(total);                  // => 5 (depends on history; harder to test)

// — 2. Immutability – don't mutate, return new data —————
const nums = [1, 2, 3];

// ❌ mutating:
// nums.push(4);  nums.sort();
```

```
//  immutable: produce NEW arrays/objects instead
const added = [...nums, 4]; // => [1,2,3,4], nums untouched
const doubled = nums.map((n) => n * 2); // => [2,4,6]
const sorted = [...nums].sort((a, b) => b - a); // copy before sort
console.log(nums, added, doubled, sorted); // => [1,2,3] [1,2,3,4] [2,4,6] [3,2,1]

const user = { name: 'Sam', age: 25 };
const older = { ...user, age: 26 }; // new object, original unchanged
console.log(user.age, older.age); // => 25 26

// — 3. Higher-order functions (take or return functions) —————
// map/filter/reduce are HOFs (lesson 13). You can write your own:
const withLogging = (fn) => (...args) => {
  console.log(`calling with ${args}`);
  return fn(...args);
};
const loggedAdd = withLogging(addPure);
console.log(loggedAdd(4, 5)); // => calling with 4,5 / 9

// — 4. Currying – turn f(a, b, c) into f(a)(b)(c) —————
// Lets you "fix" arguments one at a time and build specialized functions.
const multiply = (a) => (b) => a * b;
console.log(multiply(3)(4)); // => 12
const triple = multiply(3); // pre-fill the first argument
console.log(triple(10)); // => 30

// A generic curry helper:
function curry(fn) {
  return function curried(...args) {
    if (args.length >= fn.length) return fn(...args);
    return (...more) => curried(...args, ...more);
  };
}
const sum3 = curry((a, b, c) => a + b + c);
console.log(sum3(1)(2)(3)); // => 6
console.log(sum3(1, 2)(3)); // => 6
console.log(sum3(1)(2, 3)); // => 6

// — 5. Partial application – pre-fill SOME arguments —————
const greet = (greeting, name) => `${greeting}, ${name}!`;
const sayHello = greet.bind(null, 'Hello'); // fix the first arg
console.log(sayHello('Ana')); // => Hello, Ana!

// — 6. Composition – pipe data through a series of functions —————
```



```
// pipe: left → right (read top to bottom). compose: right → left (math style).
const pipe = (...fns) => (input) => fns.reduce((acc, fn) => fn(acc), input);
const compose = (...fns) => (input) => fns.reduceRight((acc, fn) => fn(acc), input);

const inc = (n) => n + 1;
const double = (n) => n * 2;
const square = (n) => n * n;

const transform = pipe(inc, double, square); // ((n+1)*2)^2
console.log(transform(3)); // => 64    (3→4→8→64)

const transform2 = compose(square, double, inc); // same result, reversed order
console.log(transform2(3)); // => 64

// — 7. Putting it together (declarative data pipeline) —————
const orders = [
  { item: 'pen', price: 5, qty: 3 },
  { item: 'pad', price: 8, qty: 2 },
  { item: 'ink', price: 12, qty: 1 },
];
const grandTotal = orders
  .map(o => o.price * o.qty)    // line totals: [15, 16, 12]
  .filter(t => t >= 12)        // keep big lines: [15, 16, 12]
  .reduce((sum, t) => sum + t, 0);
console.log(grandTotal); // => 43

/* WHY FP? -----
 *   • Pure + immutable = predictable, easy to test, fewer "spooky" bugs.
 *   • Small composable functions read like a recipe.
 *   • You don't have to go 100% FP – adopting purity & immutability where it
 *     helps already makes most codebases cleaner.
 * ----- */

/* PRACTICE -----
 *   1. Rewrite a loop that mutates an array as a pure map/filter chain.
 *   2. Curry a function divide(a)(b) and make a halve = divide-by-2 helper.
 *   3. Use pipe to: trim → lowercase → replace spaces with "-" on a string.
 * ----- */
```

## 31 · TRICKY CONCEPTS & INTERVIEW GOTCHAS

```
/* =====
 * 31 · TRICKY CONCEPTS & INTERVIEW GOTCHAS
 * =====
 * Run:  node lessons/31-tricky-concepts.js
```

```

*
* WHAT YOU'LL LEARN
*   • The classic puzzles that trip up everyone (and show up in interviews)
*   • WHY each one behaves the way it does
*
* For each: read the setup, GUESS the output, then check the // => comment.
* Understanding the "why" means you've truly understood the language.
* ===== */

// — 1. var in a loop with setTimeout (the #1 classic) —————
// var is function-scoped: all callbacks share ONE i, which is 3 by the time
// they run. let is block-scoped: each iteration gets its OWN i.
console.log('--- var vs let in loops ---');
for (var i = 0; i < 3; i++) {
  setTimeout(() => console.log('var i =', i), 0); // => 3, 3, 3
}
for (let j = 0; j < 3; j++) {
  setTimeout(() => console.log('let j =', j), 10); // => 0, 1, 2
}

// — 2. Hoisting: function declarations vs expressions —————
console.log('--- hoisting ---');
console.log(declared()); // => "I work!" (declarations hoist fully)
function declared() { return 'I work!'; }
try {
  expressed(); // ❌ TypeError: expressed is not a function
} catch (e) {
  console.log('expression:', e.constructor.name); // => TypeError
}
var expressed = function () { return 'nope'; }; // var = undefined when called above

// — 3. Reference equality —————
console.log('--- reference vs value ---');
console.log({} === {}); // => false (two different objects)
console.log([] === []); // => false
const x = {}; const y = x;
console.log(x === y); // => true (same reference)
console.log(0.1 + 0.2 === 0.3); // => false (floating point – lesson 06)

// — 4. == coercion landmines (why we use ===) —————
console.log('--- loose equality ---');
console.log([] == ![]); // => true 🤯 (![] is false → [] == false → '' == 0 → true)
console.log(null == undefined); // => true
console.log(null == 0); // => false (null only loosely equals undefined)
console.log(NaN == NaN); // => false (use Number.isNaN instead)

```

```
// — 5. this depends on the CALL SITE —————
console.log('--- this binding ---');
const obj = {
  name: 'Sam',
  regular() { return this?.name; },
  arrow: () => (typeof globalThis.name === 'string' ? globalThis.name : '<no this>'),
};
console.log(obj.regular());          // => Sam (called as obj.method → this = obj)
const detached = obj.regular;
console.log(detached());             // => undefined (called standalone → this lost)
console.log(obj.arrow());            // => <no this> (arrow has no own this)
```

```
// — 6. Array holes & common surprises —————
console.log('--- array quirks ---');
console.log(typeof NaN);             // => number (NaN is a number!)
console.log([1, 2, 3].map(String));  // => [ '1', '2', '3' ]
console.log(['1', '7', '11'].sort()); // => [ '1', '11', '7' ] (string sort!)
console.log([1, 7, 11].sort((a, b) => a - b)); // => [ 1, 7, 11 ] (numeric sort)
console.log(parseInt('08'));         // => 8 (modern; older engines: beware!)
```

```
// — 7. Closures capturing a shared variable —————
console.log('--- closure capture ---');
function makeFns() {
  const fns = [];
  for (let k = 0; k < 3; k++) fns.push(() => k); // let → each captures its own k
  return fns;
}
console.log(makeFns().map((f) => f())); // => [ 0, 1, 2 ]
// (With var k, this would be [3, 3, 3] – same trap as #1.)
```

```
// — 8. Object key coercion —————
console.log('--- object keys are strings ---');
const o = {};
o[1] = 'number one';
o['1'] = 'string one'; // SAME key – keys are coerced to strings
console.log(o[1]);      // => string one (the second write won)
console.log(Object.keys(o)); // => [ '1' ] (only one key)
```

```
// — 9. Truthy/falsy in conditions —————
console.log('--- truthy/falsy ---');
if ([]) console.log('empty array is truthy'); // prints (empty array → truthy)
if ('0') console.log('"0" string is truthy'); // prints (non-empty string)
if (!0) console.log('0 is falsy');           // prints
// The 8 falsy values: false, 0, -0, 0n, "", null, undefined, NaN. Else → truthy.
```

```
// — 10. The mutation-by-reference bug —————
console.log('--- shared reference bug ---');
function addItem(item, list = []) { list.push(item); return list; }
console.log(addItem('a')); // => [ 'a' ]
console.log(addItem('b')); // => [ 'b' ] (default param is fresh each call – safe)
// △ But a SHARED outer array would accumulate:
const shared = [];
const buggy = (item) => { shared.push(item); return shared; };
buggy('x'); buggy('y');
console.log(shared); // => [ 'x', 'y' ] (mutated across calls – often a bug)

/* INTERVIEW TIPS -----
*   • Always prefer let/const over var (kills the loop-closure trap).
*   • Always use === ; reserve == only for `== null` (catches null+undefined).
*   • Remember: this is set by HOW a function is called, not where it's defined.
*   • Objects/arrays are passed by REFERENCE – copy before mutating.
*   • Know the 8 falsy values cold.
* ----- */

/* PRACTICE -----
*   1. Fix a "3,3,3" var loop two ways: with let, and with an IIFE closure.
*   2. Explain out loud why [] == ![] is true, token by token.
*   3. Write a function that safely copies before mutating its array argument.
* ----- */
```

## 32 · DATES & TIME

---

```
/* =====
* 32 · DATES & TIME
* =====
* Run: node lessons/32-dates-time.js
*
* WHAT YOU'LL LEARN
*   • Creating Date objects
*   • Reading & changing parts of a date
*   • Timestamps and date math (durations)
*   • Formatting dates for humans with Intl.DateTimeFormat
*
* NOTE: Date is quirky (months are 0-based!). Read the comments carefully.
* ===== */

// — 1. Creating dates —————
const now = new Date(); // current date & time
const fromISO = new Date('2026-06-04'); // from an ISO string (recommended format)
```

```

const fromParts = new Date(2026, 5, 4); // ⚠ month is 0-based! 5 = JUNE
const fromStamp = new Date(0);           // from a timestamp (ms since 1970-01-01 UTC)

console.log(fromISO.toISOString());      // => 2026-06-04T00:00:00.000Z
console.log(fromParts.getFullYear(), fromParts.getMonth()); // => 2026 5 (June)
console.log(fromStamp.toISOString());    // => 1970-01-01T00:00:00.000Z

// — 2. Reading parts of a date —————
const d = new Date('2026-06-04T09:30:00');
console.log(d.getFullYear()); // => 2026
console.log(d.getMonth());    // => 5    (0=Jan ... 5=Jun – add 1 for humans)
console.log(d.getDate());     // => 4    (day of month, 1-based)
console.log(d.getDay());      // => 4    (day of week, 0=Sun ... 4=Thu)
console.log(d.getHours(), d.getMinutes()); // => 9 30
// UTC variants exist too: getUTCFullYear(), getUTCHours(), ...

// — 3. Timestamps & "now" —————
const ms = Date.now();        // milliseconds since 1970 (an integer)
console.log(typeof ms);       // => number
console.log(d.getTime());     // => the timestamp of date d
// Timestamps make dates easy to compare and subtract.

// — 4. Date math – durations —————
const start = new Date('2026-06-01');
const end = new Date('2026-06-04');
const diffMs = end - start;   // subtracting dates → milliseconds
const diffDays = diffMs / (1000 * 60 * 60 * 24);
console.log(diffDays);        // => 3

// Add time by working in ms (or with setDate):
const tomorrow = new Date(start);
tomorrow.setDate(start.getDate() + 1); // setDate handles month rollovers
console.log(tomorrow.toISOString().slice(0, 10)); // => 2026-06-02

// A reusable "days between" helper:
const daysBetween = (a, b) => Math.round((b - a) / 86_400_000);
console.log(daysBetween(start, end)); // => 3

// — 5. Formatting for humans – Intl.DateTimeFormat (the modern way) —————
const date = new Date('2026-06-04T15:05:00');

console.log(new Intl.DateTimeFormat('en-US').format(date)); // => 6/4/2026
console.log(new Intl.DateTimeFormat('en-GB').format(date)); // => 04/06/2026

const long = new Intl.DateTimeFormat('en-US', {

```

```

    weekday: 'long', year: 'numeric', month: 'long', day: 'numeric',
  });
  console.log(long.format(date)); // => Thursday, June 4, 2026

  const withTime = new Intl.DateTimeFormat('en-US', {
    hour: '2-digit', minute: '2-digit',
  });
  console.log(withTime.format(date)); // => 03:05 PM

  // Quick built-ins (less control, but easy):
  console.log(date.toDateString()); // => Thu Jun 04 2026
  console.log(date.toLocaleDateString()); // => locale-specific short date

  // — 6. "Time ago" – a common real-world function —————
  function timeAgo(date, from) {
    const seconds = Math.floor((from - date) / 1000);
    const units = [
      ['year', 31536000], ['month', 2592000], ['day', 86400],
      ['hour', 3600], ['minute', 60], ['second', 1],
    ];
    for (const [name, secs] of units) {
      const value = Math.floor(seconds / secs);
      if (value >= 1) return `${value} ${name}${value > 1 ? 's' : ''} ago`;
    }
    return 'just now';
  }
  const past = new Date('2026-06-01T12:00:00');
  const reference = new Date('2026-06-04T12:00:00');
  console.log(timeAgo(past, reference)); // => 3 days ago

  /* GOTCHAS -----
  *   • Months are 0-based (January = 0). Days of month are 1-based. Confusing!
  *   • Always prefer ISO strings ('2026-06-04') – other formats are ambiguous.
  *   • Dates carry a time zone; '2026-06-04' is parsed as UTC midnight.
  *   • For heavy date work, libraries like date-fns or Day.js save headaches.
  * ----- */

  /* PRACTICE -----
  *   1. Log today's date as "Weekday, Month Day, Year" using Intl.
  *   2. Compute how many days until a target date.
  *   3. Build a function addDays(date, n) that returns a new date n days later.
  * ----- */

```

## 33 · BROWSER STORAGE ⚠ MOSTLY BROWSER

```
/* =====
* 33 · BROWSER STORAGE ⚠ MOSTLY BROWSER
* =====
* Run: node lessons/33-browser-storage.js (runs a Node-safe simulation)
*      OR load via index.html to use REAL localStorage in the browser.
*
* WHAT YOU'LL LEARN
* • localStorage vs sessionStorage vs cookies – when to use each
* • Storing & reading data (it's always strings → use JSON)
* • A typical "save/load app state" pattern
*
* localStorage doesn't exist in Node, so this file runs a tiny in-memory
* stand-in when needed, and shows the REAL browser API in comments.
* ===== */

// — 0. Get a storage object (real in browser, simulated in Node) —————
const storage =
  typeof localStorage !== 'undefined'
    ? localStorage
    : (() => {
        // Minimal in-memory mock so the lesson runs in Node:
        const map = new Map();
        return {
          setItem: (k, v) => map.set(k, String(v)),
          getItem: (k) => (map.has(k) ? map.get(k) : null),
          removeItem: (k) => map.delete(k),
          clear: () => map.clear(),
        };
      })();

// — 1. The basics: setItem / getItem / removeItem —————
// ⚠ Storage only holds STRINGS. Numbers/booleans get converted to strings.
storage.setItem('username', 'Sam');
console.log(storage.getItem('username')); // => Sam
console.log(storage.getItem('missing')); // => null (not undefined!)

storage.setItem('count', 5);
console.log(typeof storage.getItem('count')); // => string ("5", not 5!)

// — 2. Storing objects/arrays → JSON.stringify on the way in, parse on out —
const user = { name: 'Ana', theme: 'dark', favorites: ['js', 'css'] };

storage.setItem('user', JSON.stringify(user)); // object → string
const loaded = JSON.parse(storage.getItem('user')); // string → object
console.log(loaded.favorites); // => [ 'js', 'css' ]
```

```
// — 3. A safe save/load helper (handles missing keys & bad JSON) —————
function save(key, value) {
  storage.setItem(key, JSON.stringify(value));
}
function load(key, fallback = null) {
  const raw = storage.getItem(key);
  if (raw === null) return fallback;
  try {
    return JSON.parse(raw);
  } catch {
    return fallback; // corrupted data → return the default instead of crashing
  }
}

save('settings', { volume: 0.8, muted: false });
console.log(load('settings')); // => { volume: 0.8, muted: false }
console.log(load('nothing', 'default')); // => default


// — 4. Removing & clearing —————
storage.removeItem('count'); // delete one key
console.log(storage.getItem('count')); // => null
// storage.clear(); // △ wipes EVERYTHING for this site


/* — 5. localStorage vs sessionStorage vs cookies —————
*
*   localStorage → persists FOREVER (until cleared). Same API as above.
*                   Use for: theme, settings, draft content, cached data.
*
*   sessionStorage → cleared when the TAB closes. Identical API:
*                   sessionStorage.setItem('k', 'v')
*                   Use for: one-visit state, multi-step form progress.
*
*   cookies → small (~4KB), sent to the SERVER on every request.
*             Read/write the raw string:
*             document.cookie = 'token=abc; max-age=3600; path=/';
*             document.cookie // "token=abc; other=..."
*             Use for: auth tokens / things the server must see.
*             (For real apps, set auth cookies from the server as
*             HttpOnly so JS can't read them – safer.)
*
*   LIMITS: localStorage/sessionStorage ~5MB, synchronous, strings only,
*           per-origin (one site can't read another's storage).
*/
```



```

/* — 6. Real browser example (uncomment when loaded via index.html) —————
 * // Persist a counter across page reloads:
 * const n = Number(localStorage.getItem('visits') || 0) + 1;
 * localStorage.setItem('visits', n);
 * console.log(`You've visited ${n} times`);
 */

/* PRACTICE -----
 * 1. Save a to-do array to storage, reload it, and add one item.
 * 2. Build loadTheme()/saveTheme() that default to 'light' when unset.
 * 3. In the browser, make a counter that survives page refreshes.
 * ----- */

```

## 34 · BOM & TIMERS (the Browser Object Model)

---

```

/* =====
 * 34 · BOM & TIMERS (the Browser Object Model)
 * =====
 * Run: node lessons/34-bom-timers.js (timers run; BOM parts are shown safely)
 *
 * WHAT YOU'LL LEARN
 * • setTimeout / setInterval and how to cancel them
 * • The BOM: window, location, history, navigator, screen
 * • A practical countdown timer pattern
 *
 * The DOM (lessons 23–24) is the PAGE. The BOM is the BROWSER around it –
 * the window, the URL, history, the device. Timers work everywhere.
 * ===== */

// — 1. setTimeout – run once after a delay —————
const id = setTimeout(() => {
  console.log('runs after 50ms');
}, 50);
// Cancel it before it fires with clearTimeout(id):
// clearTimeout(id);

// — 2. setInterval – run repeatedly every N ms —————
let ticks = 0;
const intervalId = setInterval(() => {
  ticks++;
  console.log('tick', ticks);
  if (ticks >= 3) {
    clearInterval(intervalId); // △ ALWAYS stop intervals or they run forever
    console.log('interval stopped');
  }
}

```

```
}, 20);
```

```
// — 3. A practical countdown —————
```

```
function countdown(from, onTick, onDone) {
  let n = from;
  const t = setInterval(() => {
    onTick(n);
    if (n === 0) {
      clearInterval(t);
      onDone();
    }
    n--;
  }, 15);
  return t;
}
countdown(3, (n) => console.log(`T-minus ${n}`), () => console.log('🚀 liftoff'));
```

```
// — 4. setTimeout(0) – defer work to "after the current code" —————
```

```
// Not actually 0ms – it runs after the current synchronous code & microtasks.
console.log('sync 1');
setTimeout(() => console.log('deferred (macrotask)'), 0);
console.log('sync 2'); // logs before "deferred" – see lesson 29's event loop
```

```
/* — 5. The BOM objects (browser only – shown as comments) —————
```

```
*
* window – the global object in browsers. window.x === x at top level.
*       window.innerWidth / innerHeight → viewport size
*       window.alert() / confirm() / prompt() → dialog boxes
*       window.scrollTo(0, 0) → scroll the page
*
* location – the current URL (read AND navigate):
*       location.href // "https://site.com/page?q=1#top"
*       location.pathname // "/page"
*       location.search // "?q=1"
*       location.hash // "#top"
*       location.href = 'https://example.com' // navigate to a new page
*       location.reload() // refresh
*
* history – the back/forward stack:
*       history.back(); history.forward(); history.go(-2);
*       history.pushState({}, '', '/new-url'); // change URL, no reload (SPAs)
*
* navigator – info about the browser/device:
*       navigator.language // "en-US"
*       navigator.onLine // true / false
*       navigator.clipboard.writeText('copied!')
*       navigator.geolocation.getCurrentPosition(...)
```

```

*
* screen – the physical display: screen.width, screen.height
*/

// — 6. Reading query params (works in browser; demo'd with URL here) —————
// In a browser you'd use: new URLSearchParams(location.search)
const params = new URLSearchParams('?user=sam&page=2');
console.log(params.get('user')); // => sam
console.log(params.get('page')); // => 2
console.log([...params]);        // => [ [ 'user', 'sam' ], [ 'page', '2' ] ]

/* globalThis – the global object EVERYWHERE -----
*   In browsers the global is `window`; in Node it's `global`. `globalThis`
*   works in both, so portable code uses it.
*/
console.log(typeof globalThis); // => object

/* PRACTICE -----
*   1. Build a clock that logs the time every second, then stops after 5.
*   2. Parse "?theme=dark&lang=en" with URLSearchParams and read both values.
*   3. (Browser) Log location.pathname and navigator.language in the console.
*   ----- */

```

## 35 • DEBUGGING & THE CONSOLE

---

```

/* =====
* 35 • DEBUGGING & THE CONSOLE
* =====
* Run:  node lessons/35-debugging-console.js
*
* WHAT YOU'LL LEARN
*   • The console methods beyond console.log
*   • Reading error messages & stack traces
*   • The `debugger` statement and breakpoints
*   • A practical debugging mindset
*
* Debugging is HALF of programming. Knowing your tools turns "why won't this
* work?!" into a calm, 2-minute investigation.
* ===== */

// — 1. Beyond console.log – the full console toolkit —————
console.log('plain message');
console.info('📄 informational');
console.warn('⚠ a warning (yellow in browsers)');

```

```

console.error('✗ an error (red – does NOT stop the program)');

// console.table – gorgeous for arrays of objects:
console.table([
  { name: 'Ana', age: 25 },
  { name: 'Bob', age: 30 },
]);

// Grouping related logs (indented; collapsible in the browser):
console.group('User load');
console.log('fetching...');
console.log('done');
console.groupEnd();

// Counting how many times a line runs:
console.count('loop'); // => loop: 1
console.count('loop'); // => loop: 2

// Timing how long something takes:
console.time('work');
let sum = 0;
for (let i = 0; i < 1e6; i++) sum += i;
console.timeEnd('work'); // => work: 1.2ms (your number will vary)

// Assertions – log ONLY if the condition is false (great for sanity checks):
console.assert(sum > 0, 'sum should be positive'); // (prints nothing – it passed)

// — 2. The log-the-label trick (see WHICH variable is which) —————
const x = 5, y = 10;
// Wrap variables in an object → labels come for free:
console.log({ x, y }); // => { x: 5, y: 10 } (clearer than console.log(x, y))

// — 3. Reading a stack trace —————
function level3() {
  throw new Error('something failed deep down');
}
function level2() { level3(); }
function level1() { level2(); }

try {
  level1();
} catch (err) {
  console.log('message:', err.message); // => something failed deep down
  // err.stack shows the call path TOP-to-bottom (where it threw → who called it):
  console.log('first stack line:', err.stack.split('\n')[1].trim());
  // Read stacks from the TOP: that's where the error actually happened.
}

```

```
// — 4. The `debugger` statement & breakpoints —————
// Drop `debugger;` into your code. When DevTools (browser) or an inspector
// (node --inspect) is open, execution PAUSES there so you can inspect variables
// step by step. It's ignored when no debugger is attached.
function buggyAverage(nums) {
  // debugger; // ← uncomment, run `node --inspect-brk` or use VS Code's debugger
  const total = nums.reduce((a, b) => a + b, 0);
  return total / nums.length;
}
console.log(buggyAverage([2, 4, 6])); // => 4

/* — 5. A debugging MINDSET (more useful than any tool) —————
* 1. Reproduce it reliably – know the exact steps that trigger the bug.
* 2. Read the FULL error message + the FIRST line of the stack trace.
* 3. Isolate: log values right BEFORE the failing line. What's unexpected?
* 4. Check assumptions: log the TYPE too – console.log(typeof x, x).
*    (Most bugs are "I thought this was a number but it's a string/undefined.")
* 5. Narrow it down: comment out half, see if the bug remains. Repeat.
* 6. Fix the CAUSE, not the symptom. Then re-run to confirm.
*
* COMMON ERROR MESSAGES, DECODED:
* "x is not defined"           → typo, or used before declaring (TDZ).
* "Cannot read properties of   → you accessed .foo on null/undefined.
*   undefined (reading 'foo')"   Log the object first; use ?. (lesson 03).
* "x is not a function"        → it's not what you think (typo, or undefined).
* "Unexpected token"           → a syntax error (missing }, ), or comma).
*/

/* PRACTICE -----
* 1. Use console.table to print an array of 3 product objects.
* 2. Trigger a "Cannot read properties of undefined" on purpose, then fix it
*    with optional chaining.
* 3. Wrap a slow loop in console.time/timeEnd and read the duration.
* ----- */
```

## 36 · WEAKMAP, MEMORY & FINAL LANGUAGE DETAILS

```
/* =====
* 36 · WEAKMAP, MEMORY & FINAL LANGUAGE DETAILS
* =====
* Run: node lessons/36-weakmap-memory.js
*
* WHAT YOU'LL LEARN
```

```

*   • How JS memory & garbage collection work (the short version)
*   • WeakMap / WeakSet and why "weak" matters
*   • WeakRef & FinalizationRegistry (advanced GC hooks)
*   • Strict mode
*   • Tagged template literals
*
* The last few pieces to round out your JavaScript knowledge.
* ===== */

// — 1. Garbage collection in 60 seconds —————
// JS manages memory FOR you. An object stays in memory as long as something
// can still REACH it (a variable, an array, a property...). When nothing
// references it anymore, the garbage collector frees it automatically.
let data = { big: 'lots of stuff' };
data = null; // the object is now unreachable → eligible to be cleaned up
// You don't free memory manually – but you CAN cause "leaks" by holding
// references you forgot about (e.g. an ever-growing array, or listeners you
// never remove). The fix is usually: stop referencing what you're done with.


// — 2. Map vs WeakMap —————
// A normal Map holds its keys STRONGLY – they can never be garbage-collected
// while the Map exists. A WeakMap holds keys WEAKLY: if the key object is no
// longer referenced anywhere else, it (and its value) can be cleaned up.

const weak = new WeakMap();
let user = { name: 'Sam' };
weak.set(user, { lastLogin: '2026-06-04' }); // associate metadata with an object
console.log(weak.get(user)); // => { lastLogin: '2026-06-04' }
console.log(weak.has(user)); // => true

user = null; // now nothing else references the user object →
             // the WeakMap entry becomes eligible for garbage collection.

// RESTRICTIONS (the price of being "weak"):
//   • Keys MUST be objects (not strings/numbers).
//   • NOT iterable – no .size, no for...of, no .keys(). You can't list entries.
// USE CASES: attach private/extra data to objects without preventing their
// cleanup – caches, metadata, "have I processed this object?" tracking.


// — 3. WeakSet – same idea, for a set of objects —————
const seen = new WeakSet();
let task = { id: 1 };
seen.add(task);
console.log(seen.has(task)); // => true
task = null; // entry can be garbage-collected; you don't have to clean up
// Great for "have I already handled this object?" without leaking memory.

```

```

// — 3b. WeakRef & FinalizationRegistry (advanced – use rarely) —————
// A WeakRef holds a reference that does NOT keep its target alive: the GC may
// still collect the object. Call .deref() to get it back – or `undefined` if
// it's already been collected. Useful for memory-sensitive caches.
let cached = { data: 'expensive to compute' };
const ref = new WeakRef(cached);
console.log(ref.deref()?.data); // => expensive to compute (still alive here)
// After `cached = null` AND a GC cycle, ref.deref() would return undefined.
// ⚠ You can't predict WHEN the GC runs, so never rely on timing.

// A FinalizationRegistry lets you register a cleanup callback that MAY run after
// an object is collected – e.g. to release an external resource tied to it.
const registry = new FinalizationRegistry((heldValue) => {
  console.log('cleanup ran for:', heldValue); // may fire later, or never
});
registry.register(cached, 'cache-entry-1'); // associate a label with the object
// ⚠ Finalizers are NOT guaranteed to run (e.g. on exit) and timing is
// unpredictable. Treat them as a best-effort safety net, never your main logic.
// RULE OF THUMB: prefer WeakMap/WeakSet; reach for WeakRef/FinalizationRegistry
// only for genuine memory caches or wrapping external (C/WASM) resources.

// — 4. Strict mode —————
// 'use strict' opts into a safer JS: it turns silent mistakes into errors.
// ES modules (lesson 22) and class bodies are ALWAYS strict automatically.
function sloppyVsStrict() {
  'use strict';
  // mistyped = 5; // ❌ in strict mode: ReferenceError (without it: silently
  //              // creates a global – a classic hard-to-find bug!)
  return 'strict mode catches accidental globals & other footguns';
}
console.log(sloppyVsStrict());
// Strict mode also: forbids duplicate params, makes `this` undefined in plain
// function calls (lesson 11), and disallows deleting variables. Prefer it
// (you get it free with modules/classes).

// — 5. Tagged template literals —————
// A function placed before a template literal receives the string PIECES and
// the interpolated VALUES separately – letting you process them.
function highlight(strings, ...values) {
  // strings: the text chunks; values: the ${...} results
  return strings.reduce(
    (out, str, i) => out + str + (i < values.length ? `[${values[i]}]` : ''),
    ''
  );
}
const name = 'Sam';
const score = 95;

```

```

console.log(highlight`User ${name} scored ${score} points`);
// => User [Sam] scored [95] points

// Real-world tags: styled-components (CSS-in-JS), String.raw, safe HTML/SQL
// escaping, i18n. String.raw is a built-in tag that ignores escape sequences:
console.log(String.raw`C:\new\test`); // => C:\new\test  (\n NOT a newline)

/* MEMORY-LEAK AVOIDANCE TIPS -----
 *   • Remove event listeners when you're done (removeEventListener).
 *   • Clear timers you no longer need (clearInterval / clearTimeout).
 *   • Don't let global arrays/objects grow forever – cap or clear them.
 *   • Use WeakMap/WeakSet when associating data with objects you don't own.
 * ----- */

/* 🎉 THAT'S THE WHOLE CORE LANGUAGE – basics, intermediate, advanced, and the
 *   browser. From here the course shifts gears:
 *   • Part 10  (37)      – modern JS (ES2021–ES2026)
 *   • Part 11  (38–47)  – professional & production engineering
 *   • Part 12  (48–50)  – computer-science core (data structures,
 *                       algorithms) and binary data
 *   A great next step right now: build a small app (to-do, weather, quiz)
 *   using lessons 23–25 + 33. Building is what makes it all stick. */

/* PRACTICE -----
 *   1. Use a WeakMap to cache a computed result keyed by an object argument.
 *   2. Add 'use strict' to a function and trigger an accidental-global error.
 *   3. Write a tag function that uppercases every interpolated value.
 *   4. Wrap an object in a WeakRef, log .deref()?.x, then set the original to
 *       null and explain why .deref() MIGHT (eventually) return undefined.
 * ----- */

```

## 37 · MODERN JAVASCRIPT (ES2021 → ES2026)

---

```

/* =====
 * 37 · MODERN JAVASCRIPT (ES2021 → ES2026)
 * =====
 * Run:  node lessons/37-modern-js.js   (needs Node 22+ ; you have v24 ✅)
 *
 * WHAT YOU'LL LEARN
 *   The newest, genuinely useful additions to JavaScript – grouped by year.
 *   These make a lot of older patterns cleaner. Several REPLACE workarounds
 *   you saw earlier in the course (noted with "↔ replaces ...").
 *
 * NOTE: very new features need an up-to-date runtime. Everything here runs in
 *       Node 22/24 and current browsers (2024+). Older environments may not
 *       support the ES2024/ES2025 items yet.

```



```

* ===== */

console.log('==== ES2021 ====');

// — Logical assignment: ??= ||= &&= —————
// Combine a logical check with assignment. Super handy for defaults.
let config = { theme: null, retries: 0 };
config.theme ??= 'light'; // assign ONLY if null/undefined → 'light'
config.retries ||= 3; // assign if falsy (0 counts!) → 3
config.theme &&= 'dark'; // assign only if already truthy → 'dark'
console.log(config); // => { theme: 'dark', retries: 3 }
// ↪ replaces: if (config.theme == null) config.theme = 'light'

// — Promise.any – first to SUCCEED wins —————
// (vs Promise.race which settles on the first to finish, success OR fail)
const fastest = await Promise.any([
  Promise.reject('server A down'),
  Promise.resolve('server B responded'),
]);
console.log(fastest); // => server B responded

console.log('==== ES2022 ====');

// — Top-level await —————
// In ES modules you can `await` at the top level – no async IIFE wrapper.
// (This file uses it; that's why awaits above work without an async function.)
const ready = await Promise.resolve('no IIFE needed');
console.log(ready); // => no IIFE needed

// — Object.hasOwn(obj, key) – the safe "has own property" check —————
const obj = { a: 1 };
console.log(Object.hasOwn(obj, 'a')); // => true
console.log(Object.hasOwn(obj, 'toString')); // => false (inherited, not own)
// ↪ replaces the clunky: Object.prototype.hasOwnProperty.call(obj, 'a')

// — Error cause – chain errors without losing the original —————
try {
  try {
    throw new Error('low-level network fail');
  } catch (original) {
    throw new Error('Could not load user', { cause: original });
  }
} catch (err) {
  console.log(err.message, '| caused by:', err.cause.message);
  // => Could not load user | caused by: low-level network fail
}

```

```

console.log('===== ES2023 =====');

// — findLast / findLastIndex – search from the END —————
const nums = [1, 2, 3, 4, 5];
console.log(nums.findLast((n) => n % 2 === 1)); // => 5 (last odd)
console.log(nums.findLastIndex((n) => n % 2 === 1)); // => 4

// — Immutable array methods – return a NEW array, no mutation ★ —————
// These are the modern fix for the "copy-before-sort" dance from lessons 12/13.
const original = [3, 1, 2];
console.log(original.toSorted((a, b) => a - b)); // => [1, 2, 3]
console.log(original.toReversed()); // => [2, 1, 3]
console.log(original.with(0, 99)); // => [99, 1, 2] (replace index 0)
console.log(original.toSpliced(1, 1)); // => [3, 2] (remove 1 at idx 1)
console.log(original); // => [3, 1, 2] UNCHANGED ✓
// ↳ replaces: [...arr].sort(...) and [...arr].reverse()

console.log('===== ES2024 =====');

// — Object.groupBy – group array items by a key ★ —————
const people = [
  { name: 'Ana', dept: 'eng' },
  { name: 'Bob', dept: 'sales' },
  { name: 'Cy', dept: 'eng' },
];
const byDept = Object.groupBy(people, (p) => p.dept);
console.log(byDept.eng.map((p) => p.name)); // => [ 'Ana', 'Cy' ]
// ↳ replaces a whole reduce() into an object (you wrote one in lesson 13!)

// Map.groupBy is the same, but returns a Map (keys can be any type):
const byParity = Map.groupBy([1, 2, 3, 4], (n) => (n % 2 ? 'odd' : 'even'));
console.log(byParity.get('odd')); // => [ 1, 3 ]

// — Promise.withResolvers – get a promise + its resolve/reject together —————
// Cleaner than capturing them inside the executor with outer `let`s.
const { promise, resolve } = Promise.withResolvers();
setTimeout(() => resolve('done!'), 10);
console.log(await promise); // => done!

// — Array.fromAsync – build an array from an async iterable —————
async function* drip() { yield 1; yield 2; yield 3; }
console.log(await Array.fromAsync(drip())); // => [ 1, 2, 3 ]

console.log('===== ES2025 (newest) =====');

// — Iterator helpers – map/filter/take LAZILY, without arrays ★ —————
// You can now chain methods directly on iterators (e.g. from .values(),

```

```

// generators, Map/Set entries) – processed lazily, one item at a time.
function* naturals() { let i = 1; while (true) yield i++; }
const firstThreeEvens = naturals()
  .filter((n) => n % 2 === 0)
  .map((n) => n * 10)
  .take(3) // stop after 3 – works on an INFINITE source
  .toArray();
console.log(firstThreeEvens); // => [ 20, 40, 60 ]
// ↪ before, .map/.filter only existed on arrays (you had to materialize first)

// — New Set methods – real set algebra ★ —————
const a = new Set([1, 2, 3, 4]);
const b = new Set([3, 4, 5, 6]);
console.log([...a.union(b)]); // => [ 1, 2, 3, 4, 5, 6 ]
console.log([...a.intersection(b)]); // => [ 3, 4 ]
console.log([...a.difference(b)]); // => [ 1, 2 ]
console.log([...a.symmetricDifference(b)]); // => [ 1, 2, 5, 6 ]
console.log(new Set([1, 2]).isSubsetOf(a)); // => true

// — Promise.try – run a function and ALWAYS get a promise back —————
// Wraps sync OR async functions uniformly; sync throws become rejections.
const result = await Promise.try(() => 42);
console.log(result); // => 42

// — RegExp.escape – safely put arbitrary text into a regex —————
const userInput = 'a.b*c';
const safe = new RegExp(RegExp.escape(userInput)); // dots/stars treated literally
console.log(safe.test('a.b*c')); // => true
console.log(safe.test('aXbYc')); // => false (the . and * are NOT wildcards now)

console.log('==== ES2026 & THE CUTTING EDGE ====');

// — using / await using – automatic cleanup (explicit resource management) ———
// A `using` declaration auto-calls the resource's [Symbol.dispose]() when the
// block ends – like try/finally, but automatic. `await using` calls the async
// [Symbol.asyncDispose]() . Perfect for files, DB connections, locks, sockets.
// (Stage 3 / ES2026; needs a very recent runtime. Shown so you recognize it.)
//
// function openResource(name) {
//   return { [Symbol.dispose]() { console.log(`closing ${name}`); } };
// }
// {
//   using db = openResource('db'); // no manual close needed
//   using file = openResource('file');
//   // ...use them...
// } // ← here both are disposed automatically, in REVERSE order
// // ↪ replaces the try { } finally { db.close(); file.close(); } dance.

// — Import attributes – import non-JS modules safely —————

```

```

// Tell the engine what kind of module you're importing (currently JSON):
//   import config from './config.json' with { type: 'json' };
//   const data = await import('./data.json', { with: { type: 'json' } });
// ↪ replaces reading + JSON.parse-ing a file by hand for static config.

// — Decorators – annotate classes/methods with @reusable behavior —————
// A decorator is a function that wraps a class/method to add behavior (logging,
// validation, memoization) declaratively. (Stage 3; TypeScript/Angular already
// use them; frameworks lean on them heavily.)
//
//   function logged(originalMethod, ctx) {
//     return function (...args) {
//       console.log(`calling ${ctx.name}`);
//       return originalMethod.call(this, ...args);
//     };
//   }
//   class Api { @logged getUser(id) { /* ... */ } } // @logged wraps the method
// ↪ a standardized version of the "decorator" pattern from lesson 40.

// — Float16Array & Math.f16round – half-precision numbers (ES2025/26) —————
// 16-bit floats: half the memory, enough precision for ML weights, graphics,
// and GPU data. A typed array (lesson 50) for compact decimal data.
if (typeof Float16Array !== 'undefined') {
  const half = new Float16Array([1.5, 2.25]);
  console.log('Float16Array:', half[0]); // => 1.5
} else {
  console.log('Float16Array: not in this runtime yet (very new)');
}

console.log('===== ON THE HORIZON =====');

/* — Temporal API – the modern replacement for Date (NOT shipped yet) —————
 * Date is famously awkward (0-based months, mutable, no time-zone clarity).
 * Temporal fixes all of it. It's at Stage 3 and not yet in Node/most browsers,
 * so this is shown as a comment. Use the @js-temporal/polyfill package to try it.
 *
 *   const today = Temporal.Now.plainDateISO();           // 2026-06-04 (immutable)
 *   const next = today.add({ days: 7 });                 // clean date math
 *   const meeting = Temporal.ZonedDateTime.from('2026-06-04T15:00[Asia/Kolkata]');
 *   today.month           // 6 (1-based – finally sane!)
 *   today.until(next).days // 7
 *
 * Until it ships, use lesson 32's Date + Intl, or a library (date-fns, Day.js).
 */
console.log('Temporal: coming soon – see comments. For now use lesson 32.');
```

  

```

/* WHAT CHANGED FOR YOUR EVERYDAY CODE -----
 * • Sorting without mutating ..... arr.toSorted()      (was [...arr].sort())

```

```

*   • Grouping data ..... Object.groupBy(...)    (was a reduce)
*   • Defaults ..... x ??= value
*   • Own-property check ..... Object.hasOwn(o, k)
*   • Lazy pipelines ..... iterator.map().filter().take()
*   • Set math ..... a.union(b), a.intersection(b)
*   These are all "nice to have" – the fundamentals you already learned still
*   work everywhere. Reach for these when your runtime supports them.
*   ----- */

/* PRACTICE -----
*   1. Rewrite lesson 13's letter-counting reduce using Object.groupBy.
*   2. Replace a [...arr].sort() somewhere in your code with arr.toSorted().
*   3. Use Set.intersection to find common items in two arrays of tags.
*   ----- */

```

## 38 • TESTING — proving your code works

---

```

/* =====
* 38 • TESTING – proving your code works
* =====
* Run:  node --test lessons/38-testing.js    (or just: node lessons/38-testing.js)
*
* WHAT YOU'LL LEARN
*   • Why we write automated tests
*   • Node's BUILT-IN test runner (node:test) + assert – zero install
*   • Unit tests, grouping, async tests, and mocking
*   • The testing pyramid: unit → integration → end-to-end (E2E)
*   • Code coverage
*   • The TDD loop (red → green → refactor)
*
* Node 18+ ships a real test runner. The same ideas map 1:1 onto Jest/Vitest
* (describe/it/expect) – only the import and matcher names differ.
* ===== */

import { test, describe, mock, before, after } from 'node:test';
import assert from 'node:assert/strict'; // "strict" = uses === by default

// — The code we want to test (normally this lives in another file) —————
function add(a, b) {
  return a + b;
}
function divide(a, b) {
  if (b === 0) throw new Error('cannot divide by zero');
  return a / b;
}
async function fetchUser(id) {
  // pretend this hits a network; resolves after a tick
  return { id, name: 'Sam' };
}

```

```
}
```

```
// — 1. A single unit test —————
```

```
test('add() sums two numbers', () => {  
  assert.equal(add(2, 3), 5);          // pass: 2 + 3 === 5  
  assert.notEqual(add(2, 3), 6);  
});
```

```
// — 2. Grouping related tests with describe —————
```

```
describe('divide()', () => {  
  test('divides normally', () => {  
    assert.equal(divide(10, 2), 5);  
  });  
  
  test('throws on divide-by-zero', () => {  
    // assert.throws checks that the function DOES throw (and the message matches)  
    assert.throws(() => divide(1, 0), /cannot divide by zero/);  
  });  
});
```

```
// — 3. Common assertions —————
```

```
test('assertion styles', () => {  
  assert.ok(1 < 2);                // truthy check  
  assert.equal('a', 'a');          // === for primitives  
  assert.deepEqual({ a: 1 }, { a: 1 }); // structural equality for objects  
  assert.deepEqual([1, 2], [1, 2]);  // ...and arrays  
});
```

```
// — 4. Async tests – just make the test function async and await —————
```

```
test('fetchUser() resolves with a user', async () => {  
  const user = await fetchUser(7);  
  assert.equal(user.id, 7);  
  assert.equal(user.name, 'Sam');  
});
```

```
// — 5. Mocking – replace a real dependency with a fake you control —————
```

```
describe('mocking', () => {  
  test('mock.fn records calls and returns what you tell it', () => {  
    const sendEmail = mock.fn(() => 'sent'); // fake function  
  
    function notify(send, to) {  
      return send(`hello ${to}`);  
    }  
  })  
});
```

```

    const result = notify(sendEmail, 'ana@example.com');
    assert.equal(result, 'sent');
    assert.equal(sendEmail.mock.callCount(), 1); // called once
    assert.equal(sendEmail.mock.calls[0].arguments[0], 'hello ana@example.com');
  });
});

```

// — 6. Setup / teardown hooks —————

```

describe('hooks (before/after)', () => {
  let db;
  before(() => { db = { rows: [] }; }); // runs once before the tests below
  after(() => { db = null; }); // cleanup after

  test('starts empty', () => {
    assert.equal(db.rows.length, 0);
  });
});

```

/\* — 7. THE TESTING PYRAMID – unit / integration / E2E —————

```

* Different tests catch different bugs. The PYRAMID = lots of fast unit tests,
* fewer integration tests, a handful of slow end-to-end tests.
*
*   UNIT (most)      – one function/module in isolation (everything above).
*                     Fast, run on every save. Mock the outside world.
*   INTEGRATION (some) – several pieces TOGETHER (e.g. a route + its database).
*                     Catches "they each work but don't fit" bugs.
*   E2E (few)        – drive the REAL app in a REAL browser like a user would:
*                     click, type, assert what's on screen. Slow but proves
*                     the whole thing actually works.
*
* E2E is done with a browser-automation tool – PLAYWRIGHT (or Cypress):
*   // npm i -D @playwright/test && npx playwright install
*   import { test, expect } from '@playwright/test';
*   test('user can log in', async ({ page }) => {
*     await page.goto('https://localhost:3000');
*     await page.fill('#email', 'ana@example.com');
*     await page.fill('#password', 'secret');
*     await page.click('button[type=submit]');
*     await expect(page.getByText('Welcome, Ana')).toBeVisible(); // real UI check
*   });
* Playwright can also test across Chromium/Firefox/WebKit and take screenshots.
*/

```

/\* — 8. CODE COVERAGE – how much of your code the tests run —————

```

* Coverage shows which lines/branches your tests actually exercise. Node has it
* built in – no install:
*

```

```

*   node --test --experimental-test-coverage lessons/38-testing.js
*
* It prints a table of % lines/branches/functions covered, and which lines were
* MISSED. (Jest/Vitest: `--coverage`) ▲ 100% coverage ≠ bug-free – it means
* "executed", not "asserted correctly". Use it to FIND untested code, not as a
* trophy.
*/

/* THE TDD LOOP -----
*   1. RED   – write a failing test for the behavior you want.
*   2. GREEN – write the SIMPLEST code that makes it pass.
*   3. REFACTOR – clean up, tests still green. Repeat.
*   Tests are a safety net: they let you change code fearlessly.
*
* JEST / VITEST EQUIVALENT (same idea, different names):
*   import { describe, it, expect } from 'vitest';
*   it('adds', () => { expect(add(2,3)).toBe(5); });
* ----- */

/* PRACTICE -----
*   1. Write a test for a `capitalize(str)` function (first letter upper).
*   2. Test that it throws (or returns '') on an empty string.
*   3. Mock a `logger` and assert it was called with the right message.
*   4. Run this file with --experimental-test-coverage and read the report.
*   5. Sketch (in comments) one unit, one integration, and one E2E test for a
*       login feature – note what each would and wouldn't catch.
* ----- */

```

## 39 • SECURITY ESSENTIALS — don't get hacked

---

```

/* =====
* 39 • SECURITY ESSENTIALS – don't get hacked
* =====
* Run:   node lessons/39-security.js
*
* WHAT YOU'LL LEARN
*   • XSS and how to render user input safely
*   • CSRF, CORS, and CSP in plain English
*   • Input validation, auth tokens, and secure cookies
*   • Prototype pollution & dependency safety
*
* Security is about NOT TRUSTING input. Most web vulnerabilities come from
* treating user-controlled data as if it were safe code or markup.
* ===== */

// — 1. XSS (Cross-Site Scripting) – the #1 frontend vulnerability —————
// XSS happens when user input is injected into the page AS HTML/script.

```



```
// ❌ DANGEROUS: el.innerHTML = userInput ← a <script> or onerror runs!
// ✅ SAFE:      el.textContent = userInput ← rendered as plain text, never run.
const userInput = '<img src=x onerror="stealCookies()">';

// If you MUST build HTML, escape the dangerous characters first:
function escapeHTML(str) {
  return str
    .replaceAll('&', '&amp;')
    .replaceAll('<', '&lt;')
    .replaceAll('>', '&gt;')
    .replaceAll('"', '&quot;')
    .replaceAll("'", '&#39;');
}
console.log(escapeHTML(userInput));
// => &lt;img src=x onerror=&quot;stealCookies()&quot;&gt; (harmless text now)
// In real apps, prefer textContent, or a vetted sanitizer like DOMPurify for
// rich HTML. Frameworks (React/Vue) escape by default – don't bypass them.

// — 2. CSRF (Cross-Site Request Forgery) —————
// A malicious site tricks the user's browser into sending a request to YOUR
// site using their logged-in cookies (e.g. a hidden form that transfers money).
// Defenses:
//   • CSRF tokens – a secret per-session value the server checks on writes.
//   • SameSite cookies – `Set-Cookie: token=...; SameSite=Strict` (or Lax).
//   • Check the Origin/Referer header on state-changing requests.

// — 3. CORS (Cross-Origin Resource Sharing) —————
// Browsers BLOCK JS from reading responses across origins UNLESS the server
// opts in with headers. CORS is the SERVER granting permission – it is NOT a
// way to "fix" a blocked request from the client side.
// Server sends: Access-Control-Allow-Origin: https://your-app.com
// "No 'Access-Control-Allow-Origin' header" = the API must allow your origin.

// — 4. CSP (Content Security Policy) —————
// An HTTP header that whitelists where scripts/styles/images may load from.
// A strong CSP turns many XSS bugs into no-ops (inline scripts won't run).
// Content-Security-Policy: default-src 'self'; script-src 'self'

// — 5. Validate & sanitize ALL input —————
// Never trust the client – validate on the SERVER too (client checks are UX).
function validateSignup({ email, age }) {
  const errors = [];
  if (!/^[@\s]+\.[^\s]+\.[^\s]+$/.test(email)) errors.push('invalid email');
  if (!Number.isInteger(age) || age < 13 || age > 120) errors.push('invalid age');
  return { ok: errors.length === 0, errors };
}
```

```

}
console.log(validateSignup({ email: 'bad', age: 5 }));
// => { ok: false, errors: [ 'invalid email', 'invalid age' ] }
console.log(validateSignup({ email: 'a@b.co', age: 30 })); // => { ok: true, errors: [] }

// — 6. Auth tokens & cookies – rules of thumb —————
// • NEVER store secrets/passwords in plain text – hash with bcrypt/argon2 (server).
// • A JWT is 3 base64 parts: header.payload.signature. The payload is NOT
//   encrypted – anyone can read it. Don't put secrets in it; verify the signature.
const fakeJwt = 'eyJhbGciOiJIUzI1NiJ9.eyJ1c2VyIjoiaW5hIn0.sig';
const [, payload] = fakeJwt.split('.');
console.log(JSON.parse(Buffer.from(payload, 'base64').toString())); // => { user: 'ana' }
// • Store session tokens in cookies set: HttpOnly; Secure; SameSite=Strict
//   (HttpOnly = JS can't read it → safe from XSS theft; Secure = HTTPS only.)

// — 7. Prototype pollution —————
// Merging untrusted JSON that contains "__proto__" can poison EVERY object.
// Guard against keys named __proto__ / constructor / prototype when merging,
// or use Object.create(null) / Map for untrusted key-value data.
const safeMap = new Map();
safeMap.set('__proto__', 'harmless here'); // a Map key is just data, not the proto
console.log(safeMap.get('__proto__')); // => harmless here

// — 8. Dependency & supply-chain safety —————
// • Run `npm audit` and keep dependencies patched.
// • Pin versions (lockfile) and review what you install – packages run code.
// • Don't commit secrets; use environment variables (see lesson 44/45).

/* PRACTICE -----
 * 1. Write a safeRender(el, text) helper that uses textContent.
 * 2. Add password-strength validation (min length, has number) to validateSignup.
 * 3. Explain in a comment why CORS errors can't be fixed from the browser.
 * ----- */

```

## 40 • DESIGN PATTERNS — reusable solutions with shared names

---

```

/* =====
 * 40 • DESIGN PATTERNS – reusable solutions with shared names
 * =====
 * Run:  node lessons/40-design-patterns.js
 *
 * WHAT YOU'LL LEARN
 *   The classic patterns every team uses – knowing their NAMES lets you

```

```

*   communicate designs in seconds ("let's make it an Observer").
*
*   • Module • Singleton • Factory • Observer (Pub/Sub) • Strategy
*   • Decorator • Facade • Adapter • Command • Dependency Injection
*   • A note on MVC / MVVM
*
* A pattern is just a well-known shape of code, not a library. Use them when
* they fit – not everywhere.
* ===== */

// — 1. MODULE – group related code, hide the internals —————
// Expose a small public API; keep everything else private (via closures).
const Counter = (() => {
  let count = 0; // private – unreachable from outside
  return {
    increment() { count++; return count; },
    reset() { count = 0; },
    get value() { return count; },
  };
})();
Counter.increment();
Counter.increment();
console.log(Counter.value); // => 2
// (Modern ES modules – lesson 22 – are the file-level version of this.)

// — 2. SINGLETON – exactly ONE shared instance —————
// Useful for things there should only be one of: a config, a cache, a DB pool.
const Settings = (() => {
  let instance;
  function create() { return { theme: 'dark', loadedAt: 'once' }; }
  return {
    get() {
      if (!instance) instance = create(); // created lazily, only the first time
      return instance;
    },
  };
})();
console.log(Settings.get() === Settings.get()); // => true (same object every time)

// — 3. FACTORY – a function that builds objects for you —————
// Centralizes creation logic so callers don't use `new` or know the details.
function createUser(role) {
  const base = { role, createdAt: 'now' };
  const perks = {
    admin: { canDelete: true, canEdit: true },
    editor: { canDelete: false, canEdit: true },
    viewer: { canDelete: false, canEdit: false },
  };
};

```

```

    return { ...base, ...(perks[role] ?? perks.viewer) };
  }
  console.log(createUser('admin')); // => { role: 'admin', createdAt: 'now', canDelete: true,
  canEdit: true }
  console.log(createUser('viewer')); // => { ..., canDelete: false, canEdit: false }

```

// — 4. OBSERVER (Pub/Sub) – broadcast events to many listeners —————  
 // Subscribers register callbacks; the subject notifies them all on change.  
 // This is how event systems, state stores, and reactive UIs work under the hood.

```

class EventBus {
  #listeners = {};
  on(event, fn) {
    (this.#listeners[event] ??= []).push(fn);
    return () => this.off(event, fn); // return an unsubscribe function
  }
  off(event, fn) {
    this.#listeners[event] = (this.#listeners[event] ?? []).filter((f) => f !== fn);
  }
  emit(event, data) {
    (this.#listeners[event] ?? []).forEach((fn) => fn(data));
  }
}
const bus = new EventBus();
const unsub = bus.on('login', (user) => console.log('welcome', user));
bus.emit('login', 'Ana'); // => welcome Ana
unsub(); // stop listening
bus.emit('login', 'Bob'); // (nothing – unsubscribed)

```

// — 5. STRATEGY – swap an algorithm at runtime —————  
 // Instead of a big if/else, store interchangeable behaviors and pick one.

```

const shipping = {
  standard: (w) => w * 1.0,
  express: (w) => w * 2.5 + 5,
  pickup: () => 0,
};
function quote(method, weight) {
  const strategy = shipping[method] ?? shipping.standard;
  return strategy(weight);
}
console.log(quote('express', 2)); // => 10
console.log(quote('pickup', 99)); // => 0

```

// — 6. DECORATOR – wrap something to add behavior, same interface —————  
 // Take a function/object and return an enhanced version WITHOUT editing the  
 // original. (You've already met decorators: memoize and debounce wrap a fn.)

```

function withLogging(fn) {
  return (...args) => {

```

```

    console.log(`calling with ${JSON.stringify(args)}`);
    const result = fn(...args);
    console.log(`→ returned ${result}`);
    return result;
  };
}

const add = (a, b) => a + b;
const loudAdd = withLogging(add); // same (a, b) interface, extra behavior
loudAdd(2, 3); // logs the call + result, still returns 5


// — 7. FACADE – a simple front over a complicated subsystem —————
// Hide several messy steps behind ONE friendly method. (jQuery, axios, and most
// SDKs are facades over fetch/XHR/etc.)
const subsystemA = { connect: () => 'A ready' };
const subsystemB = { auth: () => 'B authed' };
const subsystemC = { load: () => 'C loaded' };
const app = {
  start() {
    // callers don't need to know the 3 steps
    return [subsystemA.connect(), subsystemB.auth(), subsystemC.load()].join(' | ');
  },
};
console.log('facade:', app.start()); // => A ready | B authed | C loaded


// — 8. ADAPTER – make an incompatible interface fit —————
// Wrap an object so it matches the shape YOUR code expects (e.g. integrating a
// third-party API whose field names differ from yours).
const oldApiUser = { user_name: 'ana', years_old: 30 }; // their shape
function userAdapter(raw) { // → your shape
  return { name: raw.user_name, age: raw.years_old };
}
console.log('adapter:', userAdapter(oldApiUser)); // => { name: 'ana', age: 30 }


// — 9. COMMAND – wrap an action as an object (enables undo/redo, queues) ———
// Each command knows how to do() and undo() itself. A history stack (lesson 48)
// then gives you undo for free – this is how editors implement Ctrl+Z.
class History {
  #done = [];
  run(command) { command.execute(); this.#done.push(command); }
  undo() { this.#done.pop()?.undo(); }
}
const doc = { text: '' };
const typeCmd = (s) => ({ execute: () => (doc.text += s), undo: () => (doc.text = doc.text.slice(0, -s.length)) });
const history = new History();
history.run(typeCmd('Hello'));
history.run(typeCmd(' World'));
console.log('command:', doc.text); // => Hello World

```

```

history.undo();
console.log('after undo:', doc.text); // => Hello

// — 10. DEPENDENCY INJECTION – pass collaborators IN, don't hard-code them —
// Instead of a function reaching for a global, give it what it needs as an
// argument. Result: easy to swap a real logger for a fake one in tests (lesson 38).
function createOrderService(logger, payments) { // dependencies injected here
  return {
    checkout(amount) {
      payments.charge(amount);
      logger.log(`charged ${amount}`);
    },
  };
}
const fakeLogger = { log: (m) => console.log('DI log:', m) };
const fakePayments = { charge: (n) => {} };
createOrderService(fakeLogger, fakePayments).checkout(50); // => DI log: charged 50

/* — 11. MVC / MVVM (architectural patterns) – the big picture —————
* Bigger than the above; they organize a WHOLE app, not one object:
* MVC – Model (data/rules) · View (UI) · Controller (handles input, updates
*       the model, picks the view). Classic server frameworks use this.
* MVVM – Model · View · ViewModel (exposes data the View binds to). The
*       ViewModel + data-binding is what React/Vue/Angular popularized.
* Takeaway: separate WHAT your app knows (model) from HOW it's shown (view) so
* each can change independently. Frameworks give you this structure for free.
*/

/* WHEN TO REACH FOR EACH -----
* Module .... hide internals, expose a clean API
* Singleton .. one shared resource (config/cache) – use sparingly
* Factory .... complex/conditional object creation
* Observer ... "when X happens, tell everyone who cares" (events, state)
* Strategy ... many interchangeable algorithms behind one call
* Decorator .. add behavior by wrapping, keep the same interface
* Facade ..... one simple method over a complex subsystem
* Adapter .... make a foreign interface fit your code
* Command .... actions as objects → undo/redo, queues, logging
* Dep. Inject  pass collaborators in → testable, swappable code
* ----- */

/* PRACTICE -----
* 1. Build a Logger module with private log history and a public add()/last().
* 2. Add a 'logout' event to the EventBus and subscribe two listeners.
* 3. Add a 'discount' strategy set (none/student/senior) chosen at runtime.
* 4. Write a withTiming(fn) DECORATOR that logs how long fn took (lesson 41).
* 5. Add a 'delete line' command to the History demo and test undo on it.

```

```
* 6. Refactor createOrderService to inject a THIRD dependency (an emailer).
* ----- */
```

## 41 · PERFORMANCE — make it fast (and know what "fast" means)

---

```
/* =====
* 41 · PERFORMANCE — make it fast (and know what "fast" means)
* =====
* Run: node lessons/41-performance.js
*
* WHAT YOU'LL LEARN
* • Big-O intuition — why algorithm choice beats micro-tweaks
* • Memoization & caching
* • Debounce/throttle recap (lesson 29)
* • Browser rendering: reflow/repaint, lazy loading, code splitting
* • Core Web Vitals — how performance is actually measured in 2026
* • Measuring & profiling: performance marks, PerformanceObserver
* • List virtualization (windowing) for huge lists
*
* Rule: measure before optimizing. "Premature optimization is the root of all
* evil" — make it correct, measure, then fix the real bottleneck.
* ===== */

// — 1. Big-O intuition — how cost grows with input size —————
// O(1)      constant ..... array[i], map.get(key)
// O(log n)   logarithmic ... binary search
// O(n)       linear ..... a single loop
// O(n log n) ..... good sorts
// O(n²)      quadratic .... nested loop over the same data — avoid at scale
//
// Picking the right data structure changes the Big-O. "Is X in this list?":
const big = Array.from({ length: 100_000 }, (_, i) => i);
const asSet = new Set(big);
console.log(big.includes(99_999)); // O(n) — scans up to 100k items
console.log(asSet.has(99_999));    // O(1) — instant lookup ✅

// — 2. Memoization — cache expensive results by their inputs —————
function memoize(fn) {
  const cache = new Map();
  return (...args) => {
    const key = JSON.stringify(args);
    if (cache.has(key)) return cache.get(key); // cache HIT — skip the work
    const result = fn(...args);
    cache.set(key, result);
    return result;
  };
}
```

```
let calls = 0;
const slowSquare = (n) => { calls++; return n * n; };
const fastSquare = memoize(slowSquare);
fastSquare(9);
fastSquare(9);           // served from cache – fn body NOT re-run
fastSquare(9);
console.log(fastSquare(9), 'computed', calls, 'time(s)'); // => 81 computed 1 time(s)
```

```
// — 3. Debounce & throttle (limit how OFTEN code runs) – see lesson 29 ————
// debounce: wait until activity STOPS (search-as-you-type, resize).
// throttle: run at most once per interval (scroll, mousemove).
// Both prevent doing heavy work on every single event.
```

```
// — 4. Browser rendering performance (concept) —————
// • REFLOW (layout): the browser recomputes geometry – expensive. Triggered by
//   changing size/position or reading layout props (offsetHeight) mid-mutation.
// • REPAINT: redrawing pixels (e.g. color change) – cheaper than reflow.
// Tips: batch DOM writes; build nodes off-DOM then insert once (lesson 23);
//       animate `transform`/`opacity` (GPU) instead of top/left/width.
```

```
// — 5. Loading performance (concept) —————
// • LAZY LOADING: load things only when needed.
//   <img loading="lazy">           // images
//   const mod = await import('./chart.js'); // dynamic import → CODE SPLITTING
//   Code splitting ships a smaller initial bundle; the rest loads on demand.
// • Defer non-critical JS: <script defer src="app.js"></script>
// • Compress & cache: gzip/brotli, HTTP caching headers, a CDN.
```

```
// — 6. Core Web Vitals – the metrics that matter (2026) —————
//   LCP (Largest Contentful Paint) – load speed → aim < 2.5s
//   INP (Interaction to Next Paint) – responsiveness → aim < 200ms
//   CLS (Cumulative Layout Shift) – visual stability → aim < 0.1
// Measure with Lighthouse, the Performance panel, or the web-vitals library.
```

```
// — 7. Measuring in code —————
const t0 = performance.now();
let sum = 0;
for (let i = 0; i < 1_000_000; i++) sum += i;
const ms = performance.now() - t0;
console.log(`loop took ${ms.toFixed(1)}ms`); // varies by machine
// (console.time/timeEnd from lesson 35 do the same thing more conveniently.)
```

```
// — 8. Profiling with marks & measures (the Performance API) —————
```



```

// performance.now() is one stopwatch. The Performance API lets you drop named
// MARKS and MEASURE the span between them – those show up in browser DevTools'
// Performance panel too, so you can see them on a real timeline.
performance.mark('work-start');
let total = 0;
for (let i = 0; i < 500_000; i++) total += i;
performance.mark('work-end');
const measure = performance.measure('heavy-work', 'work-start', 'work-end');
console.log(`measured "${measure.name}": ${measure.duration.toFixed(1)}ms`);

// A PerformanceObserver reacts to entries as they're recorded – this is how the
// `web-vitals` library and monitoring tools collect LCP/INP/CLS in real users'
// browsers (then report them, lesson 47):
const obs = new PerformanceObserver((list) => {
  for (const entry of list.getEntries()) {
    console.log(`observed ${entry.entryType} "${entry.name}": ${entry.duration.toFixed(1)}ms`);
  }
});
obs.observe({ entryTypes: ['measure'] });
performance.measure('observed-span', 'work-start', 'work-end');
// In the browser you'd also observe: 'largest-contentful-paint', 'event'
// (for INP), 'layout-shift' (for CLS), 'longtask', 'resource', 'navigation'.
setTimeout(() => obs.disconnect(), 50); // stop observing (avoid leaks, lesson 36)

/* — 9. List virtualization (windowing) – render only what's visible —————
 * Rendering 10,000 rows = 10,000 DOM nodes = a frozen page. VIRTUALIZATION
 * renders only the ~20 rows currently in view (a "window") and swaps their
 * contents as you scroll. The list FEELS infinite but the DOM stays tiny.
 *
 *   const ROW_H = 30, visible = Math.ceil(container.clientHeight / ROW_H);
 *   container.onscroll = () => {
 *     const start = Math.floor(container.scrollTop / ROW_H);
 *     render(items.slice(start, start + visible + 2)); // only the window
 *   };
 *
 * Libraries: TanStack Virtual, react-window. This is THE fix for slow long
 * lists/tables/feeds – it turns O(n) DOM nodes into O(window).
 */

/* PRACTICE -----
 * 1. Memoize a slow fibonacci(n) and compare call counts with/without it.
 * 2. Rewrite an O(n²) "find duplicates" using a Set to get O(n).
 * 3. Name which Web Vital each fixes: slow hero image, janky button, jumpy layout.
 * 4. Wrap two code blocks in performance.mark/measure and log which is slower.
 * 5. In words: why does virtualization keep a 100k-row list fast? (DOM size.)
 * ----- */

```

## 42 · POLYFILLS — implement the built-ins yourself

---

```
/* =====
 * 42 · POLYFILLS — implement the built-ins yourself
 * =====
 * Run:  node lessons/42-polyfills.js
 *
 * WHAT YOU'LL LEARN
 *   Re-building the methods you use every day. This is the single best way to
 *   *truly* understand them — and it's the most common JS interview exercise.
 *
 *   We'll re-implement: map, filter, reduce, call/apply/bind, debounce,
 *   and a minimal Promise.
 *
 * (A real "polyfill" also checks `if (!Array.prototype.x)` before defining, so
 * it only fills the gap in old engines. We focus on the implementations.)
 * ===== */

// — 1. Array.prototype.map —————
function myMap(arr, fn) {
  const out = [];
  for (let i = 0; i < arr.length; i++) out.push(fn(arr[i], i, arr));
  return out;
}
console.log(myMap([1, 2, 3], (n) => n * 2)); // => [ 2, 4, 6 ]

// — 2. Array.prototype.filter —————
function myFilter(arr, predicate) {
  const out = [];
  for (let i = 0; i < arr.length; i++) {
    if (predicate(arr[i], i, arr)) out.push(arr[i]);
  }
  return out;
}
console.log(myFilter([1, 2, 3, 4], (n) => n % 2 === 0)); // => [ 2, 4 ]

// — 3. Array.prototype.reduce —————
function myReduce(arr, reducer, initial) {
  let acc = initial;
  let start = 0;
  if (acc === undefined) { acc = arr[0]; start = 1; } // no seed → use first item
  for (let i = start; i < arr.length; i++) acc = reducer(acc, arr[i], i, arr);
  return acc;
}
console.log(myReduce([1, 2, 3, 4], (a, b) => a + b, 0)); // => 10
```

```
// — 4. Function.prototype.call / apply / bind —————
// The trick: put the function on the target object so `this` becomes that object.
function myCall(fn, context, ...args) {
  const key = Symbol('fn');
  context = context ?? globalThis;
  context[key] = fn;
  const result = context[key](...args);
  delete context[key];
  return result;
}
function myBind(fn, context, ...preset) {
  return (...later) => myCall(fn, context, ...preset, ...later);
}
const person = { name: 'Ana' };
function greet(greeting) { return `${greeting}, ${this.name}`; }
console.log(myCall(greet, person, 'Hi')); // => Hi, Ana
const sayHello = myBind(greet, person, 'Hello');
console.log(sayHello()); // => Hello, Ana
```

```
// — 5. debounce – delay until calls stop (lesson 29, built from scratch) ———
function debounce(fn, delay) {
  let timer;
  return (...args) => {
    clearTimeout(timer); // cancel the previous pending call
    timer = setTimeout(() => fn(...args), delay);
  };
}
const log = debounce((msg) => console.log('debounced:', msg), 50);
log('a'); log('b'); log('c'); // only 'c' fires, ~50ms later
```

```
// — 6. A minimal Promise —————
// Simplified, but shows the core: state machine + queued callbacks + then().
class MyPromise {
  constructor(executor) {
    this.state = 'pending';
    this.value = undefined;
    this.callbacks = [];
    const resolve = (value) => this.#settle('fulfilled', value);
    const reject = (reason) => this.#settle('rejected', reason);
    try { executor(resolve, reject); } catch (e) { reject(e); }
  }
  #settle(state, value) {
    if (this.state !== 'pending') return; // settle only once
    this.state = state;
    this.value = value;
    this.callbacks.forEach((cb) => cb()); // flush anyone waiting
  }
  then(onFulfilled, onRejected) {
```

```

return new MyPromise((resolve, reject) => {
  const run = () => {
    try {
      if (this.state === 'fulfilled') resolve(onFulfilled ? onFulfilled(this.value) :
this.value);
      else reject(onRejected ? onRejected(this.value) : this.value);
    } catch (e) { reject(e); }
  };
  // if already settled, run on the microtask queue; else queue until settle
  this.state === 'pending' ? this.callbacks.push(run) : queueMicrotask(run);
});
}
}
new MyPromise((resolve) => setTimeout(() => resolve(21), 10))
  .then((n) => n * 2)
  .then((n) => console.log('MyPromise result:', n)); // => 42

/* PRACTICE -----
* 1. Implement myForEach(arr, fn) (like map but returns nothing).
* 2. Implement myFlat(arr, depth) using recursion (lesson 09).
* 3. Add a .catch(fn) method to MyPromise (hint: then(undefined, fn)).
* ----- */

```

## 43 · TYPESCRIPT ON-RAMP — JavaScript with types

---

```

/* =====
* 43 · TYPESCRIPT ON-RAMP – JavaScript with types
* =====
* Run:  node lessons/43-typescript.js   (the runnable parts are plain JS)
*
* WHAT YOU'LL LEARN
*   What TypeScript adds on top of the JS you already know, why teams use it,
*   and the core syntax – so your first .ts file feels familiar.
*
* TypeScript = JavaScript + a TYPE LAYER checked BEFORE you run. The types are
* erased at build time; the browser/Node runs plain JS. It catches a whole
* class of bugs ("undefined is not a function") at your desk, not in production.
*
* Run real TS:  npm i -D typescript && npx tsc file.ts   (or use ts-node / Bun / Deno)
* Below, the TS is shown in comments; the runnable code is the JS equivalent.
* ===== */

console.log('=== TypeScript concepts (TS in comments, JS runs) ===');

/* — 1. Basic types -----
*   let name: string = 'Ana';
*   let age: number = 30;

```

```

* let active: boolean = true;
* let tags: string[] = ['js', 'ts'];
* let pair: [string, number] = ['x', 1]; // tuple
* Type INFERENCE means you rarely annotate obvious things:
* let count = 0; // inferred as number – reassigning to 'hi' errors
*/

```

## /\* — 2. Function signatures —————

```

* function add(a: number, b: number): number { return a + b; }
* const greet = (name: string): string => `Hi ${name}`;
* add(2, 'x'); // ❌ compile error: 'x' is not a number ← caught early!
*/

```

```

function add(a, b) { return a + b; } // the JS this compiles to
console.log('add(2,3):', add(2, 3)); // => 5

```

## /\* — 3. Interfaces & type aliases – shapes of objects —————

```

* interface User { id: number; name: string; email?: string } // ? = optional
* type ID = number | string; // union type
* function format(u: User): string { return `${u.id}:${u.name}`; }
*/

```

```

const user = { id: 1, name: 'Ana' }; // a value matching the User shape
console.log('user:', `${user.id}:${user.name}`); // => 1:Ana

```

## /\* — 4. Union types & narrowing —————

```

* function len(x: string | string[]): number {
*   if (typeof x === 'string') return x.length; // TS NARROWS x to string here
*   return x.length; // here x is string[]
* }

```

```

* Narrowing = TS uses your runtime checks (typeof, in, Array.isArray) to know
* the precise type inside each branch. */

```

```

function len(x) {
  return typeof x === 'string' ? x.length : x.length;
}
console.log('len:', len('hello'), len(['a', 'b'])); // => 5 2

```

## /\* — 5. Generics – reusable, type-safe containers/functions —————

```

* function first<T>(arr: T[]): T | undefined { return arr[0]; }
* first<number>([1, 2, 3]); // returns number
* first(['a', 'b']); // T inferred as string
* Think "a type parameter" – like a function argument, but for types. */

```

```

function first(arr) { return arr[0]; }
console.log('first:', first([10, 20]), first(['x'])); // => 10 x

```

## /\* — 6. Utility & helpers you'll meet —————

```

* Partial<User> // all fields optional
* Readonly<User> // all fields immutable
* Pick<User, 'id'> // just some fields
* enum Role { Admin, Editor, Viewer }
* as const // freeze a literal's type
* unknown vs any // prefer `unknown` (forces you to check before use)

```

```

*/

/* — 7. JSDoc – types WITHOUT TypeScript —————
 * You can get editor type-checking in plain .js files using JSDoc comments –
 * a nice on-ramp before adopting .ts. This is real and checkable: */
/** @param {number} n @returns {number} */
function double(n) { return n * 2; }
console.log('double:', double(21)); // => 42

/* WHY TEAMS USE IT -----
 * • Bugs caught at compile time, not runtime.
 * • Autocomplete & refactoring that actually understands your data.
 * • Types document the code and survive refactors.
 * Cost: a build step and a learning curve. For anything beyond a small
 * script, most 2026 production codebases are TypeScript.
 * ----- */

/* PRACTICE -----
 * 1. Write (in a comment) a `Product` interface: id:number, name:string,
 *    price:number, inStock?:boolean.
 * 2. Annotate a `total(items: Product[]): number` signature.
 * 3. Add JSDoc types to a real isEven(n) function in this file.
 * ----- */

```

## 44 · TOOLING & BUILD SYSTEMS — the modern JS workflow

---

```

/* =====
 * 44 · TOOLING & BUILD SYSTEMS – the modern JS workflow
 * =====
 * Run:  node lessons/44-tooling.js  (a guided tour; prints the key concepts)
 *
 * WHAT YOU'LL LEARN
 * The tools that turn your source code into a shippable app, and what each
 * one is FOR. You don't memorize commands – you learn the moving parts.
 *
 * npm/package.json · semver · bundlers · transpilers · linters · git · CI/CD
 * ===== */

console.log('=== The modern JavaScript toolchain ===\n');

// — 1. npm & package.json – the project manifest —————
// `npm init -y` creates package.json. It records dependencies and scripts.
const packageJson = {
  name: 'my-app',
  version: '1.0.0',
  type: 'module', // use ES modules (lesson 22)
  scripts: {

```

```

    dev: 'vite', // run with: npm run dev
    build: 'vite build',
    test: 'vitest',
    lint: 'eslint .',
  },
  dependencies: { /* runtime deps: npm install <pkg> */ },
  devDependencies: { /* build/test: npm install -D <pkg> */ },
};
console.log('package.json scripts → run with `npm run <name>`');
console.log(Object.keys(packageJson.scripts).join(', ')); // => dev, build, test, lint
// • node_modules/ – installed packages (never commit it; it's huge).
// • package-lock.json – pins EXACT versions so installs are reproducible.

// — 2. Semantic versioning (semver): MAJOR.MINOR.PATCH —————
// ^1.2.3 → allow 1.x.x (minor+patch updates) ← npm default
// ~1.2.3 → allow 1.2.x (patch only)
// 1.2.3 → exactly this version
console.log('semver: MAJOR(breaking).MINOR(features).PATCH(fixes)');

// — 3. Bundlers – many files → optimized output —————
// A bundler resolves your imports, bundles them, tree-shakes dead code, and
// minifies for production. Modern choices:
// • Vite – fast dev server + build (most common in 2026)
// • esbuild / Rollup – the engines under many tools
// • Webpack – older, powerful, lots of legacy projects
console.log('bundler job: resolve imports → tree-shake → minify → split');

// — 4. Transpiling – new syntax → widely-supported JS —————
// • Babel / esbuild convert modern JS (and JSX/TS) into code older browsers run.
// • A "browserslist" config decides how far back to compile.
// • This is why you can write ES2025 today and still support older targets.

// — 5. Linting & formatting – consistency + catching mistakes —————
// • ESLint – flags bugs & bad patterns (unused vars, == misuse). `npx eslint .`
// • Prettier – auto-formats code so the team never argues about style.
// Run them on save and in CI so problems never reach main.

// — 6. Git – version control basics (every project) —————
const gitFlow = ['git init', 'git add .', 'git commit -m "msg"', 'git push',
  'git branch feature', 'git merge', 'open a Pull Request → review → merge'];
console.log(`\ngit flow:`, gitFlow.join(' → '));
// • .gitignore keeps node_modules/, .env, and build output out of the repo.

// — 7. CI/CD – automate test & deploy —————
// Continuous Integration: on every push, a server (GitHub Actions, etc.) runs
// install → lint → test → build. Red build = don't merge.
// Continuous Deployment: green build auto-ships to staging/production.
console.log('CI on push: install → lint → test → build → deploy');

// — 8. Source maps – debug the original code, not the bundle —————

```

```
// Build tools emit .map files linking minified output back to your source, so
// DevTools shows YOUR code in the debugger even in production builds.

/* THE BIG PICTURE -----
 *   write code → lint/format → transpile → bundle → test → CI → deploy
 *   You rarely wire this by hand: `npm create vite@latest` scaffolds most of it.
 * ----- */

/* PRACTICE -----
 *   1. Run `npm init -y` in a folder and add a "start": "node index.js" script.
 *   2. Explain the difference between dependencies and devDependencies.
 *   3. Scaffold a real app: `npm create vite@latest` and run `npm run dev`.
 * ----- */
```

## 45 • NODE.JS — JavaScript on the server

---

```
/* =====
 * 45 • NODE.JS – JavaScript on the server
 * =====
 * Run:  node lessons/45-nodejs.js
 *
 * WHAT YOU'LL LEARN
 *   The Node-specific APIs that the browser doesn't have: process & env,
 *   the file system, paths, events, streams, and a real HTTP server.
 *
 * Node is JavaScript outside the browser – it powers backends, CLIs, and
 * tooling. Core modules live under the `node:` prefix.
 * ===== */

import process from 'node:process';
import path from 'node:path';
import os from 'node:os';
import fs from 'node:fs/promises';
import { EventEmitter } from 'node:events';
import http from 'node:http';

// — 1. process – info about the running program —————
console.log('node version  :', process.version);           // => v24.x
console.log('platform     :', process.platform);           // => linux / darwin / win32
console.log('cwd          :', process.cwd().split('/').pop()); // current folder name
// CLI args:  node script.js foo bar → process.argv = [node, script, 'foo', 'bar']
console.log('args         :', process.argv.slice(2));       // => [] (none here)
// Environment variables – where secrets/config live (NEVER hard-code secrets):
//   run:  API_KEY=abc node lessons/45-nodejs.js
console.log('API_KEY      :', process.env.API_KEY ?? '(not set – see lesson 39)');
```



```
// — 2. path – build file paths the cross-platform way —————  
console.log('join      :', path.join('src', 'utils', 'math.js')); // src/utils/math.js  
console.log('extname   :', path.extname('report.pdf'));           // => .pdf  
console.log('basename :', path.basename('/a/b/notes.txt'));       // => notes.txt
```

```
// — 3. fs – read & write files (async/await) —————  
const tmpFile = path.join(os.tmpdir(), 'js-learn-demo.txt');  
await fs.writeFile(tmpFile, 'hello from Node\n'); // create/overwrite  
const contents = await fs.readFile(tmpFile, 'utf8'); // read back as text  
console.log('file says :', contents.trim()); // => hello from Node  
await fs.appendFile(tmpFile, 'second line\n'); // append  
await fs.unlink(tmpFile); // delete (cleanup)  
console.log('temp file cleaned up ✅');
```

```
// — 4. EventEmitter – Node's built-in observer (lesson 40 pattern) —————  
class Order extends EventEmitter {}  
const order = new Order();  
order.on('paid', (amount) => console.log(`order paid: ${amount}`)); // listener  
order.emit('paid', 99); // => order paid: $99  
// Many Node objects (streams, servers, sockets) ARE EventEmitters.
```

```
// — 5. Streams – process data piece by piece, not all at once —————  
// Streams let you handle huge files/responses with low memory by working on  
// CHUNKS. (Conceptual – fs.createReadStream(path).on('data', chunk => ...).)  
// Examples: reading a 2GB log line-by-line, or piping a download to disk.
```

```
// — 6. A real HTTP server —————  
// http.createServer gives you a callback per request. We start it on a random  
// free port (0), make one request to ourselves, print the reply, then close.  
const server = http.createServer((req, res) => {  
  res.writeHead(200, { 'Content-Type': 'application/json' });  
  res.end(JSON.stringify({ ok: true, path: req.url }));  
});
```

```
await new Promise((resolve) => server.listen(0, resolve)); // 0 = pick a free port  
const { port } = server.address();  
const reply = await fetch(`http://localhost:${port}/hello`).then((r) => r.json());  
console.log('server replied:', reply); // => { ok: true, path: '/hello' }  
server.close(); // stop listening so the program can exit  
// In real apps you'd use a framework (Express/Fastify/Hono) on top of this.
```

```
/* BROWSER vs NODE -----  
* Browser has: window, document, DOM, localStorage.  
* Node has:    process, fs, path, http, os, Buffer, EventEmitter.
```

```

*   Shared:      JS syntax, fetch, Promises, console, timers, JSON.
*   ----- */

/* PRACTICE -----
*   1. Read process.argv and greet a name passed on the command line.
*   2. Write an object to a JSON file with fs.writeFile, then read it back.
*   3. Add a '/time' route to the server that returns the current time.
*   ----- */

```

## 46 · ADVANCED BROWSER APIs (browser)

---

```

/* =====
* 46 · ADVANCED BROWSER APIs (browser)
* =====
* HOW TO RUN: these are BROWSER features. Open index.html and try them in the
* console/devtools – they don't exist in Node. This file runs harmlessly in
* Node (it just prints a notice); the real examples are in the comments.
*
* WHAT YOU'LL LEARN
*   The powerful platform APIs beyond DOM/events:
*   Web Workers · Service Workers/PWA · IndexedDB · Observers ·
*   Web Components · requestAnimationFrame · device APIs
* ===== */

// Guard so the file is harmless in Node:
if (typeof window === 'undefined') {
  console.log('⚠ Lesson 46 is BROWSER-only. Open index.html and read the code in devtools.');
```

```

}

/* — 1. Web Workers – run heavy JS off the main thread —————
* The main thread renders the UI; long loops there FREEZE the page. A Worker
* runs in parallel and talks via messages – no shared variables.
*
* // main.js
* const worker = new Worker('worker.js');
* worker.postMessage({ n: 40 });
* worker.onmessage = (e) => console.log('result', e.data);
*
* // worker.js
* onmessage = (e) => postMessage(fib(e.data.n)); // CPU work, UI stays smooth
*/

/* — 2. Service Workers & PWAs – offline + installable —————
* A Service Worker is a proxy between your app and the network. It can CACHE
* assets so the app works offline and loads instantly. Plus a manifest.json =
* an installable Progressive Web App.
*
* navigator.serviceWorker.register('/sw.js');
```

```

* // sw.js
* self.addEventListener('fetch', (e) =>
*   e.respondWith(caches.match(e.request).then((c) => c ?? fetch(e.request))));
*/

/* — 3. IndexedDB – a real database in the browser —————
* localStorage (lesson 33) holds small strings synchronously. IndexedDB stores
* large, structured, queryable data (objects, blobs) asynchronously.
*
* const req = indexedDB.open('myDB', 1);
* req.onupgradeneeded = (e) => e.target.result.createObjectStore('users', { keyPath: 'id' });
* req.onsuccess = (e) => { const db = e.target.result; ... transactions ... };
* Libraries like `idb` make its callback API far nicer.
*/

/* — 4. Observer APIs – react to changes efficiently —————
* • IntersectionObserver – "is this element on screen?" → lazy-load images,
*   infinite scroll, fade-in-on-scroll (no scroll-event spam).
*   new IntersectionObserver(es => es.forEach(e => e.isIntersecting && load(e.target)))
*     .observe(img);
* • MutationObserver – watch the DOM for added/removed/changed nodes.
* • ResizeObserver – react when an element's SIZE changes.
*/

/* — 5. Web Components – reusable custom elements (framework-free) —————
* Define your own HTML tags with encapsulated markup/style via Shadow DOM.
*
* class MyCounter extends HTMLElement {
*   connectedCallback() {
*     const root = this.attachShadow({ mode: 'open' }); // isolated DOM/CSS
*     root.innerHTML = `<button>+</button><span>0</span>`;
*   }
* }
*
* customElements.define('my-counter', MyCounter); // <my-counter></my-counter>
*/

/* — 6. requestAnimationFrame – smooth 60fps animation —————
* Don't animate with setInterval. rAF runs your callback right before the next
* paint, syncing to the display and pausing in background tabs.
*
* function tick(t) { move(); requestAnimationFrame(tick); }
* requestAnimationFrame(tick);
*/

/* — 7. Device & misc APIs (ask permission; HTTPS only) —————
* navigator.geolocation.getCurrentPosition(pos => ...); // location
* await navigator.clipboard.writeText('copied!'); // clipboard
* new Notification('Hi!'); // notifications
* matchMedia('(prefers-color-scheme: dark)').matches; // dark mode
* navigator.onLine; // online status

```

```

*/

/* PRACTICE (in the browser) -----
* 1. Use IntersectionObserver to log when a bottom element scrolls into view.
* 2. Define a <hello-world> Web Component that renders your name.
* 3. Animate a box across the screen with requestAnimationFrame.
* ----- */

```

## 47 · REAL-TIME NETWORKING & PRODUCTION OPS

```

/* =====
* 47 · REAL-TIME NETWORKING & PRODUCTION OPS
* =====
* Run:  node lessons/47-realtime-and-production.js
*
* WHAT YOU'LL LEARN
*   • Pushing data in real time: WebSockets vs Server-Sent Events vs polling
*   • Running software in production: logging, monitoring, config, flags
*   • Accessibility (ally) & internationalization (i18n) basics
*
* "It works on my machine" isn't done. Production code must be observable,
* configurable, resilient, and usable by everyone.
* ===== */

// — 1. Real-time: three ways to get fresh data —————
// • POLLING ..... ask repeatedly on a timer. Simple, but wasteful/laggy.
//   setInterval(() => fetch('/api/messages'), 5000);
// • SSE (EventSource) ... server → client, ONE-WAY stream over plain HTTP.
//   const es = new EventSource('/stream');
//   es.onmessage = (e) => console.log('update:', e.data); // great for feeds
// • WEBSOCKETS ..... full DUPLEX (both directions), low latency. Chat, games,
//   live cursors. A persistent connection, not request/response.
//   const ws = new WebSocket('wss://example.com');
//   ws.onopen    = () => ws.send('hello');
//   ws.onmessage = (e) => console.log('got:', e.data);
//   On the server (Node) you'd use the `ws` library or Socket.IO.
console.log('real-time: poll (simple) < SSE (one-way) < WebSocket (two-way)');

// Quick decision guide:
function chooseTransport(need) {
  if (need === 'two-way') return 'WebSocket';
  if (need === 'server-push') return 'SSE';
  return 'polling';
}

console.log(chooseTransport('two-way')); // => WebSocket
console.log(chooseTransport('server-push')); // => SSE

```

```
// — 2. Logging – structured, leveled —————
// Don't ship console.log spam. Use levels and structured (JSON) logs so they're
// searchable in a log platform. (Libraries: pino, winston.)
const LEVELS = { error: 0, warn: 1, info: 2, debug: 3 };
function log(level, msg, meta = {}) {
  if (LEVELS[level] <= LEVELS.info) { // current threshold = info
    console.log(JSON.stringify({ level, msg, ...meta }));
  }
}
log('info', 'user signed up', { userId: 42 });
// => {"level":"info","msg":"user signed up","userId":42}
log('debug', 'noisy detail'); // suppressed (below threshold)

// — 3. Monitoring & error tracking —————
// • Catch and REPORT errors from real users (Sentry, etc.) instead of losing them:
//   window.addEventListener('error', (e) => report(e));
//   window.addEventListener('unhandledrejection', (e) => report(e.reason));
//   process.on('uncaughtException', report); // Node
// • Track metrics (latency, error rate, throughput) and set alerts.
//   "If you can't see it, you can't fix it."

// — 4. Configuration & secrets —————
// • Config comes from ENVIRONMENT VARIABLES, not hard-coded values (lesson 45).
// • Secrets (API keys, DB passwords) live in a secret manager / .env that is
//   GIT-IGNORED – never committed (lesson 39).
const config = {
  apiUrl: process.env.API_URL ?? 'http://localhost:3000',
  logLevel: process.env.LOG_LEVEL ?? 'info',
};
console.log('config:', config);

// — 5. Feature flags & graceful failure —————
// • Feature flags toggle features without redeploying (gradual rollout, kill-switch).
const flags = { newCheckout: false };
console.log(flags.newCheckout ? 'new checkout' : 'old checkout'); // => old checkout
// • Resilience: retries with backoff (lesson 29), timeouts, circuit breakers,
//   and fallback UI so one failed call doesn't crash the whole app.

// — 6. Accessibility (ally) – usable by everyone —————
// • Semantic HTML first (<button>, <nav>, <label>) – it's accessible by default.
// • Provide alt text, keyboard navigation, visible focus, sufficient contrast.
// • Use ARIA only to fill gaps semantic HTML can't (aria-label, roles, live regions).
// • Test with a keyboard only and a screen reader. It's also often a legal requirement.
```

```
// — 7. Internationalization (i18n) —————
// • Don't concatenate translated strings; use a message catalog per locale.
// • Format with Intl (lessons 06/32) – numbers, currency, dates, plurals:
const price = new Intl.NumberFormat('de-DE', { style: 'currency', currency: 'EUR' });
console.log('localized price:', price.format(1234.5)); // => 1.234,50 €

/* THE PRODUCTION CHECKLIST -----
 * tested · linted · typed · secured · observable (logs+metrics+alerts) ·
 * configurable (env) · resilient (retries/timeouts) · accessible · documented
 * ----- */

/* PRACTICE -----
 * 1. Add a 'warn' log call and confirm it prints but 'debug' does not.
 * 2. Extend chooseTransport for a 'one-off request' → 'fetch'.
 * 3. Read LOG_LEVEL from env and make the log() threshold respect it.
 * ----- */
```

## 48 · DATA STRUCTURES — the containers behind every program

```
/* =====
 * 48 · DATA STRUCTURES – the containers behind every program
 * =====
 * Run: node lessons/48-data-structures.js
 *
 * WHAT YOU'LL LEARN
 * The classic computer-science data structures, built in plain JS:
 * • Stack (LIFO) and Queue (FIFO)
 * • Linked list
 * • Hash map (how Map/objects work under the hood)
 * • Binary tree + binary search tree
 * • Graph (adjacency list)
 *
 * WHY THIS MATTERS
 * Arrays and objects (lessons 12–16) cover 90% of daily work. But knowing
 * these structures – when each is fast, and WHY – is the difference between
 * O(n) and O(1) code, and it's the backbone of every coding interview.
 * Pair this with lesson 49 (algorithms) and lesson 41 (Big-O).
 * ===== */

// — 1. STACK (LIFO – Last In, First Out) —————
// Think a stack of plates: you add and remove from the TOP. Used for: undo,
// browser back, call stack, balancing brackets, depth-first traversal.
class Stack {
  #items = [];
  push(value) { this.#items.push(value); return this; } // add to top
  pop() { return this.#items.pop(); } // remove from top
  peek() { return this.#items.at(-1); } // look, don't remove
}
```

```

    get size() { return this.#items.length; }
    get isEmpty() { return this.#items.length === 0; }
  }
  const undo = new Stack();
  undo.push('type A').push('type B').push('type C');
  console.log('stack peek:', undo.peek()); // => type C
  console.log('stack pop:', undo.pop()); // => type C (last in, first out)
  console.log('stack pop:', undo.pop()); // => type B

```

// Classic use: are the brackets balanced? "([])"  "([)]" 

```

function balanced(str) {
  const pairs = { ')': '(', ']': '[', '}': '{' };
  const stack = [];
  for (const ch of str) {
    if (ch === '(' || ch === '[' || ch === '{') stack.push(ch);
    else if (ch in pairs) {
      if (stack.pop() !== pairs[ch]) return false; // mismatch
    }
  }
  return stack.length === 0; // nothing left unclosed
}
console.log('balanced ([]):', balanced('([])')); // => true
console.log('balanced ([):', balanced('([)]')); // => false

```

// — 2. QUEUE (FIFO – First In, First Out) —————

// Think a line at a shop: first to arrive is first served. Used for: task queues, print spoolers, breadth-first traversal, rate limiting.

// NOTE: array.shift() is O(n) (it re-indexes). For big queues use two stacks or // a ring buffer. We use shift() here for clarity.

```

class Queue {
  #items = [];
  enqueue(value) { this.#items.push(value); return this; } // add to back
  dequeue() { return this.#items.shift(); } // remove from front
  peek() { return this.#items[0]; }
  get size() { return this.#items.length; }
}
const line = new Queue();
line.enqueue('Ana').enqueue('Bob').enqueue('Cy');
console.log('queue dequeue:', line.dequeue()); // => Ana (first in, first out)
console.log('queue peek :', line.peek()); // => Bob

```

// — 3. LINKED LIST —————

// A chain of nodes; each node holds a value + a pointer to the NEXT node.

// Strength: O(1) insert/remove at the front (no re-indexing like arrays).

// Weakness: O(n) to reach the middle (no random access). Great teaching tool.

```

class LinkedList {
  #head = null;
  #size = 0;

```

```

prepend(value) {                                // add to front – O(1)
  this.#head = { value, next: this.#head };
  this.#size++;
  return this;
}
append(value) {                                  // add to end – O(n)
  const node = { value, next: null };
  if (!this.#head) { this.#head = node; }
  else {
    let cur = this.#head;
    while (cur.next) cur = cur.next;    // walk to the last node
    cur.next = node;
  }
  this.#size++;
  return this;
}
toArray() {                                      // walk the chain into an array
  const out = [];
  for (let cur = this.#head; cur; cur = cur.next) out.push(cur.value);
  return out;
}
get size() { return this.#size; }
}

const list = new LinkedList();
list.append('b').append('c').prepend('a');
console.log('linked list:', list.toArray()); // => [ 'a', 'b', 'c' ]

```

```

// — 4. HASH MAP (how Map / objects work under the hood) —————
// A hash map turns a KEY into an array index via a hash function, giving ~O(1)
// lookup. Collisions (two keys → same bucket) are handled by "chaining": each
// bucket holds a list of [key, value] pairs. (In real code, just use `Map`.)
class HashMap {
  #buckets = Array.from({ length: 8 }, () => []);
  #hash(key) {                                // turn a string into a bucket index
    let h = 0;
    for (const ch of String(key)) h = (h * 31 + ch.charCodeAt(0)) % this.#buckets.length;
    return h;
  }
  set(key, value) {
    const bucket = this.#buckets[this.#hash(key)];
    const pair = bucket.find((p) => p[0] === key);
    if (pair) pair[1] = value;                // update existing
    else bucket.push([key, value]);           // or add new (collision → same bucket)
    return this;
  }
  get(key) {
    const bucket = this.#buckets[this.#hash(key)];
    return bucket.find((p) => p[0] === key)?.[1];
  }
}

```



```

}
const ages = new HashMap();
ages.set('Ana', 30).set('Bob', 25);
console.log('hashmap get Ana:', ages.get('Ana')); // => 30
console.log('hashmap get Zz:', ages.get('Zz')); // => undefined

```

## // — 5. BINARY SEARCH TREE (BST) —————

```

// A tree where every node has up to two children, and for each node:
//   left subtree < node < right subtree.
// That ordering makes search/insert  $O(\log n)$  on a balanced tree – like binary
// search (lesson 49) but in a structure you can keep adding to.

```

```

class BST {
  #root = null;
  insert(value) {
    this.#root = this.#insert(this.#root, value);
    return this;
  }
  #insert(node, value) {
    if (!node) return { value, left: null, right: null };
    if (value < node.value) node.left = this.#insert(node.left, value);
    else if (value > node.value) node.right = this.#insert(node.right, value);
    return node; // duplicates ignored
  }
  has(value) {
    let node = this.#root;
    while (node) {
      if (value === node.value) return true;
      node = value < node.value ? node.left : node.right; // go one way only
    }
    return false;
  }
  inOrder() { // in-order walk → SORTED values
    const out = [];
    (function walk(node) {
      if (!node) return;
      walk(node.left); out.push(node.value); walk(node.right);
    })(this.#root);
    return out;
  }
}

const tree = new BST();
[8, 3, 10, 1, 6, 14].forEach((n) => tree.insert(n));
console.log('BST sorted   :', tree.inOrder()); // => [ 1, 3, 6, 8, 10, 14 ]
console.log('BST has 6    :', tree.has(6));    // => true
console.log('BST has 7    :', tree.has(7));    // => false

```

## // — 6. GRAPH (adjacency list) —————

```

// A set of nodes ("vertices") connected by edges. The adjacency-list form maps

```

```
// each node → an array of its neighbours. Models networks: friends, roads,
// dependencies, the web. (Traversal – BFS/DFS – is in lesson 49.)
class Graph {
  #adj = new Map();
  addNode(node) {
    if (!this.#adj.has(node)) this.#adj.set(node, []);
    return this;
  }
  addEdge(a, b) { // undirected: link both ways
    this.addNode(a).addNode(b);
    this.#adj.get(a).push(b);
    this.#adj.get(b).push(a);
    return this;
  }
  neighbours(node) { return this.#adj.get(node) ?? []; }
  get nodes() { return [...this.#adj.keys()]; }
}

const social = new Graph();
social.addEdge('Ana', 'Bob').addEdge('Ana', 'Cy').addEdge('Bob', 'Dee');
console.log("graph Ana's friends:", social.neighbours('Ana')); // => [ 'Bob', 'Cy' ]
```

```
/* WHICH STRUCTURE, WHEN? -----
*   Stack ..... undo, parsing, depth-first work, the call stack
*   Queue ..... scheduling, buffering, breadth-first work
*   Linked list ... O(1) front insert/remove; teaching pointers
*   Hash map ..... O(1) key-value lookup (you'll usually just use Map)
*   BST ..... kept-sorted data with O(log n) search/insert (balanced)
*   Graph ..... relationships/networks; pair with BFS/DFS (lesson 49)
*
* Big-O recap (lesson 41): array index = O(1), array search = O(n),
*   Set/Map lookup = O(1), balanced BST search = O(log n).
* ----- */

/* PRACTICE -----
*   1. Add a `reverse()` method to LinkedList that flips the chain in place.
*   2. Use your Stack to reverse the characters of a string.
*   3. Add `min()`/`max()` to the BST (hint: go all the way left / right).
*   4. Add a directed-edge option to Graph (link only a → b, not both ways).
* ----- */
```

## 49 • ALGORITHMS — the problem-solving toolkit

---

```
/* =====
* 49 • ALGORITHMS – the problem-solving toolkit
* =====
* Run: node lessons/49-algorithms.js
* 
```

```

* WHAT YOU'LL LEARN
*   The algorithm patterns that show up in real code AND every interview:
*   • Searching: linear vs binary search
*   • Sorting: bubble (to learn) vs merge & quick (to use)
*   • Two-pointer & sliding-window techniques
*   • Recursion & dynamic programming (memoization)
*   • Graph traversal: BFS and DFS (uses lesson 48's structures)
*
* Read each one with its Big-O (lesson 41) in mind: the GOAL is rarely "make it
* work" – it's "make it work without blowing up as the input grows."
* ===== */

// — 1. LINEAR SEARCH – O(n) —————
// Check items one by one. Works on ANY array. Slow on large data.
function linearSearch(arr, target) {
  for (let i = 0; i < arr.length; i++) if (arr[i] === target) return i;
  return -1; // not found
}
console.log('linear:', linearSearch([5, 3, 8, 1], 8)); // => 2

// — 2. BINARY SEARCH – O(log n) —————
// MUCH faster, but the array MUST be sorted. Halve the search range each step:
// look at the middle, then throw away the half that can't contain the target.
function binarySearch(sorted, target) {
  let lo = 0, hi = sorted.length - 1;
  while (lo <= hi) {
    const mid = Math.floor((lo + hi) / 2);
    if (sorted[mid] === target) return mid;
    if (sorted[mid] < target) lo = mid + 1; // target is in the right half
    else hi = mid - 1; // target is in the left half
  }
  return -1;
}
console.log('binary:', binarySearch([1, 3, 5, 8, 13, 21], 13)); // => 4
// 1M items: linear ≈ 1,000,000 checks; binary ≈ 20. That's the power of O(log n).

// — 3. BUBBLE SORT – O(n²) – learn it, don't ship it —————
// Repeatedly swap adjacent out-of-order pairs until sorted. Easy to understand,
// too slow for real data – but it teaches what sorting *is*.
function bubbleSort(arr) {
  const a = [...arr]; // copy – don't mutate the caller's array (lesson 12)
  for (let i = 0; i < a.length; i++) {
    for (let j = 0; j < a.length - 1 - i; j++) {
      if (a[j] > a[j + 1]) [a[j], a[j + 1]] = [a[j + 1], a[j]]; // swap
    }
  }
  return a;
}

```

```
console.log('bubble:', bubbleSort([5, 2, 9, 1, 5, 6])); // => [1, 2, 5, 5, 6, 9]
```

// — 4. MERGE SORT –  $O(n \log n)$  – a real "divide & conquer" sort —————

// Split the array in half, sort each half (recursively), then MERGE the two  
// sorted halves. Reliable  $O(n \log n)$  – the kind of algorithm engines use.

```
function mergeSort(arr) {
  if (arr.length <= 1) return arr; // base case
  const mid = Math.floor(arr.length / 2);
  const left = mergeSort(arr.slice(0, mid)); // sort left half
  const right = mergeSort(arr.slice(mid)); // sort right half
  return merge(left, right);
}
function merge(left, right) {
  const out = [];
  let i = 0, j = 0;
  while (i < left.length && j < right.length) { // take the smaller front item
    out.push(left[i] <= right[j] ? left[i++] : right[j++]);
  }
  return out.concat(left.slice(i), right.slice(j)); // drain whatever's left
}
console.log('merge:', mergeSort([5, 2, 9, 1, 5, 6])); // => [1, 2, 5, 5, 6, 9]
// (In practice: arr.toSorted() – lesson 37. These show HOW sorting works.)
```

// — 5. QUICK SORT –  $O(n \log n)$  average – pick a pivot, partition —————

// Choose a "pivot", put smaller items left and larger right, recurse on each.

```
function quickSort(arr) {
  if (arr.length <= 1) return arr;
  const [pivot, ...rest] = arr;
  const smaller = rest.filter((n) => n < pivot);
  const larger = rest.filter((n) => n >= pivot);
  return [...quickSort(smaller), pivot, ...quickSort(larger)];
}
console.log('quick:', quickSort([5, 2, 9, 1, 5, 6])); // => [1, 2, 5, 5, 6, 9]
```

// — 6. TWO-POINTER technique —————

// Walk two indices toward each other. Turns many  $O(n^2)$  problems into  $O(n)$ .

// Example: does a SORTED array contain two numbers that sum to a target?

```
function hasPairSum(sorted, target) {
  let lo = 0, hi = sorted.length - 1;
  while (lo < hi) {
    const sum = sorted[lo] + sorted[hi];
    if (sum === target) return [sorted[lo], sorted[hi]];
    if (sum < target) lo++; // need a bigger sum → move left pointer up
    else hi--; // need a smaller sum → move right pointer down
  }
  return null;
}
```

```
console.log('two-pointer:', hasPairSum([1, 3, 4, 7, 9], 11)); // => [4, 7]
```

```
// — 7. SLIDING WINDOW —————
```

```
// Keep a moving "window" over the data instead of recomputing from scratch.
```

```
// Example: max sum of any k consecutive numbers –  $O(n)$  instead of  $O(n*k)$ .
```

```
function maxWindowSum(arr, k) {
  let windowSum = 0;
  for (let i = 0; i < k; i++) windowSum += arr[i]; // first window
  let best = windowSum;
  for (let i = k; i < arr.length; i++) {
    windowSum += arr[i] - arr[i - k]; // slide: add new, drop oldest
    best = Math.max(best, windowSum);
  }
  return best;
}
console.log('sliding window:', maxWindowSum([2, 1, 5, 1, 3, 2], 3)); // => 9 (5+1+3)
```

```
// — 8. DYNAMIC PROGRAMMING (memoization) —————
```

```
// Naive recursive fibonacci is  $O(2^n)$  – it recomputes the same values endlessly.
```

```
// DP = remember sub-results so each is computed ONCE. Drops it to  $O(n)$ .
```

```
function fib(n, memo = new Map()) {
  if (n <= 1) return n;
  if (memo.has(n)) return memo.get(n); // already solved → reuse
  const result = fib(n - 1, memo) + fib(n - 2, memo);
  memo.set(n, result);
  return result;
}
console.log('fib(40):', fib(40)); // => 102334155 (instant, thanks to memoization)
```

```
// — 9. GRAPH TRAVERSAL – BFS & DFS —————
```

```
// Visit every node reachable from a start. BFS uses a QUEUE (lesson 48) and
```

```
// explores level by level (→ shortest path in unweighted graphs). DFS uses a
```

```
// STACK / recursion and dives deep first.
```

```
const graph = {
  A: ['B', 'C'],
  B: ['A', 'D', 'E'],
  C: ['A', 'F'],
  D: ['B'], E: ['B', 'F'], F: ['C', 'E'],
};

function bfs(graph, start) {
  const visited = new Set([start]);
  const queue = [start]; // FIFO
  const order = [];
  while (queue.length) {
    const node = queue.shift(); // take from the FRONT
    order.push(node);
```

```

    for (const next of graph[node]) {
      if (!visited.has(next)) { visited.add(next); queue.push(next); }
    }
  }
  return order;
}

function dfs(graph, start, visited = new Set(), order = []) {
  visited.add(start);
  order.push(start);
  for (const next of graph[start]) {
    if (!visited.has(next)) dfs(graph, next, visited, order); // dive deep first
  }
  return order;
}

console.log('BFS from A:', bfs(graph, 'A')); // => [ 'A', 'B', 'C', 'D', 'E', 'F' ]
console.log('DFS from A:', dfs(graph, 'A')); // => [ 'A', 'B', 'D', 'E', 'F', 'C' ]

/* COMPLEXITY CHEAT SHEET -----
*   linear search ... O(n)           binary search ... O(log n) (needs sorted)
*   bubble sort .... O(n2)         merge/quick .... O(n log n)
*   two-pointer .... O(n)           sliding window .. O(n)
*   naive fib ..... O(2n)         memoized fib .... O(n)
*   BFS / DFS ..... O(V + E)       (vertices + edges)
*
* INTERVIEW PLAYBOOK:
*   • Sorted input? → think binary search / two-pointer.
*   • "Subarray/substring of length k"? → sliding window.
*   • Overlapping sub-problems? → memoize (DP).
*   • Graph/tree/grid? → BFS (shortest path) or DFS (explore all).
*   • Always state the Big-O of your solution out loud.
* ----- */

/* PRACTICE -----
*   1. Write isPalindrome(str) with the two-pointer technique.
*   2. Implement insertionSort and compare its output to mergeSort.
*   3. Use BFS to find the SHORTEST path (list of nodes) between two graph nodes.
*   4. Solve "climbing stairs" (1 or 2 steps at a time) with memoized recursion.
* ----- */

```

## 50 · BINARY DATA — bytes, buffers, and files

---

```

/* =====
* 50 · BINARY DATA – bytes, buffers, and files
* =====
* Run: node lessons/50-binary-data.js
*       (the Blob/File/FormData parts are BROWSER APIs – shown as comments;

```

```

*      the ArrayBuffer/TypedArray parts run in Node too.)
*
* WHAT YOU'LL LEARN
*   How JavaScript handles RAW BYTES – not just strings and numbers:
*   • ArrayBuffer – a fixed block of memory
*   • Typed arrays (Uint8Array, Float64Array, ...) – views over that memory
*   • DataView – read/write mixed types at exact byte offsets
*   • Blob / File – chunks of binary data in the browser
*   • FormData – sending files and fields to a server
*   • TextEncoder / TextDecoder – text ↔ bytes
*   • SharedArrayBuffer + Atomics – memory shared across threads
*
* WHY THIS MATTERS
*   File uploads/downloads, images, audio, WebSockets (lesson 47), streaming
*   fetch responses, WebGL, and WebAssembly all move BYTES. Strings can't
*   represent that efficiently – typed arrays can.
* ===== */

// — 1. ArrayBuffer – a raw block of memory —————
// An ArrayBuffer is just N bytes. You can't read it directly – you need a VIEW.
const buffer = new ArrayBuffer(8); // 8 bytes, all zero
console.log('buffer byteLength:', buffer.byteLength); // => 8

// — 2. Typed arrays – views over a buffer —————
// A typed array reads the buffer's bytes as numbers of a fixed size/type.
//   Uint8Array   → 1 byte,   0..255           (the workhorse for raw bytes)
//   Int32Array   → 4 bytes, signed integers
//   Float64Array → 8 bytes, decimals (same precision as a normal JS number)
const bytes = new Uint8Array(buffer); // view the 8-byte buffer as 8 × uint8
bytes[0] = 255;
bytes[1] = 256; // wraps! 256 doesn't fit in a byte → becomes 0
bytes[2] = 42;
console.log('uint8 view:', bytes); // => Uint8Array(8) [ 255, 0, 42, 0, 0, 0, 0, 0 ]

// Multiple views can share ONE buffer – they read the same bytes differently:
const ints = new Int32Array(buffer); // 8 bytes → two 32-bit ints
console.log('same bytes as int32:', ints); // values depend on byte order

// Typed arrays have most array methods (map/filter/reduce, lesson 13) but a
// FIXED length and a single number type – that's what makes them fast & compact.
const doubled = Uint8Array.from([1, 2, 3], (n) => n * 2);
console.log('typed map:', doubled); // => Uint8Array(3) [ 2, 4, 6 ]

// — 3. DataView – precise, mixed-type access —————
// When a binary format packs different types at known offsets (file headers,
// network protocols), DataView reads/writes them exactly – and lets you control
// "endianness" (byte order).
const dv = new DataView(new ArrayBuffer(8));

```

```

dv.setInt16(0, 300);           // write a 16-bit int at byte 0
dv.setFloat32(2, 3.14, true); // write a 32-bit float at byte 2 (true = little-endian)
console.log('DataView int16 :', dv.getInt16(0));           // => 300
console.log('DataView f32   :', dv.getFloat32(2, true).toFixed(2)); // => 3.14

```

```

// — 4. Text ↔ bytes (TextEncoder / TextDecoder) —————
// Strings are UTF-16 in memory; on the wire/disk text is usually UTF-8 bytes.
const encoder = new TextEncoder();
const encoded = encoder.encode('Hi 🙌'); // string → Uint8Array of UTF-8 bytes
console.log('encoded bytes:', encoded); // emoji takes 4 bytes

const decoder = new TextDecoder();
console.log('decoded back:', decoder.decode(encoded)); // => Hi 🙌

```

```

// — 5. Blob & File (BROWSER) – chunks of binary data —————
/* A Blob is an immutable bag of bytes with a MIME type. A File is a Blob with a
 * name (what you get from <input type="file">). Both are how the browser holds
 * images, downloads, and uploads.
 *
 * // Make a Blob and trigger a download:
 * const blob = new Blob(['name,score\nAna,10'], { type: 'text/csv' });
 * const url = URL.createObjectURL(blob);
 * const a = Object.assign(document.createElement('a'),
 *                          { href: url, download: 'data.csv' });
 * a.click();
 * URL.revokeObjectURL(url); // free the memory when done
 *
 * // Read a file the user picked:
 * input.addEventListener('change', async () => {
 *   const file = input.files[0];           // a File (Blob + name + size)
 *   const text  = await file.text();       // read as string
 *   const bytes = await file.arrayBuffer(); // read as raw bytes
 *   console.log(file.name, file.size, text);
 * });
 */

```

```

// — 6. FormData (BROWSER) – send files + fields to a server —————
/* FormData builds a multipart request body – the standard way to upload files
 * alongside text fields. fetch (lesson 25) sets the right headers automatically.
 *
 * const form = new FormData();
 * form.append('username', 'ana');
 * form.append('avatar', fileInput.files[0]); // a File/Blob
 * await fetch('/upload', { method: 'POST', body: form });
 * // ^ do NOT set Content-Type yourself – the browser adds the boundary.
 *
 * // FormData can also read an entire <form> at once:

```



```

*   const form2 = new FormData(document.querySelector('form'));
*   for (const [key, value] of form2) console.log(key, value);
*/

// — 7. Streaming binary (concept) —————
/* Large responses don't have to load all at once. fetch bodies are streams of
 * Uint8Array chunks – read them progressively (great for progress bars):
 *
 *   const res = await fetch('/big-file');
 *   const reader = res.body.getReader();      // a stream of byte chunks
 *   let received = 0;
 *   while (true) {
 *     const { done, value } = await reader.read(); // value = Uint8Array chunk
 *     if (done) break;
 *     received += value.length;
 *     console.log('received', received, 'bytes');
 *   }
 *   (In Node, see streams in lesson 45 – same idea, different API.)
 */

// — 8. SharedArrayBuffer + Atomics – memory shared between threads —————
// A normal ArrayBuffer is copied (transferred) when sent to a Web Worker
// (lesson 46) / Node worker_thread (lesson 45). A SharedArrayBuffer is the SAME
// memory seen by BOTH threads at once – true shared state, no copying.
// (Browsers require special COOP/COEP headers to enable it, for security.)
const shared = new SharedArrayBuffer(8); // 8 bytes, shareable across threads
const sharedView = new Int32Array(shared);

// When two threads touch the same memory you get RACE CONDITIONS. `Atomics`
// provides operations that are guaranteed to complete without interruption:
Atomics.store(sharedView, 0, 100);      // safely write slot 0
Atomics.add(sharedView, 0, 5);           // atomic read-modify-write → 105
console.log('Atomics value:', Atomics.load(sharedView, 0)); // => 105

// Atomics.wait / Atomics.notify let one thread SLEEP until another signals it –
// the building block of locks and cross-thread coordination:
// // worker: Atomics.wait(sharedView, 0, 105); // block while slot 0 === 105
// // main:   Atomics.store(sharedView, 0, 0); Atomics.notify(sharedView, 0);
// △ This is advanced/rare. You only need it for high-performance parallel work
// (image/audio processing, simulations, WASM threads). For normal worker
// communication, plain postMessage (lesson 46) is simpler and safer.

/* MENTAL MODEL -----
 *   ArrayBuffer = the memory (bytes).   Typed array / DataView = how you READ it.
 *   SharedArrayBuffer = memory shared across threads; Atomics = safe access to it.
 *   Blob/File      = bytes + a type/name (browser).   FormData = a way to SEND them.
 */

```

```

*   TextEncoder/Decoder = the bridge between text and UTF-8 bytes.
*
*   Reach for these only when you actually handle binary: uploads, media,
*   protocols, WebGL/WASM. For everyday data, JSON + objects (lessons 14/25)
*   are simpler and correct.
*   ----- */

/* PRACTICE -----
*   1. Make a Uint8Array of [72,105] and TextDecoder.decode it (what word?).
*   2. Use a DataView to store two Float32 values in one 8-byte buffer, read back.
*   3. (Browser) Build a CSV string, wrap it in a Blob, and download it.
*   4. (Browser) Add a file <input>, read the chosen file with file.text().
*   ----- */

```

## 51 • BIGINT & LANGUAGE CORNERS

```

/* =====
* 51 • BIGINT & LANGUAGE CORNERS
* =====
* Run:  node lessons/51-bigint-and-language-corners.js
*
* WHAT YOU'LL LEARN
*   The last few language features that round out "I know ALL of JavaScript":
*   • BigInt – integers bigger than Number can safely hold
*   • Object.fromEntries – build an object from key/value pairs
*   • Array.prototype.flatMap – map + flatten in one step
*   • Labeled statements – break/continue an OUTER loop
*   • eval & the Function constructor – running code from a string (and why
*     you almost never should)
*
* These are niche – you won't reach for them daily – but knowing they exist
* (and their gotchas) is the difference between "most of JS" and "all of JS".
* ===== */

// — 1. BigInt – arbitrary-precision integers —————
// Regular numbers are 64-bit floats. They can only represent integers EXACTLY
// up to Number.MAX_SAFE_INTEGER. Past that, math silently goes wrong:
console.log(Number.MAX_SAFE_INTEGER);      // => 9007199254775807 (2^53 - 1)
console.log(9007199254740991 + 1);         // => 9007199254740992 ✅
console.log(9007199254740991 + 2);         // => 9007199254740992 ❌ (should be ...93!)

// BigInt fixes this – write `n` after an integer literal, or call BigInt():
const big = 9007199254740991n;
console.log(big + 2n);                     // => 9007199254740993n ✅ exact
console.log(BigInt(42), typeof 42n);       // => 42n 'bigint'

// Arithmetic works as usual – but division TRUNCATES (BigInt has no decimals):
console.log(10n / 3n);                     // => 3n (not 3.333...)

```

```

console.log(2n ** 64n); // => 18446744073709551616n (huge, exact)

// ⚠ You CANNOT mix BigInt and Number in math – it throws on purpose:
try { 1n + 1; } catch (e) { console.log('mix error:', e.message); } // TypeError
console.log(1n + BigInt(1)); // => 2n (convert first)

// Comparisons DO work across types (no error):
console.log(2n > 1, 2n === 2, 2n == 2); // => true false true
// ^ === is false: different TYPES (bigint vs number)

// Gotchas: no Math.* (Math.sqrt(4n) throws); JSON.stringify can't serialize it:
try { JSON.stringify({ id: 5n }); } catch (e) { console.log('json error:', e.message); }
// Fix: convert to string in a replacer → JSON.stringify(obj, (k,v) =>
//     typeof v === 'bigint' ? v.toString() : v)
// USE CASES: database BIGINT ids, Twitter/Discord "snowflake" ids, nanosecond
// timestamps, cryptography – anywhere integers exceed 2^53.

// — 2. Object.fromEntries – [[k,v]] → { k: v } —————
// The inverse of Object.entries (lesson 14). Turns pairs back into an object.
const pairs = [['name', 'Ana'], ['age', 30]];
console.log(Object.fromEntries(pairs)); // => { name: 'Ana', age: 30 }

// Killer combo: transform an object by going object → entries → map → object.
const prices = { apple: 1, pear: 2 };
const doubled = Object.fromEntries(
  Object.entries(prices).map(([k, v]) => [k, v * 2])
);
console.log(doubled); // => { apple: 2, pear: 4 }

// Also converts a Map (lesson 16) or URLSearchParams (lesson 25) into an object:
console.log(Object.fromEntries(new Map([['a', 1]]))); // => { a: 1 }

// — 3. Array.prototype.flatMap – map, then flatten one level —————
// Equivalent to arr.map(fn).flat(1), but in a single pass. Great when each item
// maps to ZERO, ONE, or MANY results.
console.log([1, 2, 3].flatMap((n) => [n, n * 10])); // => [1, 10, 2, 20, 3, 30]

// Return [] to DROP an item, [x] to keep it → map + filter in one step:
console.log([1, -2, 3, -4].flatMap((n) => (n > 0 ? [n] : []))); // => [1, 3]

// Split sentences into words across an array:
console.log(['a b', 'c d'].flatMap((s) => s.split(' '))); // => ['a','b','c','d']
// (Note: flatMap only flattens ONE level; use .flat(Infinity) for deep nesting.)

// — 4. Labeled statements – break/continue an OUTER loop —————
// By default break/continue affect the NEAREST loop. A label lets you target an

```

```
// outer one – handy for breaking out of nested loops without a flag variable.
outer: for (let i = 0; i < 3; i++) {
  for (let j = 0; j < 3; j++) {
    if (i + j === 3) {
      console.log(`breaking outer at i=${i}, j=${j}`);
      break outer;          // jumps ALL the way out, not just the inner loop
    }
  }
}
// `continue label` similarly skips to the next iteration of the labeled loop.
// Use sparingly – often a helper function with an early `return` is clearer.
```

```
// — 5. eval & the Function constructor – code from strings (avoid!) —————
// eval() runs a string as code IN THE CURRENT SCOPE. It works...
console.log(eval('2 + 3'));          // => 5
// ...but it's a security hole (runs ANY code, incl. attacker input), it's slow
// (defeats engine optimizations), and it can read/modify your local variables.
// RULE: never eval untrusted input. You almost never need eval at all –
//   JSON.parse for data, optional chaining/bracket access for dynamic props.
```

```
// The Function constructor is slightly less dangerous: it builds a function from
// a string but runs in the GLOBAL scope (it can't see your locals):
const addFn = new Function('a', 'b', 'return a + b');
console.log(addFn(4, 5));            // => 9
// Same security/perf caveats apply. Legit uses are rare (templating engines,
// sandboxes, REPLs) – for everyday code, treat both as "do not use".
```

```
/* QUICK REFERENCE -----
*   BigInt ..... 123n / BigInt(x) – exact integers past 2^53; no mixing
*                               with Number, no decimals, no Math, no direct JSON.
*   Object.fromEntries  pairs → object; pair with Object.entries to transform.
*   flatMap ..... map + flat(1); return [] to drop, [x] to keep.
*   label: loop ..... break/continue an outer loop by name.
*   eval/Function ..... run strings as code – security + perf risk. Avoid.
* ----- */
```

```
/* PRACTICE -----
*   1. Compute 100! (factorial) with BigInt – impossible to do exactly with Number.
*   2. Use Object.entries + map + Object.fromEntries to UPPERCASE every value
*      of { a: 'x', b: 'y' }.
*   3. Use flatMap to turn [{tags:['js','ts']},{tags:['css']}] into one tag list.
*   4. Use a labeled loop to find the first pair (i,j) in two arrays that sums to 7.
* ----- */
```

## 52 · INTERNATIONALIZATION (the full Intl API)

---

```
/* =====
* 52 · INTERNATIONALIZATION (the full Intl API)
* =====
* Run: node lessons/52-internationalization.js
*
* WHAT YOU'LL LEARN
*   The built-in `Intl` API formats data the way each LANGUAGE/REGION expects –
*   no libraries needed. You met NumberFormat (lesson 06) & DateTimeFormat
*   (lesson 32); here are the rest:
*   • Intl.Collator – locale-aware sorting & comparison
*   • Intl.PluralRules – pick the right plural form
*   • Intl.RelativeTimeFormat – "3 days ago", "in 2 hours"
*   • Intl.ListFormat – "A, B, and C"
*   • Intl.Segmenter – split text into words/sentences (any language)
*
* BIG IDEA: never hand-build dates, numbers, plurals, or "x ago" strings – the
* platform already knows the rules for ~hundreds of locales. (See lesson 47 for
* i18n in the bigger picture: translations, RTL, etc.)
* ===== */

// — Recap: the two you already know —————
console.log(new Intl.NumberFormat('en-IN', { style: 'currency', currency:
'INR' }).format(250000));
// => ₹2,50,000.00 (note the Indian digit grouping – Intl handles it)
console.log(new Intl.DateTimeFormat('en-GB', { dateStyle: 'full' }).format(new Date('2026-06-
05')));
// => Friday, 5 June 2026

// — 1. Intl.Collator – sort/compare the way a language does —————
// String sort with < or default .sort() is by code-point, which is WRONG for
// accents and non-English alphabets. Collator sorts correctly per locale.
const words = ['zebra', 'apple', 'Ärger', 'apple pie', 'Apfel'];
const collator = new Intl.Collator('de', { sensitivity: 'base' }); // German rules
console.log('collator sort:', [...words].sort(collator.compare));
// 'ä' sorts near 'a' (correct in German) – a naive sort would misplace it.

// `numeric: true` makes "file2" come before "file10" (human/natural order):
const files = ['file10', 'file2', 'file1'];
console.log('natural sort:', [...files].sort(new Intl.Collator(undefined, { numeric:
true }).compare));
// => [ 'file1', 'file2', 'file10' ] (not file1, file10, file2)

// — 2. Intl.PluralRules – choose the correct plural CATEGORY —————
// English has 2 forms (1 item / N items). Other languages have up to 6. Don't
// hard-code `n === 1 ? 'item' : 'items'` – ask PluralRules which form to use.
```

```

const enPlural = new Intl.PluralRules('en-US');
function items(n) {
  const form = enPlural.select(n);          // 'one' | 'other' (for English)
  const label = { one: 'item', other: 'items' }[form];
  return `${n} ${label}`;
}
console.log(items(1), '/', items(5));      // => 1 item / 5 items
// Ordinals too (1st, 2nd, 3rd, 4th):
const ordinal = new Intl.PluralRules('en-US', { type: 'ordinal' });
const suffix = { one: 'st', two: 'nd', few: 'rd', other: 'th' };
console.log([1, 2, 3, 4, 11, 21].map((n) => `${n}${suffix[ordinal.select(n)]}`).join(' '));
// => 1st 2nd 3rd 4th 11th 21st

// — 3. Intl.RelativeTimeFormat – "3 days ago" / "in 2 hours" —————
// Stop writing your own "time ago" helper – this one is locale-correct.
const rtf = new Intl.RelativeTimeFormat('en', { numeric: 'auto' });
console.log(rtf.format(-1, 'day'));        // => yesterday   ('auto' → words)
console.log(rtf.format(3, 'day'));         // => in 3 days
console.log(rtf.format(-2, 'hour'));       // => 2 hours ago

// A tiny helper: turn a past Date into a friendly relative string.
function timeAgo(date) {
  const seconds = Math.round((date - new Date('2026-06-05T12:00:00')) / 1000);
  const units = [['day', 86400], ['hour', 3600], ['minute', 60], ['second', 1]];
  for (const [unit, secs] of units) {
    if (Math.abs(seconds) >= secs || unit === 'second') {
      return rtf.format(Math.round(seconds / secs), unit);
    }
  }
}
console.log(timeAgo(new Date('2026-06-04T12:00:00'))); // => yesterday

// — 4. Intl.ListFormat – join a list the way humans write it —————
// "A, B, and C" (note the Oxford comma + "and") – and it localizes.
const fruits = ['apples', 'oranges', 'pears'];
console.log(new Intl.ListFormat('en', { type: 'conjunction' }).format(fruits));
// => apples, oranges, and pears
console.log(new Intl.ListFormat('en', { type: 'disjunction' }).format(fruits));
// => apples, oranges, or pears

// — 5. Intl.Segmenter – split text into words/sentences/graphemes —————
// "Just split on spaces" breaks for many languages (Chinese/Japanese have no
// spaces) and for emoji. Segmenter splits text correctly anywhere.
const seg = new Intl.Segmenter('en', { granularity: 'word' });
const segments = [...seg.segment('Hello, world! 🙌')];
const realWords = segments.filter((s) => s.isWordLike).map((s) => s.segment);
console.log('words:', realWords);          // => [ 'Hello', 'world' ]

```

```
// Grapheme granularity counts USER-PERCEIVED characters (emoji = 1, not many):
const graphemes = [...new Intl.Segmenter().segment('a🐼b')];
console.log('grapheme count:', graphemes.length); // => 3 ('.length' would say 8+)
```

```
/* WHEN TO REACH FOR EACH -----
 *   NumberFormat ..... currency, percent, units, digit grouping (lesson 06)
 *   DateTimeFormat .... locale dates & times (lesson 32)
 *   Collator ..... sorting/searching strings correctly (accents, numeric)
 *   PluralRules ..... "1 item" vs "5 items", ordinals (1st/2nd/3rd)
 *   RelativeTimeFormat "3 days ago", "in 2 hours"
 *   ListFormat ..... "A, B, and C"
 *   Segmenter ..... counting/splitting words & characters in ANY language
 *
 * All of Intl is built into Node and every modern browser – zero dependencies.
 * ----- */
```

```
/* PRACTICE -----
 *   1. Sort ['café','car','cab'] with an Intl.Collator and compare to a plain sort.
 *   2. Write likes(n) → "1 like" / "N likes" using Intl.PluralRules.
 *   3. Format "in 5 minutes" and "2 weeks ago" with Intl.RelativeTimeFormat.
 *   4. Join ['HTML','CSS','JS'] as "HTML, CSS, and JS" with Intl.ListFormat.
 *   5. Use Intl.Segmenter to correctly count the characters in '👍🐼🎉'.
 *   ----- */
```